PART III

ADVANCED CLINICAL TOPICS

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17 Evaluation of Outcomes

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CHAPTER OUTLINE

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Context of Evaluating Endodontic Outcomes

The development of medical and dental practices has been guided by prevailing philosophies and consensus of expert opinions. The foundations of modern endodontics were shaken when Billings[97] brought the apparent relationship between oral sepsis and bacterial endocarditis to the attention of dentistry and medicine. He had strengthened the concept further from Miller,[134] who had coined the term *focus of infection* to highlight a possible link between "mouth germs" and systemic disease. The disastrous consequences of the focal infection era were sealed when Hunter[96] delivered his famous address at McGill University. Fears of fatal oral sepsis from deficient root canal treatments led to widespread extraction of pulpless teeth. Endodontics virtually disappeared from many dental schools, whereas in some areas, treatment was restricted to anterior teeth. The focal infection theory reigned for about 50 years[83] until around 1940.

The discipline of endodontics was rescued by individual, diligent practitioners in Europe and the United States who meticulously recorded their treatments, as well as their outcomes, to demonstrate the effectiveness of the procedures in controlling root canal infection. It was through these individual endeavors of merit that the reputation of endodontic procedures was restored and the discipline granted its specialist status in 1952 in the United States.[59]

Toward the 1990s, the improvements in general health and longer survival of Western populations as well as their teeth, coupled with the rising costs of health care for longer-living individuals, prompted a host of reevaluations of the altered economic burdens on society. Among them, the *cost-effectiveness of treatment procedures* for management of diseases loomed large. Thus began the era of evidence-based medicine and dentistry, with its emphasis on costs, benefits, and outcomes of treatment. Attempts to pool outcome data for greater power brought with it the realization that studies originating from different centers varied in diverse ways, leading to very heterogeneous data that challenged formal attempts at drawing definitive conclusions. Characterization of data types and its quality prompted calls for standardization of approaches in measuring outcomes to enable more meaningful pooling.

Because of the better established science of evidence assessment, a modern revisitation of the "focal infection

era"[94] did not lead to the same threat to the endodontic discipline that was evident at the beginning of the 20th century. In fact, it was suggested as an opportunity to secure research funding for assessing and managing the importance of dental care on systemic health.

Interestingly, the modern threat to endodontics was posed by the cost-economic pressures exerted on treatment planning decisions centering on the question of whether to "save the tooth" or extract it with replacement by an implant-supported crown.[238] Once again, the science of evidence-based practice has helped to avert irrational treatment recommendations leaning toward extraction of savable teeth.[54,100,103]

TYPES OF DISEASE AND THEIR TREATMENT

The statement, "Endodontists provide endodontic treatment to manage endodontic disease" is a gross oversimplification that masks important subtleties in recognition of the nature of the diseases and how best to address them. Endodontists manage inflammation of the specialized connective tissues within and surrounding the teeth; more specifically, they deal with inflammation that generally commences in the pulp tissue and progresses to the periradicular tissues via portals of communication that convey the neurovascular supply. Incipient or established pulpitis may be managed by vital pulp therapy when judged reversible, largely because a sufficient body of pulp tissue remains healthy and unaffected. More advanced pulpitis that approaches the foramina into the periodontal ligament may require root canal treatment. When pulpal inflammation, necrosis, or infection encroaches on the apical periodontal ligament, root canal treatment is required. Apical periodontitis that persists despite technically adequate root canal treatment may require root canal retreatment, periradicular surgery, or extraction to manage the source of the persistent disease. *Endodontic treatment* is therefore a collective and nonspecific term for a range of procedures directed at managing the spread of pulpal inflammation or infection. Endodontic treatment encompasses the following procedures:

1. Vital pulp therapy (indirect pulp therapy, direct pulp capping, pulpotomy, regenerative pulp therapy)
2. Nonsurgical root canal treatment
3. Nonsurgical root canal retreatment
4. Surgical retreatment

The ideal outcome for endodontic treatment consists of the controlled reduction of inflammation, accompanied by healing through regeneration, although sometimes repair may follow instead. Unfortunately, none of the involved tissues are within the direct sight of the clinician, hidden as they are by their housings of hard tooth structure or alveolar bone and their gingival/mucosal coverings. Consequently, surrogate measures must be employed to evaluate the presence or absence of the disease process and its resolution. The evaluation process is further complicated by the lack of direct correlation between measures of the disease process and its clinical manifestation.

What Are Surrogate Outcome Measures?

Signs of acute inflammation are classically described in the "triple response" exhibited by mechanically injured skin, which includes altered color (redness), texture/contour (swelling), and sensation (pain), which are directly accessible and viewable. These changes have a direct correlation with histopathologic and molecular alterations. Chronic inflammation does not necessarily exhibit the same highly visible signs and symptoms of its histopathologic character. When hidden from view, as are pulpal and periradicular tissues, the task of recognizing the presence or absence of chronic inflammation is even more challenging. The clinician is therefore left with indirect or associated (proxy) changes by which to judge the presence or absence of disease; these are called "surrogate" measures. Some of these may have to be inferred through associations, some may be directly visualized, and others are only indirectly "viewable" through various imaging techniques (such as radiography). The process therefore calls upon the clinician to assimilate various sources of information to form a judgment about the presence or absence of disease.

TYPES OF OUTCOME MEASURES

In its broadest sense, an outcome measure for a treatment intervention may constitute any consistently anticipated and measurable consequence of the treatment. The prepared shape of the root canal system, bacterial load reduction, and technical quality of the root filling may all be regarded as outcome measures by this definition. But the ultimate *clinical* measure of a treatment outcome is assessing the prevention and resolution of disease.

The outcome of endodontic treatment may be assessed in four dimensions as in other medical disciplines.[9] The first dimension is physical/physiologic and related to presence or absence of pulpal/periapical health/disease, pain, and function. The second dimension assesses longevity or tooth survival. The third dimension relates to economics and assesses direct and indirect costs. Finally, the fourth dimension examines psychologic aspects involving perceptions of oral health–related quality of life (OHRQoL) and aesthetics.

What Is the Purpose of Evaluating Outcomes?

Apart from the importance of developing a solid foundation of evidence-based practice, it is important to evaluate treatment outcomes for a number of reasons.

EFFECTIVENESS OF PROCEDURES

First, treatment procedures must be effective. Otherwise, there is no reason to recommend them to patients as a treatment option. The patient must be properly informed as to the risks, benefits, and potential outcomes of the offered treatment. The availability of pooled outcome data and consensus guidelines offers both patients and endodontists reassurance and confidence in the validity and predictability of the offered procedure. However, the pooled average outcome data may not pertain if the clinician does not have the requisite experience, skill, and personal outcomes matching at least these average figures of performance. Personal outcome data offer the patient more precise measures for comparison and expectation. It also aids practitioners in refining

their technique and knowledge to further improve their outcomes and, ultimately, to help improve the overall pooled data.[34] The treating clinician must assess his or her own personal outcome data and personal expectation of success. If the clinician is uncertain of a result or feels that the outcome could be enhanced by treatment from another clinician, then referral to someone else more qualified is paramount.

FACTORS AFFECTING OUTCOMES

Pooling data (that is preferably homogeneous) offers the potential to evaluate and prioritize the factors that exert a dominant influence on outcomes. In this way, protocols for treatment may be improved in a progressive fashion, bringing about the perfect meld of technical, clinical, and biologic insight necessary for the highest performance and most predictable outcome. Evaluation of factors affecting treatment outcomes arguably provides the strongest influence for making changes that will improve the clinical outcomes. Weighing the relative importance of individual factors should help to identify the key biologic factors at play and how best to manage them from a clinical or technical perspective.

VALUE FOR PROGNOSTICATION

Prognostication, which could be defined as the prediction, projection, prophesizing, or foretelling the likely outcome of treatment, is not often well defined in endodontics. The overall prognosis of a tooth depends on the interaction among three individual and often independent variables, including endodontic, periodontal, and restorative prognoses. Each has a set of subsidiary factors that must be considered to derive an overall prognosis. Finally, the tooth in question must be considered from a strategic perspective, relative to its position in the dental arch and the contribution it makes in the dynamic occlusion.

A deeper analysis of the factors affecting outcomes as prognostic indicators is merited in order to clarify the degree of complexity inherent to the problem. A comparison may be made with the randomized controlled trial of a drug as a treatment intervention. The drug therapy is delivered as a clearly prescribed and standardized dose regimen delivered by specified timings to effect an anticipated blood or target tissue concentration for a specified duration. Data recording is confined to compliance with the prescription and possibly to the checking of actual blood levels achieved as well as the final outcome effect.

In stark contrast, a surgical intervention, regardless of any standardization of the described procedural protocol, is subject to enormous variation in its delivery dependent on interpretation and execution of the protocol by each operator. To add to the levels of complexity and variation, surgical protocols are often multistep, sequential procedures, each prospective step dependent on the previous for its efficacy. Even characterizing and accurately recording any variations in treatment protocols becomes a challenge because many aspects and steps must be documented. This leads to the next challenge, that of demonstrating compliance with the accuracy of such complex data gathering. Furthermore, it becomes important from an analytic perspective to not only consider the effect of individual steps (factors) alone but also any interactive effect between the multiple steps inherent in the procedure. Even when such comprehensive prospective outcome data are available, utilizing these data to individually calculate the prognosis in a particular scenario may be challenging. Clinically, there may be two distinct ways to estimate prognosis: (1) to apply heuristic principles and intuitively gauge the effect of dominant factors or, at the other end of the spectrum, (2) to mathematically input comprehensive data into an algorithmic model to calculate the estimated outcome. An even more sophisticated approach may be to use mathematical modeling to calculate iterations on variations within the system. At present, the evidence base is insufficient to allow such a sophisticated approach.

It therefore follows that evaluating outcomes and the factors that affect them are fundamental to the creation of a suitable data foundation for future application to endodontic prognostication. The endodontic treatment option is in competition with other alternative therapies and therefore for the discipline to survive and thrive, a suitable blend of efficacy, efficiency, utility, predictability, and cost-effectiveness data must be generated for its principal therapies. Achievement of such a goal will require a diligent and conscientious collection of data from many sources. Analyses of such powerful datasets will ultimately yield proper insight into outcomes of the procedures we use. Coupled with biologic insight of the problem, it will become possible to derive new therapeutic solutions necessary to deliver them.

Outcome Measures for Endodontic Treatment

Before determining the success or failure of an endodontic procedure, the parameters as to what is considered a success or a failure have to be discussed. These are termed the *outcome measures* of the procedures.

OUTCOME MEASURES FOR VITAL PULP THERAPY PROCEDURES

The techniques used to maintain the vitality and health of pulps in teeth with extensive caries or those with traumatic/mechanical exposures include (1) indirect pulp capping with one-step or stepwise caries excavation, (2) direct pulp capping of exposed pulps, and (3) partial/full pulpotomy procedures for more extensively involved pulps. The surrogate outcome measures adopted in studies include (1) clinical success (pulp sensitivity to cold test and absence of pain, soft-tissue swelling, sinus tract, periradicular radiolucency, or pathologic root resorption), (2) patient satisfaction, (3) adverse events (pain, swelling, tooth fracture), and (4) tooth extraction.[89,136] Although not providing a specific follow-up strategy, the quality guidelines of the European Society of Endodontology (ESE)[61] suggests "initial review at no longer than 6 months and thereafter at further regular intervals." They also provide the criteria for judging the favorable outcome of vital pulp therapy (Table 17.1). There are substantial differences in the criteria adopted in published clinical studies on outcomes of vital pulp therapy[89,136] with few studies adopting all of the criteria. The frequency of radiologic review adopted in published studies also varies substantially, with some recommending the first review at

Table 17.1 Criteria for Assessing the Outcome of Pulp Therapy

Quality Guidelines for Endodontic Treatment: Consensus Report of the European Society of Endodontology (2006)	Guidelines on Pulp Therapy for Primary and Immature Permanent Teeth (American Academy of Pediatric Dentistry, 2014)
1. Normal response to pulp sensitivity tests (when feasible)	1. Tooth vitality maintained
2. Absence of pain and other symptoms	2. Absence of posttreatment signs or symptoms such as sensitivity, pain, or swelling
3. Radiologic evidence of dentinal bridge formation	3. Occurrence of pulp healing and reparative dentin formation
4. Radiologic evidence of continued root formation in immature teeth	4. Absence of radiographic evidence of internal or external root resorption, periapical radiolucency, abnormal calcification, or other pathologic changes
5. Absence of clinical and radiographic signs of internal root resorption and apical periodontitis	5. Teeth with immature roots should show continued root development and apexogenesis

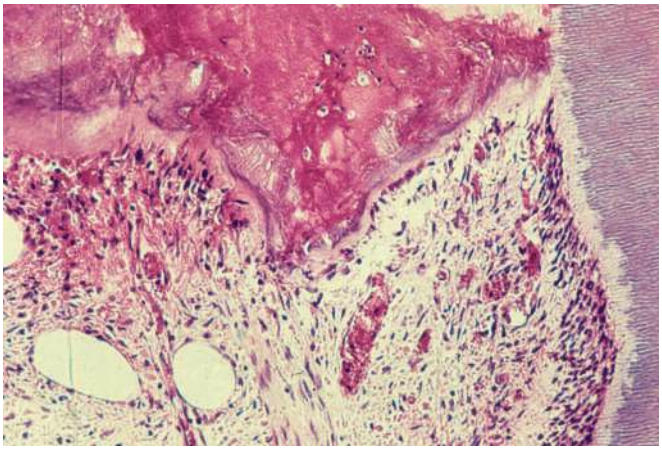


Fig. 17.1 Histologic view of intact calcific bridge formation following pulpotomy.

1 month followed by subsequent 3 monthly reviews.[172,242] The relative benefit of acquired information versus unnecessary radiation to patients has been questioned.[66] An initial assessment at 6 to 12 weeks, followed by a review 6 and 12 months after treatment, seems to have been accepted and is recommended. Some studies reported up to 10 years follow-up on case series demonstrating lower success rates, inferring the possibility of slow spread of pulpal inflammation and late failures. The conclusion is that longer follow-up periods are merited.

The process at each review consists of obtaining a history of symptoms, coupled with an examination to determine the presence or absence of tenderness to palpation of adjacent soft tissues, tenderness to pressure and percussion of the tooth, signs of radiographic pulpal, periapical changes, and responses to pulp tests. The accuracy of pulp tests may be limited in pulpotomized teeth because of the distance of the remaining pulp tissue from the tooth's surface. In the case of pulp capping and pulpotomy, additional tests include radiographic verification of the presence of the calcific barrier (Fig. 17.1) and its integrity by removal of the dressing and direct probing. Although an initial examination at 6 weeks has been suggested, this can be modified for the radiographic assessment. If there is no evidence of complete bridge formation, the treatment is considered failed and root canal treatment should be considered. In addition, in the case of incompletely formed roots there should be radiographic evidence of continued root development (Fig. 17.2).

OUTCOME MEASURES FOR NONSURGICAL ROOT CANAL TREATMENT AND RETREATMENT

Root canal treatment may be employed either to prevent or resolve periapical disease. Given that periapical lesions develop as a result of the *interaction* between bacteria (and their by-products) and the host defenses, it is clear that prevention or resolution of the disease process depends on preventing or terminating this interaction.

Prevention of apical periodontitis applies to the clinical situation where it is judged that the pulp is irreversibly inflamed, necrotic, or infected to the extent that vital pulp therapy would not resolve the problem, which therefore requires pulpectomy. The implication is that the procedure

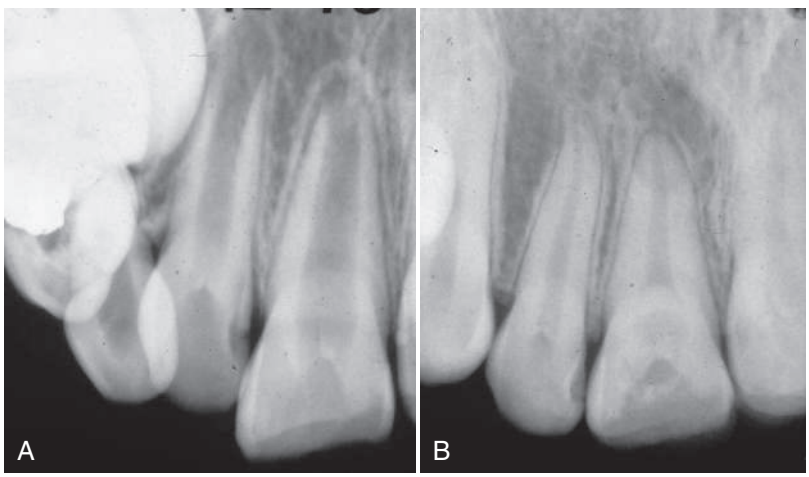


Fig. 17.2 Immature maxillary right lateral and central incisors **(A)** showing continued root development following pulpotomies **(B)**.

demands asepsis as a prime requirement. The precise technical details of the protocol may not be so important, as long as they are focused on effective and aseptic removal of the pulp tissue. This is validated by the high chance of retaining periapical health (judged by conventional radiography), regardless of the clinical protocol used.[153]

Once the periapical lesion has become established, the challenge is a different one because the purpose now is to remove the bacterial biofilm and to effect switching off of the periapical host response. The challenge seems to be greater still if the periapical lesion is larger as it is associated with a more diverse infection. A number of approaches (protocols) have been used to achieve this general aim. The periapical healing process that occurs after root canal treatment is less clear. Nevertheless, ideal healing would eventually result in regeneration and the formation of cementum over the apical termini, isolating the root canal system from the periapex (Fig. 17.3), but this is not an inevitable result. Incomplete removal of the infection will reduce but not eliminate the periradicular inflammatory reaction, and in fact this is generally what happens.[147] This implies that residual infection in the apical anatomy is typical following completion of root canal treatment and that an ongoing interaction continues between the residual infection, root filling material, and host defenses, which plays a definitive role in determining the final healing outcome.

The culture test has been used during root canal treatment as an interim measure of the efficacy of the chemomechanical procedure; however, it has fallen out of favor in contemporary practice for a variety of reasons.[139] The outcome measures that quantitate healing subsequent to root canal treatment are the absence of clinical signs and symptoms of persistent periapical disease.[17,18] The definitive outcome measure (in conjunction with the absence of signs and symptoms), however, is periapical healing, because the treatment is aimed at resolution of periapical disease (Fig. 17.4).[152] Clinical judgment of the outcome of treatment is based on the absence of signs of infection and inflammation, such as pain, tenderness to pressure/percussion of the tooth, tenderness to palpation of the related soft



Fig. 17.3 Low-power view of healing by cementum formation when Sealpex is used (black particles are residual Sealapex and root filling material). (Courtesy Prof. M. Tagger.)



Fig. 17.4 Resolution of periapical disease after root canal treatment of the mandibular left molar with preoperative periapical radiolucency associated with mesial and distal roots **(A)**, delayed healing caused by extruded filling material from the distal canal **(B)** and nearing complete healing as extruded material is resorbed **(C)**.

tissues, absence of swelling and sinus tract, and radiographic demonstration of reduction in the size of the periapical lesion (if sufficient time has lapsed), with a completely normal development of the periodontal ligament space. Although the majority of periapical lesions heal within 1 year, healing may continue for up to 4 years or longer.[229]

Absence of signs and symptoms of periapical disease with radiographic evidence of a persistent periapical radiolucency may indicate either fibrous repair (Fig. 17.5) or persistent chronic inflammation or infection. Only time and acute exacerbation would identify the latter, whereas the former should remain asymptomatic.

Longevity measures include survival of the root canal fillings or treatment[118,228,236] and tooth retention or survival.[111,149,190] The term *functional retention* was coined by Friedman in 2004[69] to mean retention of the tooth in the absence of signs and symptoms regardless of the radiographic presence of a lesion. The term *functionality* should further and more specifically cover the functional utility of the tooth—that is, some patients may complain that despite the absence of specific signs and symptoms of infection or inflammation, they find it impossible to use the tooth because it "feels" weak.

Modern definitions of health embrace a broader spectrum of measures including psychological well-being.[116] Instruments to assess these aspects have been developed in general medicine and adapted for application to dentistry, such as the Oral Health Related Quality of Life instrument.[55,87,115] There is no definitive instrument for measuring these aspects in endodontics as yet. Studies published so far have mainly adapted versions of the Oral Health Impact Profile (OHIP).

The periapical status of root-treated teeth has traditionally been assessed using two-dimensional conventional radiographic imaging. The use and limitations of two-dimensional radiographic images for assessing treatment outcome has been reviewed thoroughly and is reported to suffer from low sensitivity, particularly in the posterior parts of the mouth.[98] The development of digital imaging technology[80] brought the possibility of image manipulation, including digital subtraction,[50,158,243,259] densitometric analysis,[156] correction of gray values,[32] and the manipulation of brightness and contrast.[75] However, none of these approaches has addressed the main deficiencies of accurately elucidating and quantitating the actual existence or progression of periapical bone loss.

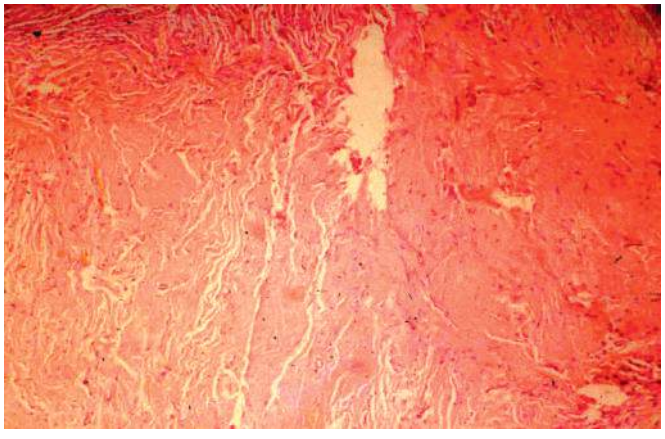


Fig. 17.5 Histologic view of fibrous periapical healing.

Cone-beam computed tomography (CBCT), a new three-dimensional imaging technique requiring only 8% of the effective dose of conventional computed tomography,[119] has been proposed as a means of overcoming the problem of superimposition of tissue layers and structures. The diagnostic values of periapical radiography and CBCT were compared on dogs' teeth using histology as the comparative method of determining apical periodontitis 180 days after root canal treatment.[44] CBCT was found to be significantly more accurate in detecting minor bone lesions compared with two-dimensional radiology.[44,174] Other evidence to support the superior sensitivity of CBCT is available from studies using a variety of simulated defects created in bone in pig[226] or human[165,221] jaws. Although the validity of clinical outcome data derived using conventional radiographic techniques has been questioned,[256] the *routine use* of CBCT is not recommended[62,91,197] owing to its higher radiation dosage.[5,91] The use of CBCT, which may be more sensitive for detection of periapical healing,[114,167] may give lower healing rates and longer durations to complete healing.

Many studies consider the treatment successful only when both radiographic and clinical criteria are satisfied.[69] It has been documented that a small proportion of cases present with persistent symptoms despite complete radiographic resolution of the periapical lesion.[176] Comparison of success rates estimated with or without clinical examination revealed no[93] or only a very small difference (1%).[150]

The widely accepted definition for endodontic success and failure by Strindberg[229] embraces both radiographic and clinical findings (Table 17.2). Friedman and Mor[69] preferred the terms *healed, healing,* and *diseased* instead of *success* and *failure* because of the potential of the latter to confuse patients. The "healed" category corresponds to "success" as defined by Strindberg,[229] whereas "healing" corresponds to "success" as defined by Bender and colleagues (see Table 17.2).[17,18]

The long duration of the periapical healing process coupled with the reduced recall rates at longer follow-up periods has provided opportunity for setting the thresholds of success at either complete healing or partial healing (reduced lesion size). The success criteria used for complete resolution have been described either as "strict"[152] or "stringent,"[69] whereas the criteria established for reduction in periapical radiolucency size has been described as either "loose"[152] or "lenient."[69] The frequency of adoption of these two thresholds has been similar in previous studies; the expected success rates using "strict" criteria would be expected to be lower than those based on "loose" criteria. The literature finds differences that vary from 4% to 48%.[153]

A periapical index (PAI) consisting of five points on the scale was devised for measuring the periapical status and used in some studies.[157,159] However, the studies only reported the increase or decrease in mean scores for the clinical factors under investigation coupled with the proportion of successful cases. This approach precluded direct comparison of their data with studies reporting conventional dichotomized outcomes. Other studies using the index dichotomized the scores into "healthy" (PAI 1 or 2) or "diseased" (PAI 3 to 5) categories,[160] thus allowing the data

Table 17.2 Criteria for Determination of Periapical Status

Strindberg (1956)	Bender et al. (1966, a and b)	Friedman and Mor (2004)
Success	**Success**	**Healed**
Clinical: No symptoms *Radiographic:* The contours, width, and structure of the periodontal margin were normal OR The periodontal contours were widened mainly around the excess filling	*Clinical:* Absence of pain/swelling; disappearance of fistula; no loss of function; no evidence of tissue destruction *Radiographic:* An eliminated or arrested area of rarefaction after a posttreatment interval of 6 months to 2 years	*Clinical:* Normal presentation *Radiographic:* Normal presentation
Failure		**Diseased**
Clinical: Presence of symptoms *Radiographic:* A decrease in the periradicular rarefaction OR Unchanged periradicular rarefaction OR An appearance of new rarefaction or an increase in the initial rarefaction		Radiolucency has emerged or persisted without change, even when the clinical presentation is normal OR Clinical signs or symptoms are present, even if the radiographic presentation is normal
Uncertain		**Healing**
Radiographic: There were ambiguous or technically unsatisfactory control radiographs that could not, for some reason, be repeated OR The tooth was extracted prior to the 3-year follow-up owing to the unsuccessful treatment of another root of the tooth		*Clinical:* Normal presentation *Radiographic:* Reduced radiolucency

to be compared directly with the more traditionally used binary outcomes of success or failure. In this system of designation, given that the PAI score 2 represents periodontal ligament widening, it effectively signals adoption of the "loose" threshold. The longitudinal follow-up[86] of 14 cases presenting with widened apical periodontal ligament (PAI score at 2) for 10 years revealed unfavorable future healing in only a small proportion of the cases (28%, 4/14).

OUTCOME MEASURES FOR PERIAPICAL SURGERY

Nonsurgical root canal treatment alone may fail to resolve apical periodontitis in a small proportion of cases because it may not allow access to the infection or resolution of the cause of the infection (i.e., root fracture or crestal bone communication to the periapical infection). Where the cause is microbial persistence, the infection may be localized intraradicularly (in the apical canal anatomy [Fig. 17.6] or in the apical dentinal tubules [Fig. 17.7] or extraradicularly [Fig. 17.8]). In these instances, a surgical approach to the periapex may be required in addition to the nonsurgical approach to removing such microbes (see Fig. 17.7).

The success of periapical surgery has been assessed with the same clinical and radiographic criteria as for nonsurgical root canal treatment. However, the radiographic criteria for successful periapical healing are different from those for nonsurgical root canal treatment (Table 17.3; Figs. 17.9–17.11).[140,186] Moreover, periodontal attachment loss in the form of marginal gingival recession is an additional criterion for measuring the outcome of periapical surgery.

Now that the parameters have been discussed as to the variables that need to be considered when assessing healing or nonhealing outcomes, the following discussion will describe the expected outcomes of the various endodontic procedures.

Outcomes of Vital Pulp Therapy Procedures

The vital pulp therapy procedures discussed in this chapter include those for managing extensive caries with a high risk of pulpal exposure or pulps exposed through caries or traumatic injuries.

INDIRECT PULP CAPPING (ONE-STEP VERSUS STEPWISE EXCAVATION)

The most conservative management for extensive caries is indirect pulp capping using either the one-step or the stepwise excavation approach. The stepwise excavation approach with initial partial caries removal was advocated to reduce the risk of pulpal exposure and is generally considered to be associated with poorer outcomes. The clinical outcomes of the one-step or stepwise approaches have been compared in three randomized controlled trials on permanent teeth with deep caries.[19,113,121,122] Stepwise excavation was found to be associated with a lower rate of pulp exposure and higher chance of long-term clinical success than the one-step excavation approach (Table 17.4).[19,113] Maltz and colleagues,[121,122] however, found the contrary, and they attributed the low long-term clinical success rate of the stepwise approach to patients being noncompliant in returning for scheduled completion of treatment. Interestingly, comparison of the cost-effectiveness of the one-step versus the stepwise excavation techniques using health-economic

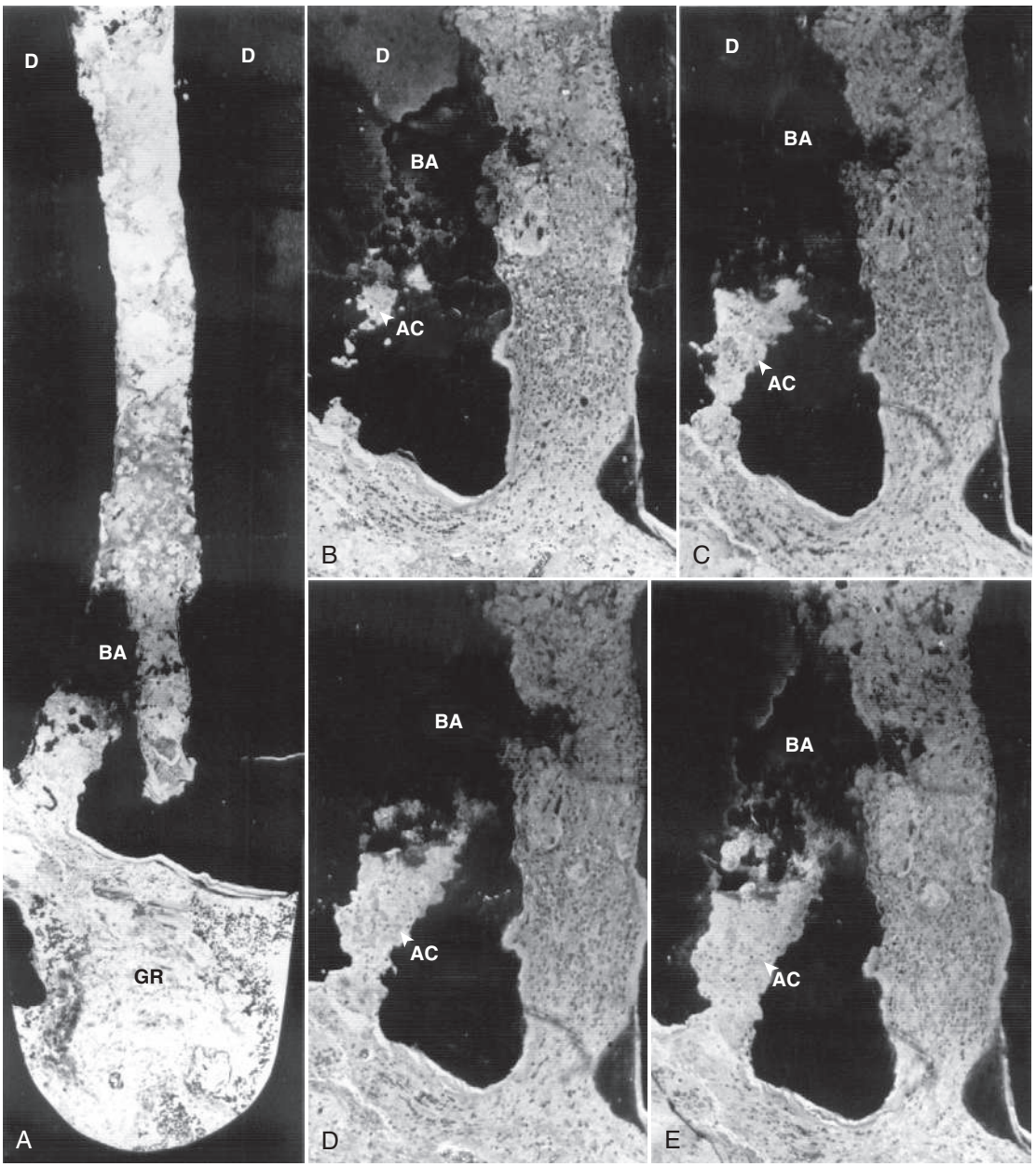


Fig. 17.6 Axial sections through the surgically removed apical portion of the root with a therapy-resistant periapical lesion *(GR)*. Note the cluster of bacteria *(BA)* visible in the root canal. Parts **(B)**–**(E)** show serial semithin sections taken at varying distances from the section plane of **(A)** to reveal the emerging **(B)** and gradually widening **(C–E)** profiles of an accessory root canal *(AC)*. Note that the accessory canal is clogged with bacteria *(BA)*. Original magnification: **A,** ×52; **B–E,** ×62. (From Nair PN, Sjogren U, Krey G, Sundqvist G: Therapy-resistant foreign body giant cell granuloma at the periapex of a root-filled human tooth, *J Endod* 16:589, 1990.)

modeling involving simulated treatment of a molar tooth with deep caries in a 15-year-old patient revealed that the one-step procedure accrues lower long-term costs as well as longer tooth retention and pulp vitality.[200]

Meta-analyses of data from studies[19,20,65,81,89,113,120-122,141] listed in Table 17.4 revealed that the weighted pooled success rate for indirect pulp capping using the one-step approach (81.7%; 95% confidence interval [CI]: 72.7%, 90.6%) (Fig. 17.12) was similar to that for the stepwise approach (81.9%; 95% CI: 72.1%, 91.7%) (Fig. 17.13).

Calcium hydroxide cement was the preferred lining material for the pulpal surface, whereas zinc oxide–eugenol–based cement was the preferred base material in most studies. Recently, resin-modified-glass-ionomer liner has also been used, but the type of lining material did not influence the outcome (see Table 17.4). The age of patients, the presence of preoperative pain, and pulpal exposure during excavation were significant negative prognostic factors.[19]

DIRECT PULP CAPPING

Direct pulp capping is performed on teeth with pulp exposures caused by caries, caries excavation, or traumatic injuries. Most studies investigating the clinical outcome of direct pulp capping on permanent teeth had excluded teeth with signs and symptoms of irreversible pulpitis and apical

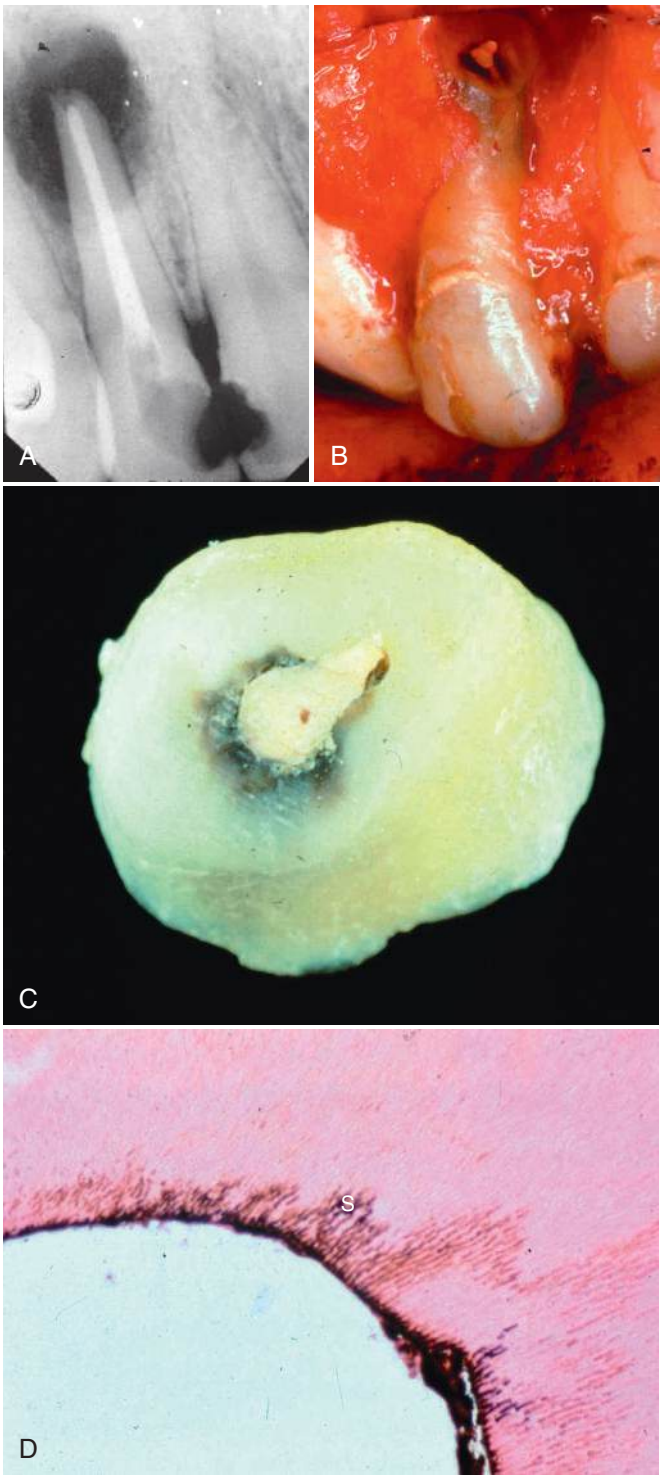


Fig. 17.7 A, Adequate root filling demonstrated on radiograph. **B,** View of the periapical surgery and root resection of the tooth shown in **(A)** shows stained root dentine. **C,** Resected root showing stained/infected dentine. **D,** Histologic view of the root end shown in **(C)** showing infected dentinal tubules *(S)*.

pathosis (Table 17.5). Saline, sodium hypochlorite, and chlorhexidine have been reportedly used to irrigate the exposed pulp and to achieve hemostasis. Calcium hydroxide paste and mineral trioxide aggregate (MTA) were the commonly used capping materials (see Table 17.5).

Table 17.3 Classification of Periapical Healing Following Apical Surgery

COMPLETE HEALING
Reformation of a Periodontal Space With
Normal width and lamina dura to be followed around the apex
Slight increase in width of apical periodontal space but less than twice the width of noninvolved parts of the root
Tiny defect in the lamina dura (maximum 1 mm) adjacent to the root filling
Complete Bone Repair With
Bone bordering the apical area does not have the same density as surrounding noninvolved bone
No apical periodontal space can be discerned
Incomplete Healing (Scar Tissue)
The Rarefaction Has Decreased in Size or Remained Stationary With
Bone structure recognized within the rarefaction
The irregular periphery of the rarefaction and demarcation by a compact bone border
The rarefaction located asymmetrically around the apex
Angular connection between the rarefaction and the periodontal space
An Isolated Scar Tissue in the Bone With Findings Above
Uncertain Healing
The Rarefaction Has Decreased in Size With
The size larger than twice the width of the periodontal space
Lamina-dura like bone structures around the border
A circular or semicircular periphery
Symmetric location around the apex as a funnel-shaped extension of the periodontal space
Bone structure discernible within the bony cavity
A collar-shaped increase in width of lamina dura coronal to the radiolucency
Unsatisfactory Healing (Failures)
The rarefaction has enlarged or is unchanged
If a case still demonstrated "uncertain healing" 4 years postoperatively, the treatment should be considered a failure[186]

From Rud J, Andreasen JO: A study of failures after endodontic surgery by radiographic, histologic and stereomicroscopic methods, *Int J Oral Surg* 1:311-328, 1972; Molven O, Halse A, Grung B: Observer strategy and the radiographic classification of healing after endodontic surgery, *Int J Oral Maxillofac Surg* 16:432-439, 1987.

A meta-analysis of data from studies[12,19,21,64,65,72,88,90,92,128,132,133,193,207,254] listed in Table 17.5 revealed the weighted pooled success rate to be 70.1% (95% CI: 59.9%, 80.2%) (Fig. 17.14). The patient's age and sex; tooth location and type; pulp exposure type, size, and its location; and the restoration type, size, and quality did not have a significant influence on success. Although the outcome of direct pulp capping of teeth with immature versus mature apices has not been systematically compared in individual studies, indirect comparison of pooled data from different studies[1] revealed that teeth with immature roots were associated with significantly more successful outcomes.

The type of capping material was another significant prognostic factor,[90,197,207] with MTA performing superiorly to calcium hydroxide in a randomized controlled trial[90] and in a systematic review.[1]

PULPOTOMY

Earlier studies on the outcome of *partial pulpotomy* (specifying varying degrees of coronal pulp removal) had only

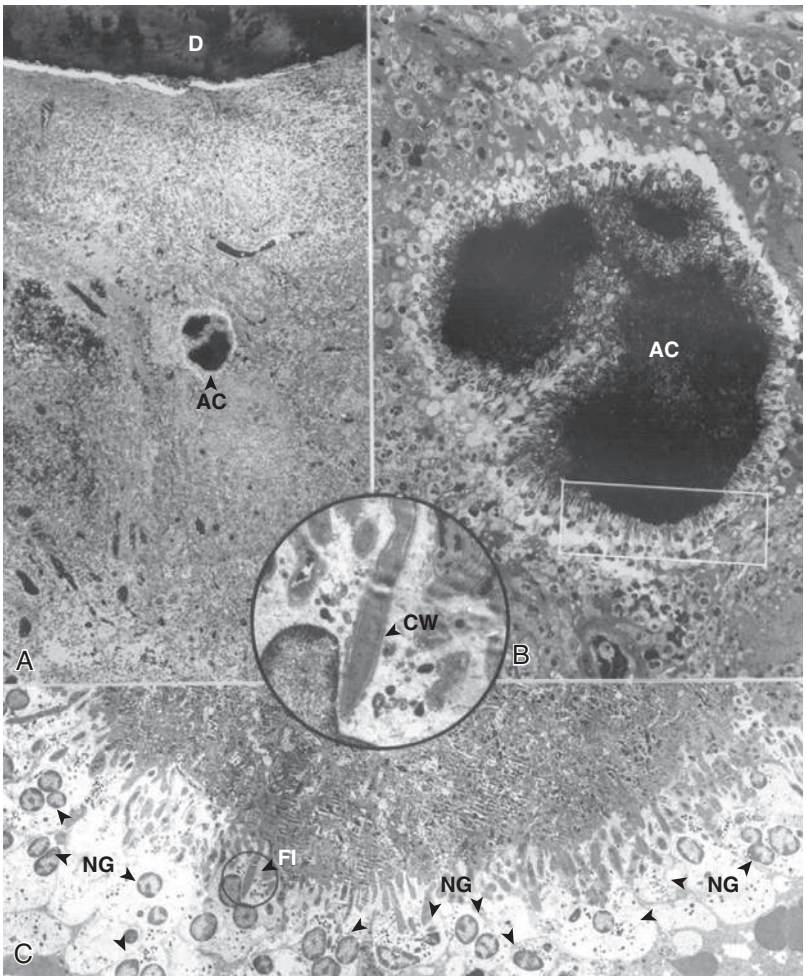


Fig. 17.8 Actinomyces in the body of a human periapical granuloma. The actinomycotic colony *(AC)* in (**A**) is magnified in (**B**). The rectangular area demarcated in **(B)** is magnified in **(C)**. Note the starburst appearance of the colony with needle-like peripheral filaments surrounded by few layers of neutrophilic granulocytes *(NG)*, some of which contain phagocytosed bacteria. A dividing peripheral filament *(FI)* is magnified in the inset. Note the typical gram-positive wall cell wall *(CW)*. *D,* Dentine. Original magnification: **A,** ×60; **B,** ×430; **C,** ×1680; inset, ×6700. (From Nair PR, Schroeder H: Periapical actinomycosis, *J Endod* 10:567, 1984.)

included teeth with vital traumatic pulp exposures; however, more recent studies have also included teeth with carious pulp exposures. Teeth with spontaneous or severe pain and signs and symptoms of apical pathosis were excluded (Table 17.6). On the other hand, the outcomes of *full pulpotomies* (completely removing the coronal pulp and retaining the radicular pulp) had only been investigated in teeth with carious exposures (Table 17.7). Saline irrigation has been preferred for achieving hemostasis, whereas calcium hydroxide, and more recently, MTA, were the preferred pulp capping materials (see Table 17.6).

A meta-analysis of data from studies listed in Tables 17.6[10,11,19,40,71,125,131,177] and 17.7[6,30,31,52,58,126,187,192,235,251,255] revealed the weighted pooled success rate to be 79.3% (95% CI: 66.7%, 91.8%) for partial pulpotomies (Fig. 17.15) and 82.4% (95% CI: 69.3%, 95.4%) for full pulpotomies (Fig. 17.16).

The effects of potential prognostic factors for pulpotomy have not been explored systematically except for the pulp capping material. Randomized controlled trials revealed that MTA achieved similar outcomes in partial[177] or full pulpotomies[58] when compared with calcium hydroxide.

SUMMARY OF PROGNOSTIC FACTORS FOR VITAL PULP THERAPY

In summary, vital pulp therapy performed under guideline standards with optimal coronal seal achieved promising long-term success in teeth with carious, mechanical, or traumatic exposures of healthy pulps.

The most important factors affecting the outcome of vital pulp therapy are preexisting health of the pulp, adequate removal of infected hard or soft tissues, careful operative technique to avoid damage to residual tissues, and elimination of microbial leakage around the final restoration. It can be difficult to gauge the health of the residual pulp as it is a matter of subjective assessment and relies on experience in pulp diagnosis. The degree of pulp bleeding upon exposure is a more reliable tool to judge the status of the pulp than the preoperative clinical signs and symptoms. Continued bleeding after 10 minutes, even after rinsing with sodium hypochlorite solution, may suggest that the residual pulp was still heavily inflamed and a complete pulpectomy may be a more effective treatment modality. Removal of infected tissue is a matter of subjective experience but may be aided by various

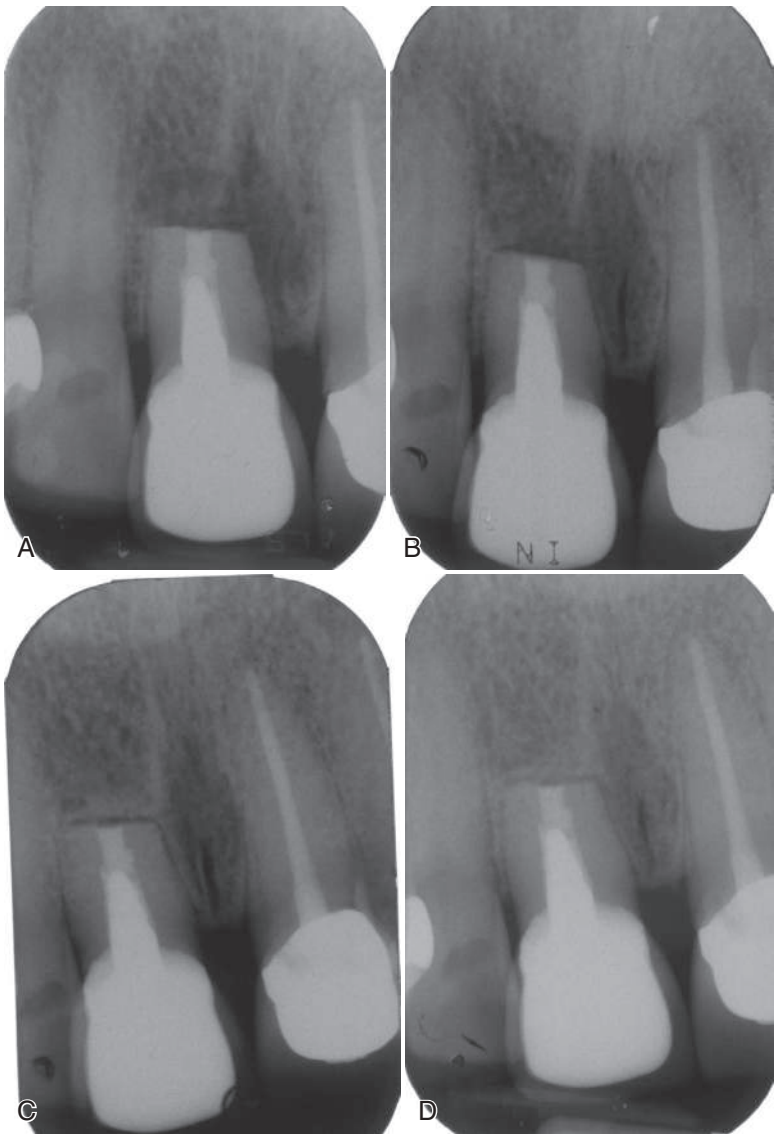


Fig. 17.9 **A,** A maxillary right central incisor having undergone periapical surgery. **B,** Incomplete periapical healing at 1 year postoperatively. Complete healing at 3 **(C)** and 4 years **(D)** postoperatively.



Fig. 17.10 Example of incomplete healing with isolated scar tissue.

dyes. The final factor relies on the correct choice of restorative material and its adequate manipulation to prevent leakage.

Factors such as the age and health of the patient, the size and nature (carious or traumatic) of pulp exposure and its duration of exposure to the oral environment (up to 48 hours) do not in themselves compromise outcomes of vital pulp therapy.

Outcomes of Nonsurgical Root Canal Treatment

In contrast to other areas of endodontics, the number of studies and extent of investigation of nonsurgical root canal treatment is more comprehensive, yielding a much greater insight even though the quality and scope of the research does not always reach the highest levels.

A systematic review and meta-analysis of the factors affecting primary root canal treatment outcome conducted

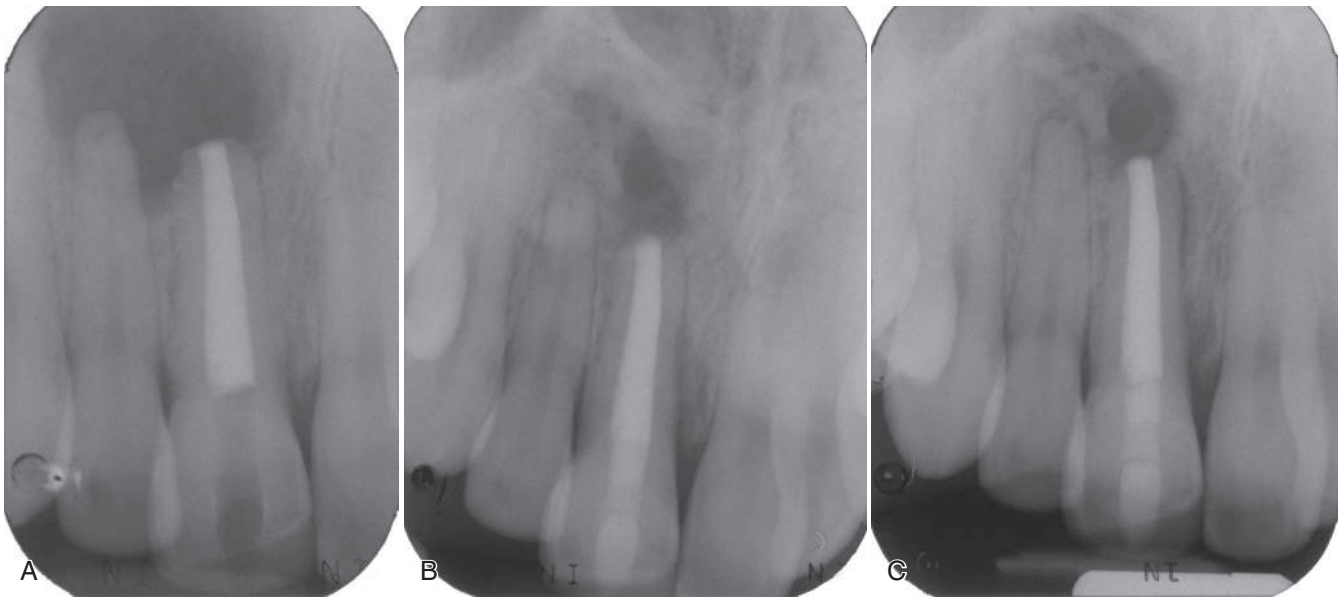


Fig. 17.11 Example of uncertain healing. Periapical radiographs of a maxillary left central incisor taken immediately after periapical surgery **(A)**, at 2 years postoperatively **(B)**, and at 3 years postoperatively **(C)**.

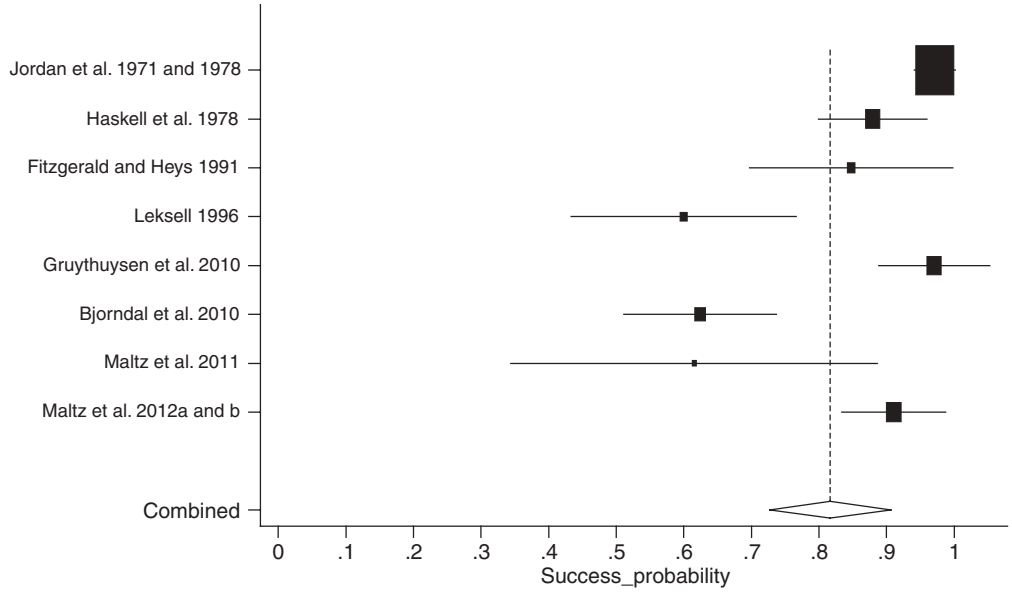


Fig. 17.12 Forest plot showing individual study and weighted pooled probabilities of success for one-step indirect pulp capping.

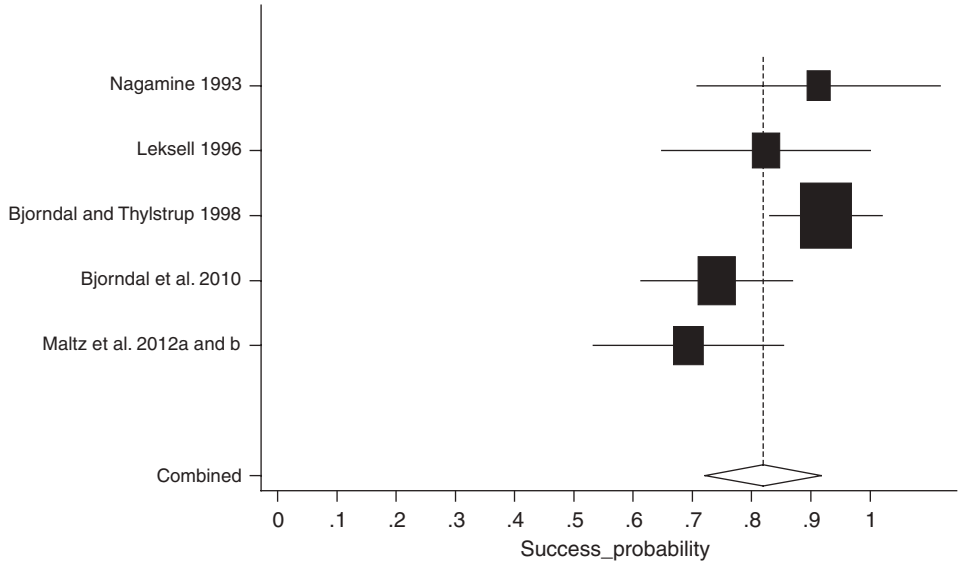


Fig. 17.13 Forest plot showing individual study and weighted pooled probabilities of success for stepwise indirect pulp capping.

Table 17.4 Studies Investigating the Clinical/Radiographic Outcome of Indirect Pulp Capping of Permanent Teeth

Study	Study Design	Type of Indirect Pulp Capping	No. of Teeth	Success (%)	Age	Pretreatment Status	Capping Material	Criteria for Success	Duration After Treatment	Notes
More, 1967	Case series	Stepwise	8	8 (100%)	11–18	Carious Vital Apical pathosis	1st visit: ZOE 2nd visit: CH/Amal	Vital/asymptomatic/ intact PDL	6–36 months	
Jordan et al., 1971 (in Hayashi et al., 2011) and 1978	Case series	1-step	243	236 (97.1%)	8–37	Deep caries without pulpitis	CH, CH + cresatin, or ZOE/ZOE + amalgam	Vital/asymptomatic/ intact PDL	Not given	24 teeth were followed up for 11 years, 11 remained vital with no evidence of pathosis
Sawusch, 1982 (in Hayashi et al., 2011)	CCT	Stepwise	48	48 (100%)	14 or younger	Deep caries without pulpitis	CH (Dycal versus improved Dycal)/ ZOE or ZnPO4	Clinical	6 months	
Fitzgerald and Heys, 1991	Case series	1-step	46	39 (84.8%)	20–60	Deep caries Vital	CH (Dycal versus Life)	Asymptomatic	1 year	The brand of CH did not have significant effect
Nagamine, 1993 (in Hayashi et al., 2011)	CCT	Stepwise	23	21 (91.3%)	17–46	Deep caries without pulpitis	Polycarboxylate cement with tannin-fluoride versus hydraulic temporary sealing material/ GIC	Vital	3 months	
Leksell, 1996	RCT	Stepwise	57	47 (82.5%)	6–16	Not given	CH/ZOE then CH/GIC	Asymptomatic/intact PDL	1–11 years (mean = 43 months)	47/57 without exposure; duration of CH dressing had no significant influence on exposure
		1-step	70	42 (60.0%)						42/70 without exposure

Table 17.4 Studies Investigating the Clinical/Radiographic Outcome of Indirect Pulp Capping of Permanent Teeth—cont'd

Study	Study Design	Type of Indirect Pulp Capping	No. of Teeth	Success (%)	Age	Pretreatment Status	Capping Material	Criteria for Success	Duration After Treatment	Notes
Bjorndal and Thylstrup, 1998	Case series	Stepwise	94	87 (92.6%)	Not given	Deep caries	CH/temporary filling	Absence of perforation, asymptomatic	1 year	88/94 without exposure
Bjorndal et al., 2010	RCT	Stepwise	143	106 (74.1%)	29 (25–38)	Deep caries Only provoked pain Vital	CH/GIC	Vital/intact PDL	1 year	Stepwise better than complete; preoperative pain, older patients reduce success
		1-step	149	93 (62.4%)						
Gruythuysen et al., 2010	Case series	1-step	34	33 (97.1%)	Up to 18	Carious (> 2/3) No spontaneous, persistent pain No apical pathosis	GIC	Survive/ asymptomatic/ intact PDL/no resorption	3 years	
Maltz et al., 2011	Case series	1-step	26	16 (61.5%)		Vital Deep caries	CH/ZOE then composite	Vital	10 years	
Maltz et al., 2012 (a and b)	RCT (multicenter)	1-step	112	111 (18 months) 102 (91.7%) (3 years)	≥ 6	Vital Carious (> 1/2) No spontaneous, persistent pain No apical pathosis	GIC/amalgam or composite	Vital/intact PDL Vital/intact PDL	18 months, 3 years (clinical and radiographic outcome)	Indirect pulp capping was associated with significantly higher success rate than stepwise, regardless of duration after treatment; the low success rate of stepwise was attributed to patients not returning for completion of treatment
		Stepwise	101	87 (18 months) 70 (69.3%) (3 years)			CH/ZOE then GIC/ amalgam or composite			

CH, Calcium hydroxide; *GIC*, glass ionomer; *PDL*, periodontal ligament; *RCT*, randomized controlled trial; *ZOE*, zinc oxide eugenol.

Table 17.5 Studies Investigating the Clinical/Radiographic Outcome of Direct Pulp Capping of Permanent Teeth

Study	Study Design	No. of Teeth	Success (%)	Age (Years)	Pretreatment Status	Hemostasis	Capping Material	Base/Restorative Material	Criteria for Success	Duration After Treatment	Notes
Weiss, 1966	Case series	160	141 (88.0%)	16–67	Not given	Not given	CH + cresatin	ZOE	Vital/asymptomatic/intact PDL	3 years	
Shovelton et al., 1971	RCT	154 (1-step) 53 (2-step)	115 (74.7%) 33 (64.7%)	15–44	Carious or traumatic exposure Vital Asymptomatic Molar/premolar	Saline	Corticosteroid + antibiotic, glycerrhetinic acid + antibiotics, ZOE, or CH	ZOE/amalgam	Vital/asymptomatic/intact PDL	24 months	There was no significant difference in success rates among the various capping materials
Haskell et al., 1978	Case series	133	117 (88.0%)	8–74	Carious exposure Asymptomatic	Not given	CH or penicillin	ZOE	Vital/asymptomatic/intact PDL	>5 years	5- to 22-year survival Success rate was not affected by age and tooth type
Gillien and Schuman, 1985	Case series	17	13 (76.4%)	6–9	Carious exposure	Not given	CH	Base material not given/amalgam or crown	Asymptomatic/intact PDL	6–12 months	
Horsted et al., 1985	Case series	510	485 (95.1%)	Not given	Exposure during cavity preparation or caries removal No periapical pathosis No pain	2% chloramine or 0.2% CHX	CH	ZOE	Vital/asymptomatic/intact PDL	5 years	Type of exposure and tooth type had no significant effects Older age had lower survival
Fitzgerald and Heys, 1991	Case series	8	6 (75.0%)	20–60	Vital Exposure during caries removal	Sterile cotton pellets	CH	$ZnPO_4$/amalgam or composite	Asymptomatic	12 months	The brand of setting CH paste did not have significant effect
Matsuo et al., 1996	Case series	44	36 (81.2%)	20–69	Carious exposure No intense pain	10% NaOCl and 3% H_2O_2	CH	ZOE/GIC	Vital/asymptomatic/intact PDL	3 years	

Table 17.5 Studies Investigating the Clinical/Radiographic Outcome of Direct Pulp Capping of Permanent Teeth—Cont'd

Study	Study Design	No. of Teeth	Success (%)	Age (Years)	Pretreatment Status	Hemostasis	Capping Material	Base/Restorative Material	Criteria for Success	Duration After Treatment	Notes
Santucci, 1999	Case series	29	15 (51.7%)	Not given	Exposure due to caries, or its removal Sensitive to cold or sweat with no other pain No periapical pathosis	Not given	CH	Composite or cast gold restoration	Asymptomatic	4.5 years	
Barthel et al., 2000	Case series	123	29 (23.6%)	10–70	Carious exposure	3% H_2O_2	CH/$ZnPO_4$ or other materials	Not given	Vital/asymptomatic/ intact PDL	5–10 years	Age, tooth type, site of exposure had no significant effects Immediate placement of permanent restoration had significantly higher success rate
Farsi et al., 2006	Case series	30	28 (93.3%)	9–12	Deep caries Potential exposure Reversible pulpitis Intact PDL	Saline	MTA	ZOE/composite	Vital/asymptomatic/ intact PDL/ continuous root development	2 years	
Bogen et al., 2008	Case series	49	48 (98.0%)	7–45	Carious exposure (0.25–2.5 mm) Reversible pulpitis	5.25/6% NaOCl	MTA	Composite	Dentine Vital/bridge/ asymptomatic/ root development/ intact PDL	1–9 years	
Bjorndal et al., 2010	RCT	22	7 (31.8%)	25–38	Exposure during caries removal Only provoked pain Vital	Saline	CH	GIC	Vital/intact PDL	1 year	

Continued

Table 17.5 Studies Investigating the Clinical/Radiographic Outcome of Direct Pulp Capping of Permanent Teeth—Cont'd

Study	Study Design	No. of Teeth	Success (%)	Age (Years)	Pretreatment Status	Hemostasis	Capping Material	Base/Restorative Material	Criteria for Success	Duration After Treatment	Notes
Mente et al., 2010	Case series	122	86 (70.5%)	8–78	Carious or mechanical exposure	0.12% CHX	MTA or CH	GIC/composite or crown	Vital/no clinical or radiographic evidence of apical pathosis	12–18 months	Use of MTA for capping and immediate placement of permanent restoration had significantly higher success rate Age; sex; tooth location and type; exposure site and type; restoration type, size, and quality did not have significant effects
Miles et al., 2010	Case series	51	23 (45.1%)	21–85	Carious exposure	2.5% NaOCl	MTA	GIC/composite or amalgam	Vital/asymptomatic/ intact PDL	12–27 months	
Hilton et al., 2013	RCT	126	81 (64.3%)	9–90	Carious, traumatic, mechanical exposure	5.25% NaOCl	CH	GIC	Vital/intact PDL/no resorption/not requiring extraction or root canal treatment	2 years	MTA was associated with significantly higher success rate than CH Patient, dentist, tooth, pulp exposure, and pulp capping characteristics did not have significant influence on the results
		144	116 (80.6%)	8–89			MTA				

CH, Calcium hydroxide; *GIC*, glass ionomer; *MTA*, mineral trioxide aggregate; *NaOCl*, sodium hypochlorite; *RCT*, randomized controlled trial.

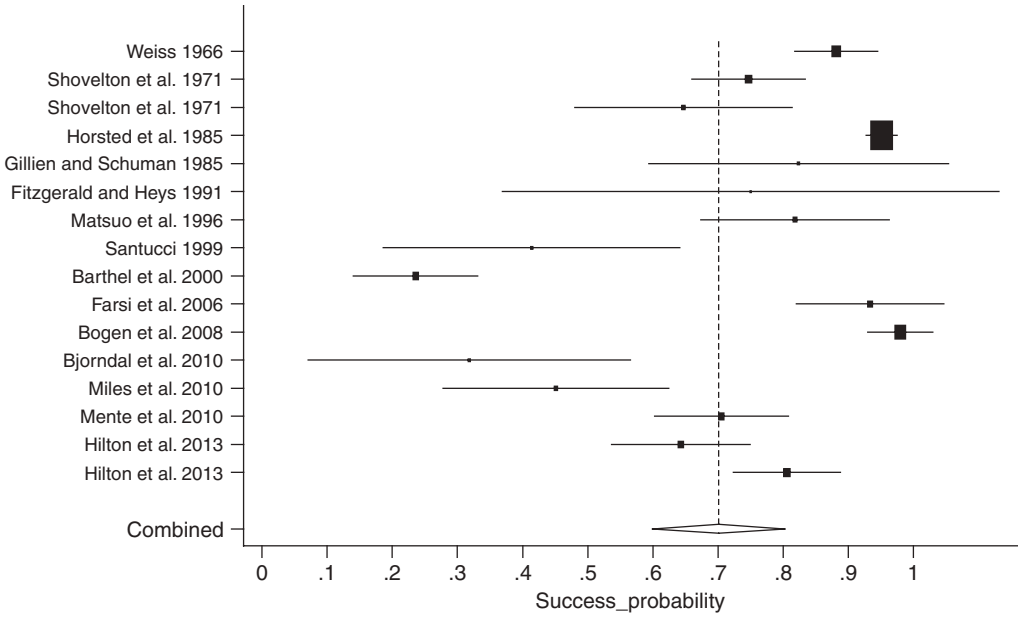


Fig. 17.14 Forest plot showing individual study and weighted pooled probabilities of success for direct pulp capping.

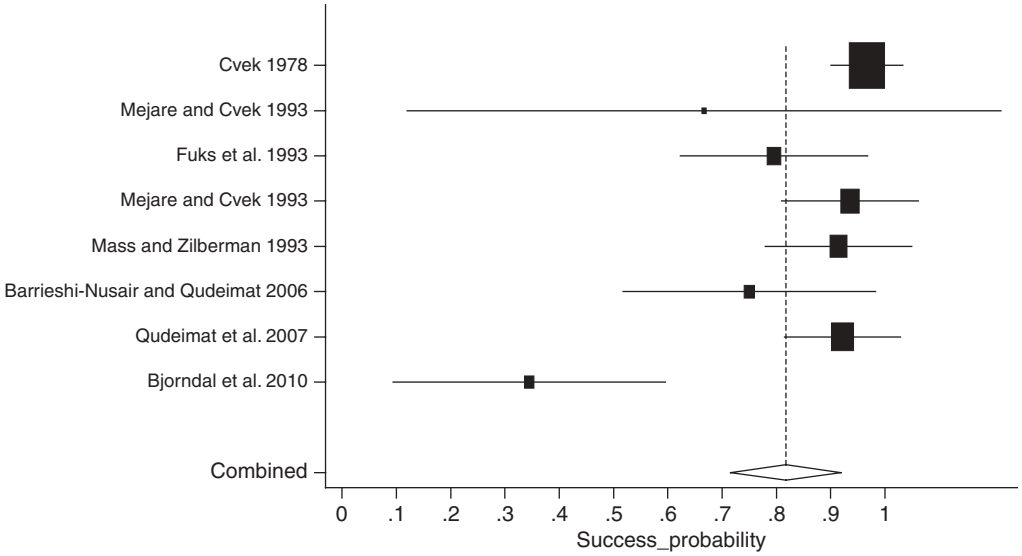


Fig. 17.15 Forest plot showing individual study and weighted pooled probabilities of success for partial pulpotomy.

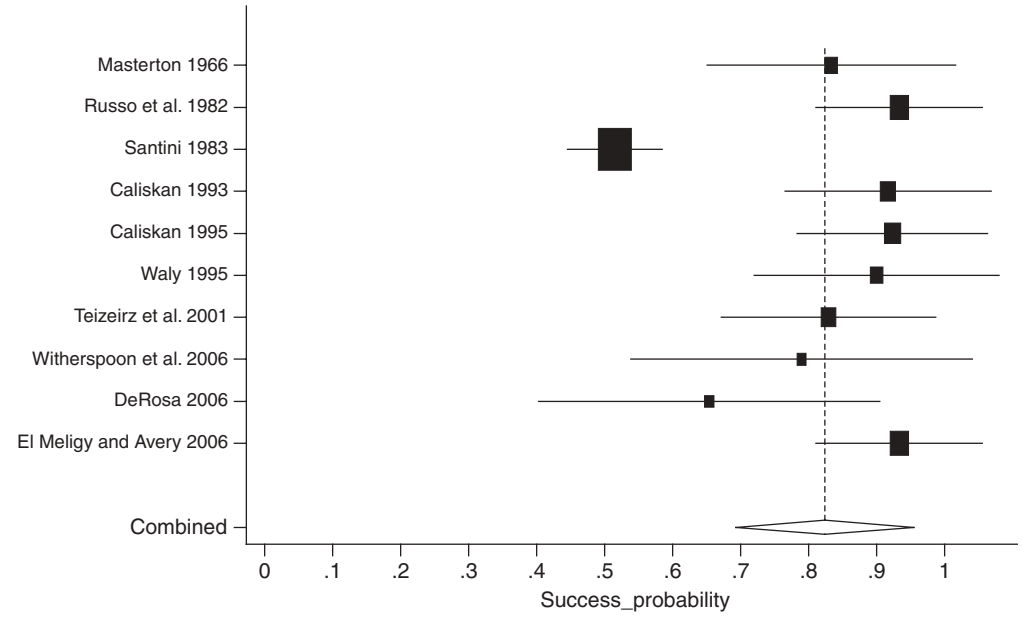


Fig. 17.16 Forest plot showing individual study and weighted pooled probabilities of success for full pulpotomy.

Table 17.6 Studies Investigating the Clinical/Radiographic Outcome of Partial Pulpotomy of Permanent Teeth

Author	No. of Teeth	Success (%)	Age	Pretreatment Status	Exposure Size	Hemostasis	Pulp Dressing Material	Base/Restorative Material	Criteria for Success	Duration After Treatment	Notes
Cvek, 1978	60	58 (96.7%)	Not given	Traumatic exposure Vital Bleeding wound	0.5–4.0 mm	Saline	1st visit: nonsetting CH 2nd visit: setting CH	1st visit: ZOE 2nd visit: Composite	EPT/asymptomatic/ intact PDL/ continued root development/ hard tissue barrier	14–60 months	Size and duration of exposure, root maturity did not affect outcome
Baratieri et al., 1989	26	26 (100%)	12–44	Exposure due to caries or caries removal Pulp tissue bleeding without signs of degeneration	Not given	Calcium hydroxide solution	CH powder then hard set CH cement	Zinc oxide cement	Asymptomatic/vital	1–2 years	
Fuks et al., 1993	44	35 (79.5%)	Not given	Traumatic exposure Vital Bleeding wound	Not given	Saline	CH	ZOE	Asymptomatic/ bridge/root development/ vital	0.5–4 years	
Mass and Zilberman, 1993	35	32 (91.4%)	7.5–25	Molar Deep caries Asymptomatic No apical pathosis	Less than 1–2 mm in diameter; 2–3 mm deep	Saline	CH	ZOE/amalgam or crown	Asymptomatic/ intact PDL/root development	1–2 years	

Table 17.6 Studies Investigating the Clinical/Radiographic Outcome of Partial Pulpotomy of Permanent Teeth—cont'd

Author	No. of Teeth	Success (%)	Age	Pretreatment Status	Exposure Size	Hemostasis	Pulp Dressing Material	Base/Restorative Material	Criteria for Success	Duration After Treatment	Notes
Mejare and Cvek, 1993	31 (2-step) 6 (1-step)	29 (93.5%) 4 (66.7%)	Not given	Carious exposure Asymptomatic No apical pathosis	N/A	Saline	CH	ZOE	Asymptomatic/ intact PDL/root development	24–140 months	
Barrieshi-Nusair and Qudeimat, 2006	28	21 (75.0%)	7.2–13.1	Carious exposure Molar Reversible pulpitis No apical pathosis	2–4 mm deep	Saline	MTA	GIC/amalgam or crown	Vital/asymptomatic/ intact PCL/root development	1–2 years	
Qudeimat et al., 2007*	23 28	21 (91.3%) 26 (92.9%)	6.8–13.3	Carious exposure Molar	2–4 mm deep		CH MTA	GIC/amalgam or crown	Absence of signs and symptoms/ intact PDL/root development	24.5–45.6 months	There was no significant difference in outcome between CH and MTA
Bjorndal et al., 2010*	29	10 (34.5%)	25–38	Deep caries Only provoked pain Vital Expose during removal	Not given	Saline	CH	GIC	Vital/intact PDL	1 year	

CH, Calcium hydroxide; *EPT*, electric pulp test; *GIC*, glass ionomer; *MTA*, mineral trioxide aggregate *PDL*, periodontal ligament; *ZOE*, zinc oxide eugenol.
NB: All the studies were case series except those labeled with *.
Qudeimat et al. (2007) is a randomized controlled trial comparing calcium hydroxide versus MTA as pulp dressing material.
Bjørndal et al. (2010) is a randomized controlled trial comparing the outcomes of 1-step versus stepwise indirect pulp capping for teeth without exposure, and direct pulp capping versus partial pulpotomy for teeth with exposure during caries removal.

Table 17.7 Studies Investigating the Clinical/Radiographic Outcome of Full Pulpotomy of Permanent Teeth

Author	No. of Teeth	Success (%)	Age	Pretreatment Status	Hemostasis	Capping Material	Base/ Restorative Material	Criteria for Success	Duration After Treatment	Notes
Masterson, 1966	30	25 (83.3%)	6–39	Not given	Not given	CH	Not given	Vital/asymptomatic	1–70 months	
Russo et al., 1982	30	28 (93.3%)	9–28	Carious exposure No apical pathosis	Not given	CH	Not given	Intact PDL	8 weeks	
Santini, 1983	373	192 (51.4%)	Not given	Carious exposure or near exposure Symptomatic	Cotton wool pellet	CH or CH + Ledermix	ZOE	Vital/bridge/ asymptomatic	6 months	Sex and medicament had no significant effect Poor healing was associated with age < 7.5
Caliskan, 1993	24	22 (91.7%)	10–22	Hyperplastic pulpitis	Saline	CH	ZOE/Amalgam or composite	Vital/asymptomatic/ dentine bridge/ intact PDL	1–4 years	
Caliskan, 1995	26	24 (92.3%)	10–24	Carious exposure Asymptomatic Apical pathosis	Saline	CH	ZOE	Asymptomatic/bridge/ root development/ vital	16–72 months	
Waly, 1995	20	18 (90.0%)		Carious exposure Molar		CH-glutaraldehyde CH	Not given	Not given	5 years	
Teizeira et al., 2001	41	34 (82.9%)	6–16	Deep caries or exposed pulp With or without apical pathosis	Not given	CH	GIC	Vital/asymptomatic/ dentine bridge/ intact PDL	24–32 weeks	
DeRosa, 2006	26	17 (65.4%)		Not given		CH	Amalgam	Asymptomatic	14–88 months	
El Meligy and Avery, 2006	15 15	13 (86.7%) 15 (100%)	6–12	Carious or traumatized teeth Immature apex No apical pathosis	Saline	CH MTA	ZOE/Amalgam or composite	Asymptomatic/intact PDL/no resorption/ root development	1 year	
Witherspoon et al., 2006	19	15 (79.0%)	7–16	Carious or traumatic exposure Irreversible pulpitis	6% NaOCl	MTA	Not given	Vital/asymptomatic/ intact PDL/root development	1 year	
Asgary and Ehsani, 2009	12	12 (100%)	14–62	Carious Irreversible pulpitis	Saline	NEC	Permanent rest	Asymptomatic/intact PDL	13–20 months	

CH, Calcium hydroxide; *GIC*, glass ionomer; *MTA*, mineral trioxide aggregate, *NaOCl*, sodium hypochlorite; *NEC*, new endodontic cement.
NB: All the studies were case series except El Meligy and Avery (2006), which was a randomized controlled trial comparing calcium hydroxide and MTA as pulp dressing material.

by the authors revealed the following: the mean success rate was 83% when vital pulpectomy was performed (Fig. 17.17), which reduced to 72% when the root canal treatment procedure was used to eradicate the established infection associated with a periapical lesion (Fig. 17.18).

FACTORS AFFECTING PERIAPICAL HEALTH OR HEALING FOLLOWING ROOT CANAL TREATMENT

The factors influencing the maintenance of periapical health or periapical healing of preexisting lesions following root canal treatment may be broadly classified into patient factors (age, sex, general health, tooth anatomy, preoperative pulpal and periapical status), treatment factors (operator variables, canal enlargement, irrigation, medication, culture test, and obturation), and restorative factors. Some factors had a profound impact on success rates, whereas others showed a negligible effect. Patient factors characterizing the nature of the disease showed the most significant effect (periapical status), whereas most of the treatment factors were found to exert a less significant effect; exceptions were the apical extent of root canal treatment relative to the root canal terminus. In addition, the quality of the postoperative restorative care also exerted a profound influence on the outcome of treatment.

Patient Factors

Patient's age and sex consistently had no significant effect on outcome, whereas some specific health conditions (diabetes,[54,67] compromised immune response[123]) apparently had a significant influence. The evidence for the effect of host immune response characterized by the general health of the patient is, however, weak. Emerging evidence indicates that the host response measured by polymorphisms of various genes involved in periapical healing may have an effect on outcomes (Table 17.8).[142,183,212,213] The importance of the host response to maintenance of periapical health or periapical healing was also supported by the statistically significant clustering effect of multiple teeth within the same patient in a prospective study.[150]

The widespread perception that single-rooted teeth with less complicated anatomy should benefit with more predictable and favorable outcomes proves to be untrue. Having accounted for potential confounding factors such as the presence of periapical disease, tooth type does not seem to exert a strong influence on success rates. This would appear to be counterintuitive but may be explained by the logical inference that canal complexities in the apical anatomy probably play a more dominant role than other complexities, such as the number and curvature of canals.

The presence and size of a periapical lesion seem to have the most negative effect on periapical health/healing; it therefore follows that these factors must be accounted for when analyzing the influence of any other factor. The profound change in success rates once a periapical lesion becomes established is interesting, as it is correlated with the establishment of infection in the apical canal anatomy. This seems to suggest that once the apical canal complexities become infected, it may be much more difficult to eradicate the infection. The negative influence of large periapical lesions has a reasonable biologic explanation: the diversity of bacteria (by number of species and their relative abundance) is greater in teeth with larger periapical lesions.[230] An endodontic infection is more likely to persist in canals with a higher number of bacteria present preoperatively.[27] In addition, larger lesions may represent longer-standing root canal infections that may have penetrated deeper into dentinal tubules and accessory anatomy in the complex canal system[206] where mechanical and chemical decontamination procedures may not be so effective. Larger lesions may also represent cystic transformation.[145] Finally, the host response may also play a part, as patients with larger lesions may innately respond less favorably to residual bacteria.[147] This speculation may crystallize into distinct questions for further biologic research into the nature of interactions among host, bacterial infection, and treatment intervention.

Most of the other investigated preoperative factors (pain, tooth tenderness to percussion, soft-tissue tenderness to palpation, soft-tissue swelling and sinus tract, periodontal probing defect of endodontic origin, root resorption) are in fact different clinical manifestations of periapical disease.[254] They may therefore act as surrogate measures or complement "presence and size of periapical lesion" in measuring the effect of severity of periapical disease within a broad continuous spectrum. Of these, only presence of preoperative pain,[68] sinus tract,[150] swelling,[150] and apical resorption[229] have been found to be significant prognostic factors that have been associated with significantly reduced success rates in root canal treatment.

The biologic explanation for the negative impact of sinus tract and swelling, either in the acute or chronic form, on periapical healing is interesting, as both represent suppuration and the proliferation of microbiota into the periapical tissues, with the inference being that the host tissues must have become locally overwhelmed. The precise reasons for reduced success rates under these conditions remain unclear but must somehow be related to the nature of the host-microbial interaction.

Treatment Factors

Operator. Although the impact of operator qualification and skill has not been specifically investigated, systematic review has shown that the involved clinicians may be grouped into undergraduate students, general practitioners, postgraduate students, and specialists. Studies show a clear trend in superior outcomes by greater experience and training. Clearly, technical skills play an important role, but this is difficult and often impossible to quantify. In addition, the technical abilities must be augmented by the overall understanding of the biologic issues and the quest for superior treatment by the operator. If clinicians do not feel that they can provide optimal treatment, it is incumbent upon them to refer the patient to a more qualified practitioner.

Isolation. The use of rubber dams in modern root canal treatment is widely accepted, and the justification seems almost empirical. One study on retreatment[244] analyzed the influence of rubber dam use compared to cotton roll isolation and found significantly higher success rates with the former approach. Another study reported a significantly higher success rate of root canal–treated teeth[76] when a rubber dam was employed during post placement compared

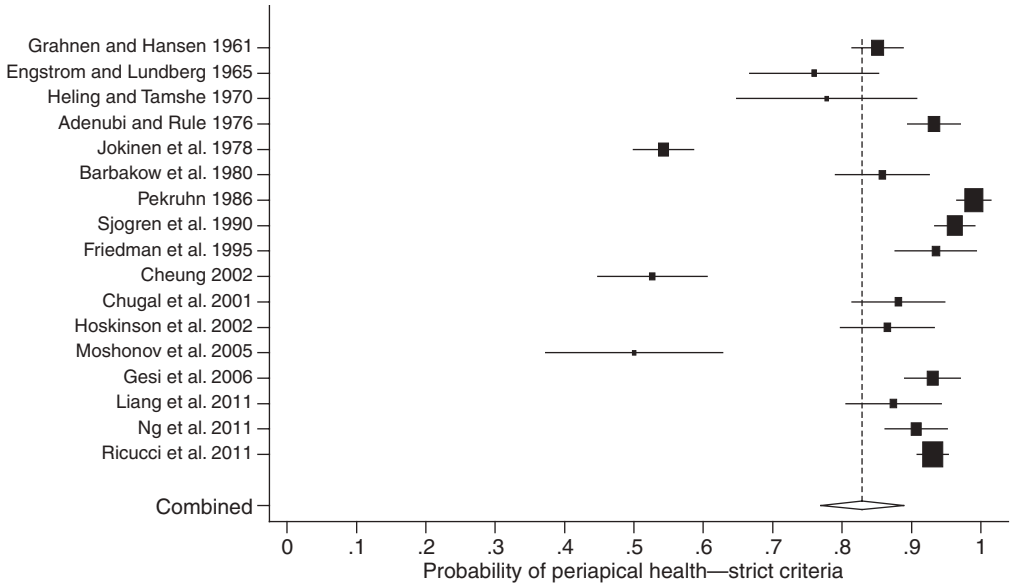


Fig. 17.17 Forest plot showing results of pooled and individual study's probability of maintained periapical health for preoperatively vital teeth undergoing root canal treatment (pooled probability = 0.83; 95% confidence interval: 0.77, 0.89).

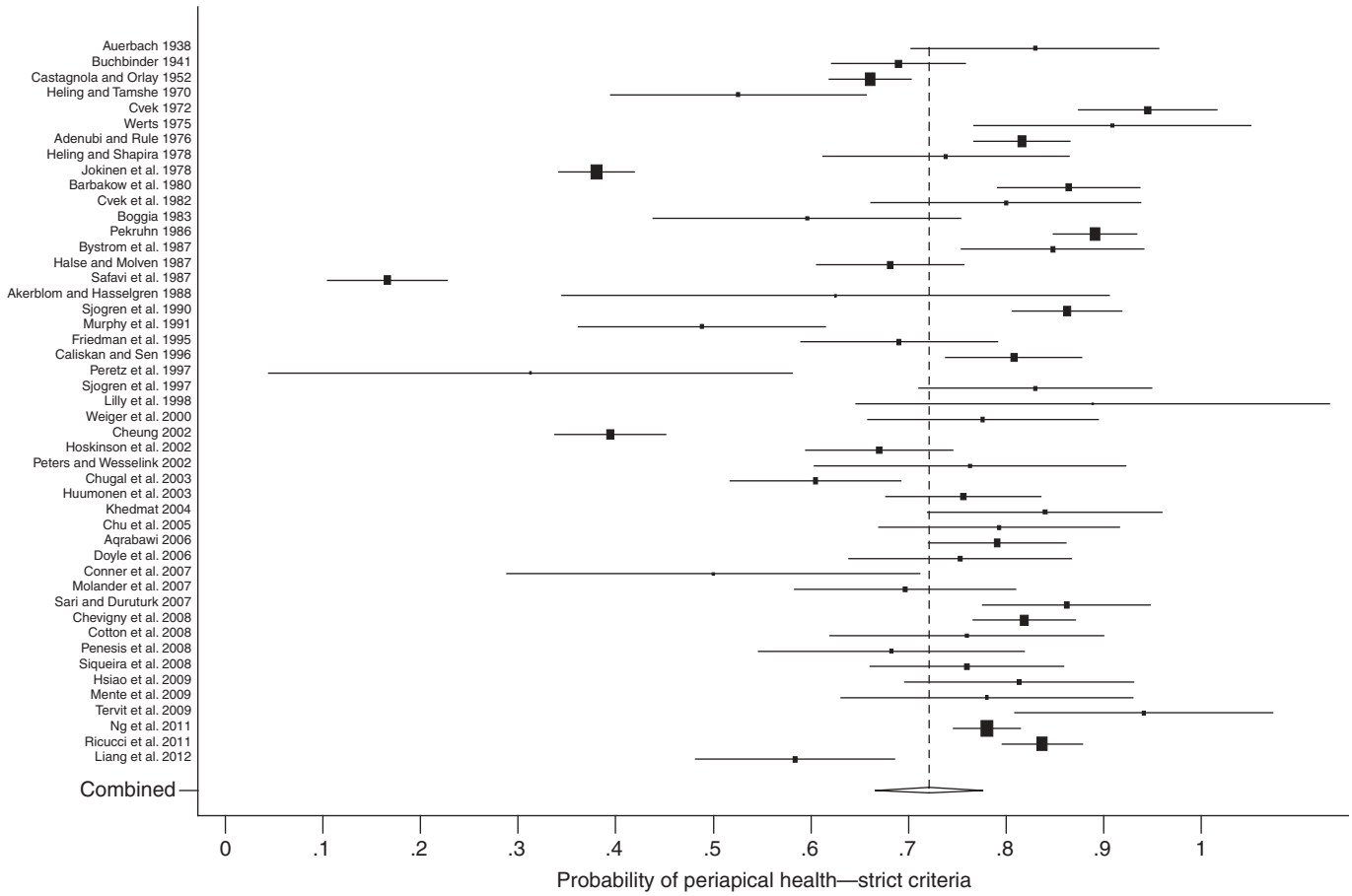


Fig. 17.18 Forest plot showing results of pooled and individual study's probability of periapical health for teeth associated with nonvital pulps and periapical radiolucencies undergoing root canal treatment (pooled probability = 0.72; 95% confidence interval: 0.67, 0.78).

Table 17.8 Type of Genes Investigated in Studies on SNPs and Periapical Healing

Study	Gene Function	Genes	Findings
Siqueira et al., 2009	Interleukin 1 Fcγ receptor	IL-1α, IL-1β FcγRIIa, FcγRIIIb	No significant association Significant association
Morsani et al., 2011	Interleukin	IL-1β	Significant association
Siqueira et al., 2011	Fcγ receptor	FcγRIIIa	No significant association
Rôças et al., 2014	Pattern recognition receptors	CD14, TLR4	No significant association

SNP, Single nucleotide polymorphism.

to when it was not. Perhaps as a consequence, the principal justification for rubber dam use is based on medicolegal implications of root canal instrument ingestion or aspiration by the patient.[61]

Magnification and Illumination. Endodontists have repeatedly reinforced the value of magnification and illumination during root canal treatment,[166] but a systematic review failed to draw any objective conclusions on their influence as no article was identified in the current literature that satisfied their inclusion criteria.[46] A prospective study investigated this factor,[150] but researchers found only an insignificant influence on the final outcome. Use of a microscope may sometimes assist location of the second mesiobuccal canal in maxillary molars, but this only made a small difference to the success rates associated with mesiobuccal roots, when a periapical lesion was present.[150] The true benefit of a microscope can only be verified through a randomized controlled trial. However, canal negotiation with less tooth structure removal and fewer procedural accidents is favorable and seems intuitively more consistent with the use of superior magnification and illumination.

Mechanical Preparation: Size, Taper, Extent, and Procedural Errors. The root canal system may be mechanically prepared to a requisite size and taper[199] using a variety of instruments of different cutting designs, tips, tapers, and materials of construction. Their efficacy is often tested in laboratory studies, and the instruments and their utility may have well-characterized properties.[95] Investigation of the influence of the type of instrument used for canal enlargement has been undertaken in one nonrandomized prospective study, but the outcome is likely subjective because of many factors, including the protocol adopted for teaching technical skills.[150] In this study, the better success rates for hand or rotary NiTi instruments compared with stainless steel instruments[150] were attributable to the fact that tactile skills training was achieved through a preliminary focus on the use of stainless steel files to develop tactile sensitivity and consistency. Only on demonstration of this competency did the trainees progress to NiTi instruments. More importantly, such senior students may also have had a better understanding of the biologic rationale for root canal treatment. The ability to gain and maintain apical patency as well as to avoid procedural errors was better instilled in the senior students, whereas in selected cases, NiTi instruments appear capable of achieving the same in primary root canal treatment undertaken by undergraduates.[175]

A key tenet of the ESE[61] guidelines is that root canal debridement must be extended to the terminus of the canal system, which is expressed variously as extension to the "apical constriction," or to "0.5 to 2 mm from the radiographic apex," or to the "cementodentinal junction." This guideline is broadly supported by the fact that outcome of treatment is compromised by canal obstruction or failure to achieve patency to the canal terminus.[150,216,229] Ng and colleagues[150] reported a twofold reduction in the success of treatment when the patency to the canal terminus was not achieved. It could be speculated that the lack of mechanical negotiability of canals may be due to the presence of obstructions caused by "denticles," tertiary dentine, acute branching or a fine plexus of apical canals, or dentine/organic debris.

The continued debate on the optimal size of apical preparation remains topical in the absence of definitive evidence; the findings from relevant in vitro and clinical studies have been previously reviewed.[15] So far, four clinical outcome studies have considered this issue or have systematically investigated the effect of apical size of canal preparation on treatment outcome.[93,102,150,188,223,229] One randomized controlled trial revealed that enlargement of the canal to three sizes larger than the first apical binding file was adequate[188] (the mean final size was ISO #30). The observational studies[95,150,229] had not designed their investigation with apical canal size as their principal focus and neither had they found a statistically significant influence from this factor; nevertheless, they all reported the same inverse trend of decreasing success rates with an increase in size of apical preparation. It was speculated that canal preparation to larger apical sizes may compromise treatment success by generation of more apical dentine debris, which in the absence of an adequate irrigation regimen serves to block apical canal exits that may still be contaminated with bacteria. Continued generation of dentine debris, in the absence of sufficient irrigation, may lead to what is termed *dentine mud,* which ultimately creates a blockage. The impatient or neophyte clinician fails to resist the temptation to force the instrument back to length, resulting in the classically described procedural errors of apical transportation, canal straightening, and perforation. An alternative mechanism is required to explain the higher failures in initially large canals; it is likely that immature roots present a different debridement challenge, where the canal shape is not amenable to planing of the main portions of the canal by conventional instruments. Perhaps an intracanal brush may be a more suitable cleaning device in such teeth. The findings from these studies therefore do not concur with views that more effective bacterial debridement may be achieved with larger apical preparations.[33,164,184]

The issue of apical preparation size should be considered together with that of the size and taper of the rest of the

canal preparation. Again, there is a paucity of sufficient direct evidence for the influence of degree of canal taper on root canal treatment outcome. The ESE guidelines[63] recommend only that canal preparation should be tapered from crown to apex without stipulating any particular degree of taper. Three studies have analyzed the influence of canal preparation taper on primary treatment and retreatment outcome, although again, none had focused their investigation primarily on this factor.[93,150,220] Smith and coworkers,[220] using loose criteria for determination of success, found that a "flared" preparation (wide taper) resulted in a significantly higher success rate compared with a "conical" preparation (narrow taper); the exact degree of taper was not reported and the effects of other treatment and nontreatment parameters were not controlled. In contrast, Hoskinson and associates[93] and Ng and colleagues,[150] using strict criteria, did not find any significant difference in treatment outcome between narrow (0.05) and wide (0.10) canal tapers. The controlled use of stainless steel instruments in a step-back technique may create 0.05 (1 mm step-back) or 0.10 (0.5 mm step-back) tapers, although, of course, uncontrolled use of such instruments may generate a variety of shapes. Ng and colleagues[150] also compared these (0.05 and 0.10) preparation tapers with 0.02, 0.04, 0.06, and 0.08 tapers (generally achieved by using greater taper nickel-titanium instruments) and found no significant effect on treatment outcome. They cautioned that their investigation of the influence of canal preparation taper without randomization could be influenced by the initial size of the canal, the type of instrument used, and operator experience.

Triangulation of the data on the effects of canal preparation size and taper on treatment outcome may intuitively lead to the conclusion that as far as current best evidence indicates, it is not necessary to overenlarge the canal to achieve periapical healing. An apical preparation size of ISO 30 with a 0.05 taper for stainless steel instrumentation or 0.06 taper for NiTi instrumentation is sufficient. Precisely what biologic and hydrodynamic mechanisms underpin such sufficiency is more difficult to define based on available evidence. Although a number of laboratory studies[4,84,112] have investigated the interaction between canal dimensions and irrigation or obturation dynamics, the precise physical, chemical, or biologic mechanisms that ultimately enable periapical healing remain unknown, although collaborations with fluid dynamics specialists[84] and (micro)biologists[82] may ultimately yield a clearer picture.

Procedural errors during root canal preparation include canal blockage, ledge formation, apical zipping and transportation, straightening of canal curvature, tooth or root perforation at the pulp chamber or radicular level, and separation of instruments. Instrument separation during treatment has been found to reduce the success rate significantly[150,229]; however, the reported prevalence of instrument separation was low (0.5% to 0.9%) in these studies, precluding an analysis of causative factors. A case-control study[225] revealed no significant difference in success rates between periapically involved teeth with or without retained separated instruments. The stage of canal debridement at which instrument separation occurred and the justification for their retention may have implications on the outcome. The coronoapical location of a separated instrument and whether the instrument was successfully bypassed were found to have no effect on treatment outcome.

Irrigant. Different chemical agents have been used as irrigants for root canal treatment, singly or in various combinations, both in clinical practice and in the studies reviewed. They have included water, saline and solutions of local anesthetic, sodium hypochlorite, iodine, chloramine, sulfuric acid, EDTA, hydrogen peroxide, organic acid, Savlon, urea peroxide, and Biosept (a quaternary ammonium compound).[153] Most of the studies had used sodium hypochlorite as an irrigant,[153] regardless of whether it was primary treatment or retreatment. This is consistent with the ESE guidelines[61] for irrigation, which recommend a solution possessing disinfectant and tissue-dissolving properties.

One prospective study[150] systematically investigated the effect of the irrigant on the success rates of root canal retreatment, which, although not a randomized controlled trial, revealed interesting new findings on the effects of irrigants. Even though a higher concentration of sodium hypochlorite made negligible difference to treatment outcome, the additional use of other specific irrigants had a significant influence on success rates.[150] The finding of a lack of improvement in periapical healing with the use of a higher concentration NaOCl solution is consistent with previous clinical/microbiologic findings.[29,41] Comparing 0.5% to 5.0% NaOCl solution for irrigation, it was found that the concentration of solution, per se, did not appear to increase the proportion of teeth rendered culture-negative[29] or associated with greater periapical healing.[41] As iodine and sodium hypochlorite are both halogen-releasing agents and attack common key protein groups,[129] the finding that the additional use of 10% povidone-iodine for irrigation had no additional influence on treatment success was as expected. Surprisingly, however, the additional use of 0.2% chlorhexidine solution for irrigation was found to reduce the success of treatment significantly.[150] This finding was in complete contrast to previous reports[211,252] on its equivalent or superior in vivo antibacterial efficacy when compared with sodium hypochlorite solution. The use of chlorhexidine as a final irrigant following sodium hypochlorite irrigation had been recommended[107] and was justified on several grounds, including its substantivity in root dentin (i.e., prolonged antibacterial effect),[185] relative lack of toxicity,[117] and broad-spectrum efficacy.[130] Not until recently has alternate irrigation with sodium hypochlorite and chlorhexidine solution raised serious concerns because of their interaction product. The interaction product is thought to be an insoluble precipitate containing para-chloroaniline, which is cytotoxic and carcinogenic.[14,24] Apart from mutually depleting the active moiety in the two solutions for bacterial inactivation, the precipitate may cause persistent irritation to the periapical tissue and block dentinal tubules and accessory anatomy, possibly explaining the observed lower success rate when chlorhexidine was used as an additional irrigant.

Ng and associates[150] also found that the additional use of EDTA had a profound effect on improving radiographically observed periapical healing associated with root canal treatment (OR = 1.5 [1.1, 2.0]). In contrast, the observed synergistic effect of sodium hypochlorite and EDTA had been previously demonstrated in terms of bacterial load reduction[26] but not periapical healing. The long-term (≥2 years) outcome of their cases stratified by canal disinfection protocols[25] did not support their microbiologic findings. Their reported success rate for alternate irrigation

with sodium hypochlorite and EDTA solutions (67%) was low when compared to the success rate for irrigation using saline (91%), 0.5% sodium hypochlorite (92%), or 5% sodium hypochlorite (86%) solutions.[162] The reported outcome data were unexpected, as preobturation negative bacterial culture was achieved in all cases. Given the complexity of their study design (clinical and microbiologic), their sample size was restricted to 11 to 15 teeth per group, limiting their outcome data. The synergistic effect of the two disinfectants has been attributed to the chelating properties of the sodium salts of EDTA, and their roles have been reviewed by Zehnder.[261] EDTA solution assists negotiation of narrow or sclerosed canals by demineralization of root dentine and helps in the removal of compacted debris from noninstrumented canal anatomy. It may also facilitate deeper penetration of sodium hypochlorite solution into dentine by opening dentinal tubules and removing the smear layer from the instrumented surface. Lastly, it may help detach or break up biofilms adhering to root canal walls.[85]

Medicament. Most previous treatment outcome studies have not standardized the type of root canal medicament used in the interappointment period, but the use of several different medicaments has been reported. The list was consistent with that recommended in the ESE guidelines for a medicament with disinfectant properties and included calcium hydroxide, creosote, and iodine solutions.[153] However, there is an absence of studies investigating the influence of this factor on treatment outcome.

The use of a mixture of calcium hydroxide and chlorhexidine has been tested based on the speculation that the mixture would be more effective against *E. faecalis.*[13,78,198]

Root Canal Bacterial Culture Results Prior to Obturation. In the past, in various centers of endodontic excellence, completion of root canal treatment by obturation would only be acceptable after a negative culture test was obtained from the canal, confirming the absence of bacteria in the part of the root canal system that could be sampled.[23,70,143] This practice has fallen out of clinical favor because of the perceived predictability and good prognosis of root canal treatment without microbiologic sampling. Sampling procedures are considered to be lengthy, difficult, and often inaccurate, requiring laboratory support and having low benefit-to-cost ratio.[138,139] A preobturation negative culture result may increase treatment success twofold (Fig. 17.19). One large study[201] helped contribute to the demise of the canal-culture test; however, even this study showed a 10% difference in success in favor of the negative culture test when periapical disease was present. The outcome is even worse when a positive culture test result combines with the presence of a periapical lesion.

Numerous studies[2,7,10,16,26-29,33,37,41,42,60,77,79,99,104,108,110,124,137,154,161,163,168,169,173,179,181,208,209,210,214,215,217,227,248,252,257,258] have evaluated the effect of different stages of root canal treatment on the intraradicular microbiota, both qualitatively and quantitatively (Table 17.9). Some studies merely report positive culture tests, whereas others have

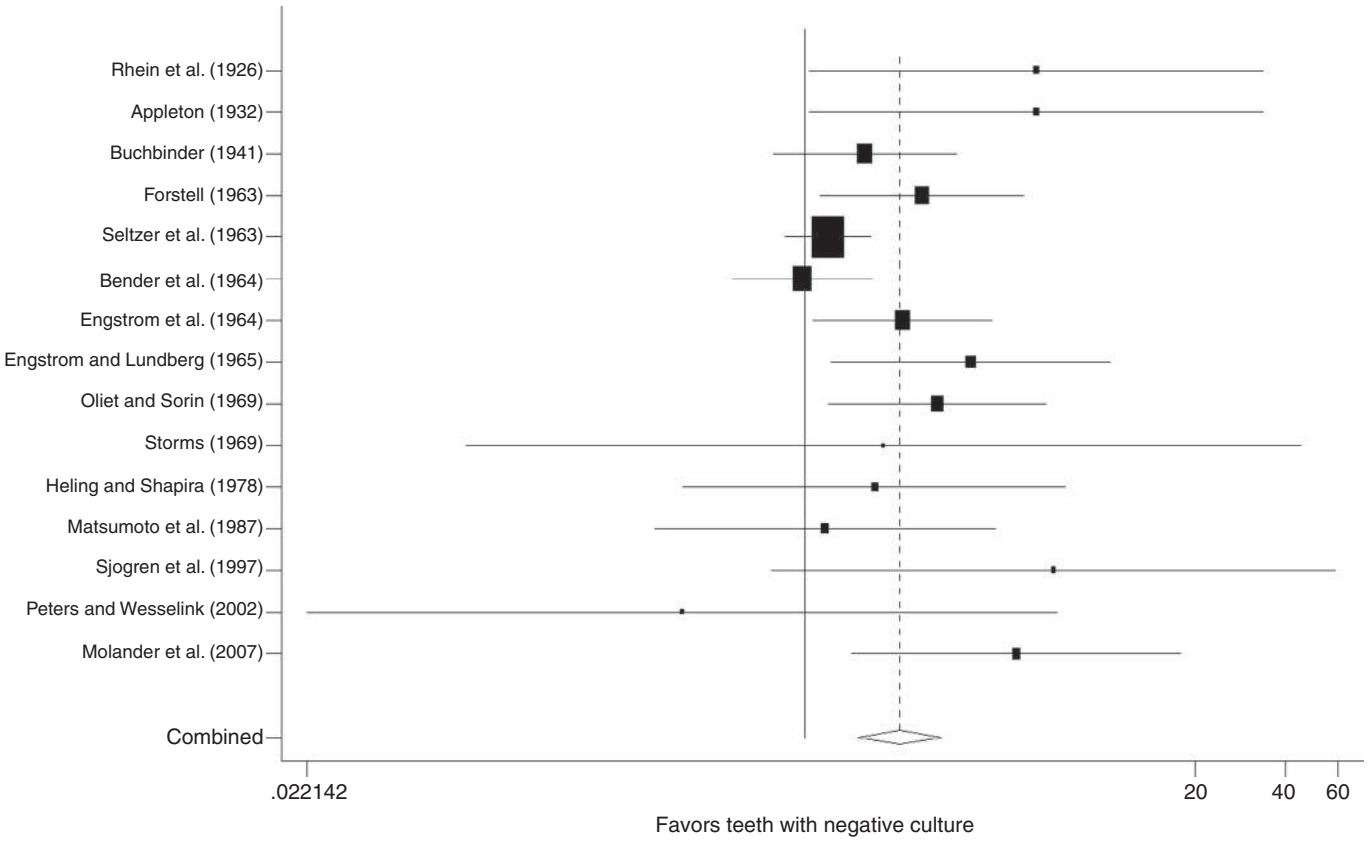


Fig. 17.19 Forest plot showing pooled and individual study's odds ratio (OR) for periapical health of teeth undergoing root canal treatment with preobturation negative versus positive culture test results (pooled OR = 2.1; 95% CI: 1.5, 2.9).

Table 17.9 Summary of Studies Evaluating the Effect of Root Canal Treatment Procedures on Bacterial Presence by Culture

			PERCENTAGE OF SAMPLES WITH BACTERIAL PRESENCE		
Study	**Year**	**Sample Size**	**At Baseline**	**After Preparation ± Irrigation**	**Next Visit (After Dressing±)**
Auerbach	1953	60 teeth	93% (56/60)	**Chlorinated soda (double strength)**: 22% (12/56)	—
Ingle and Zeldow	1958	89 teeth	73% (65/89)	**H_2O**: 70% (62/89) Some initially −ve became +ve after treatment	—
Stewart et al.	1961	77 teeth	100% (77/77)	**0.5% NaOCl + Gly-oxide:** 2% (1/44) **0.5% NaOCl + 3% H_2O_2:** 9% (3/33)	**No dressing:** 0.5% NaOCl + Gly-oxide: 34% (15/44) 0.5% NaOCl + 3% H_2O_2: 39% (17/33)
Nicholls	1962	155 teeth	100% (155/155)	**Alkaline chloramine:** 53% (39/74) **H_2O_2 and 2% NaOCl**: 50% (30/60) **H_2O and 2% NaOCl:** 71% (15/21)	—
Grahnén and Krasse	1963	97 teeth	77% (75/97)	**NaCl:** 72% (23/32) **Biocept:** 66% (21/32) **Nebacin:** 36% (12/33) Some initially −ve became +ve after Tx	**No dressing:** NaCl: 47% (15/32) Biocept: 47% (15/32) Nebacin: 18% (6/33)
Engström	1964	223 teeth (untreated or retreated)	60% (134/223)	**Biosept or Iodophor, plus alcohol, chloroform, and 0.5% NaOCl:** No data	**5% I_2 in 10% IKI:** 2nd visit: 43% (58/134); 3rd visit: 22% (29/134); 4th visit: 8% (9/134); 5th visit: 3% (4/134); 6th visit: 2% (3/134); 7th visit: 16% (22/134)
Olgart	1969	207 teeth	72% (149/207)	**H_2O_2 and 0.5% NaOCl or H_2O_2 and 1% NaOCl:** 43% (88/207)	**No dressing:** 34% (70/207)
Bence et al.	1973	33 teeth	100% (33/33)	**Pre-irritation:** 1st file: 93%, enlargement with #3: 14%, #4: 11%, #5: 21% (32% of instruments showed +ve culture, regardless of size) **5.25% NaOCl:** 48-h culture: 4% dentine, 10% pp 5-day culture: 8% dentine, 26% pp	**No dressing:** 8% dentine, 12% pp samples of teeth with negative culture after irrigation
Akpata	1976	20 extracted teeth	100% (20/20)	**NaCl:** 65% (13/20)	**38% CMCP:** 20% (2/10) When PP sample −ve, crushed tooth yielded −ve culture; When PP +ve, crushed teeth yielded +ve or −ve cultures
Cvek et al.	1976	108 teeth	NaCl group: 53% (18/34) 0.5% NaOCl group: 63% (29/46) 5% NaOCl group: 79% (22/28)	**NaCl:** 83% (15/18) **0.5% NaOCl:** 59% (17/29) **5% NaOCl:** 68% (15/22)	—
Byström and Sundqvist	1981	15 teeth	100% (15/15)	**Saline:** 100% (15/15)	**No dressing:** 47% (7/15) (5th visit) Where initial bacteria load high, difficult to eliminate
Byström and Sundqvist	1983	15 teeth	100% (15/15)	**0.5% NaOCl:** 87% (13/15)	**No dressing:** 20% (3/15) (5th visit)
Byström and Sundqvist	1985	60 teeth	100% (60/60)	**0.5% NaOCl:** No data **5% NaOCl:** No data **5% NaOCl + 15% EDTA:** No data	**No dressing:** 0.5% NaOCl: 12/20 (2nd visit); 8/20 (3rd visit) 5% NaOCl: 10/20 (2nd visit); 6/20 (3rd visit) 5% NaOCl + 15% EDTA: 11/20 (2nd visit); 3/20 (3rd visit)
Byström et al.	1985	65 teeth	100% (65/65)	**0.5% NaOCl:** No data **5.0% NaOCl:** No data	**CH:** 0/35 (1 month), 1/35 (2–4 days) **CP/CMCP** (2 weeks): 10/30

Table 17.9 Summary of Studies Evaluating the Effect of Root Canal Treatment Procedures on Bacterial Presence by Culture—cont'd

			PERCENTAGE OF SAMPLES WITH BACTERIAL PRESENCE		
Study	**Year**	**Sample Size**	**At Baseline**	**After Preparation ± Irrigation**	**Next Visit (After Dressing±)**
Sjogren and Sundqvist	1987	31 teeth	100% (31/31)	**0.5% NaOCl plus ultrasonic debridement:** No data	**No dressing:** 29% (9/31) at 2nd visit; 23% (7/31) at 3rd visit
Koontongkaew et al.	1988	15 teeth	100% (15/15)	**3% H_2O_2/5.25% NaOCl:** No data	**CMCP**: 1-day dressing: 40% (2/5); 3-day dressing: 20% (1/5); 7-day dressing 10% (1/10) **No dressing**: 60% (3/5) after 1 day, 20% (1/5) after 3 or 7 days
Reit and Dahlen	1988	35 teeth	91% (32/35)	**0.5% NaOCl:** No data	**CH**: After 14 days: 23% (8/35); after 21 days: 26% (9/35)
Molander et al.	1990	25 teeth	96% (24/25)	**0.04% iodine:** No data	**Clindamycin**: After 14 days: 16% (4/25); after 21 days: 24% (6/25)
Sjogren et al.	1991	30 teeth	100% (30/30)	**0.5% NaOCl:** 50% (15/30)	**CH:** 10 min: 50% (6/12) at 1 week later 7 day: 0% (0/18) (none after 1–5 weeks later without dressing)
Orstavik et al.	1991	23 teeth	96% (22/23)	**NaCl** irrigation and enlarged to: #20–25: 87% (20/23) further to #35–80: No data	**CH**: 34% (8/23); #35/40: 40% (6/15) #>40: 25% (2/8)
Yared and Bou Dagher	1994	60 teeth	100% (60/60)	**1% NaOCl:** Enlarged to #25: 73% (22/30) Enlarged to #40: 23% (7/30)	**CH:** 0% (0/60)
Gomes et al.	1996	42 root canals: Untreated (n = 15) Retreated (n = 27)	95% (40/42)	**2.5% NaOCl:** No data	**Empty canal** (7-10 days): 73% (29/40)
Sjogren et al.	1997	55 teeth (single canal)	100% (55/55)	**0.5% NaOCl:** 40% (22/55)	—
Dalton et al.	1998	46 teeth	100% (46/46)	NaCl + NiTi files: 68% (15/22); NaCl + K-files: 75% (18/24)	—
Reit et al.	1999	50 teeth	84% (42/50)	Enlarged to #35 (curved) or #50 (straight) with **0.5% NaOCl**: No data	**5% IKI** (5–7 days): 44% (22/50) **Empty** (7 days): 44% (22/50)
Peciuliene et al.	2000	25 teeth	80% (20/25);	**2.5% NaOCl and 17% EDTA:** No data	**Medication unknown:** 28% (7/25)
Shuping et al.	2000	42 teeth	98% (41/42)	**1.25% NaOCl:** 38% (16/42)	**CH:** 8% (3/40)
Lana et al.	2001	31 teeth	87% (27/31)	**2.5% NaOCl:** No data	**CH:** 13% (4/31) **Empty for 7 days:** 23% (7/31)
Peciuliene et al.	2001	40 teeth	83% (33/40)	**2.5% NaOCl and 17% EDTA:** 30% (10/33)	**CH (10–14 days)**: 25% (5/20) **IKI: 2% I_2 in 4% KI (10 min):** 5% (1/20)
Peters et al.	2002	42 teeth	Instrumentation to #20: 100% (42/42)	**Enlarged to #35 with 2% NaOCl:** 23% (10/42)	**CH** (4 weeks): 71% (15/21); further irrigation: 43% (9/21)
Card et al.	2002	40 mandibular teeth/canals	95% (38/40)	**1% NaOCl** Profile instrumentation (0.04 taper): 0/13 of cuspids and bicuspids, 5/27 of mesiobuccal canals Further LightSpeed instrumentation to size 57.5–65: 3/27 mesiobuccal canals of molars Only 1/16 of those mesiobuccal canals with detectable communication with the mesiolingual canals had +ve culture after the first preparation using ProFile instruments	No data

Continued

Table 17.9 Summary of Studies Evaluating the Effect of Root Canal Treatment Procedures on Bacterial Presence by Culture—cont'd

			PERCENTAGE OF SAMPLES WITH BACTERIAL PRESENCE		
Study	**Year**	**Sample Size**	**At Baseline**	**After Preparation ± Irrigation**	**Next Visit (After Dressing±)**
Kvist et al.	2004	96 teeth	98% (94/96)	**0.5% NaOCl:** 63% (60/96)	**CH** (7 days): 36% (16/44) **IPI** (10 min): 29% (15/52)
Chu et al.	2006	88 canals	99% (87/88)	**0.5% NaOCl:** No data	**CH, Septomixine forte, or Ledermix**: 36% (32/88) Exposure of pulp, tooth type, acute versus chronic condition, size of lesion, and type of medication had no significant effect
Paquette et al.	2007	22 teeth (single canal)	100% (22/22)	**2.5% NaOCl:** 68% (15/22)	**2% CHX**: 45% (10/22)
Siqueira et al.	2007a	11 teeth (single root)	100% (11/11)	**2.5% NaOCl:** 55% (6/11)	**CH/CPMC:** 9% (1/11)
Siqueira et al.	2007b	11 teeth (single root)	100% (11/11)	**2.5% NaOCl:** 45% (5/11)	**CH:** 18% (2/11)
Vianna et al.	2007	24 teeth (single root)	100% (24/24)	**Saline + 2% CHX gel:** 33% (8/24)	**2% CHX, CH or mixture:** 54% (13/24) Type of medication had no significant effect
Wang et al.	2007	43 canals	91% (39/43)	**Saline + 2% CHX gel:** 8% (4/39)	**2% CHX + CH:** 8% (3/36) Size of apical preparation (40 versus 60) had no significant effect.
Markvart et al.	2012	24 teeth	88% (21/24)	**2.5% NaOCl:** 63% (15/24)	**17% EDTA irrigation and 10 min 5% IKI medication:** 50% (12/24) Box preparation (#60): 67% (8/12) Cone preparation (#25–30): 33% (4/12)
Xavier et al.	2013	48 teeth (single canal)	100% (40/40)	**1% NaOCl:** 75% (9/12) **2% CHX:** 75% (9/12)	**CH:** 75% (18/24)

CH, Calcium hydroxide; *CP*, camphorated phenol; *CMCP*, camphorated monochlorophenol; *NaOCl*, sodium hypochlorite; *PP*, paper point sample; *r culture*, culture test before obturation.

identified and quantified intraradicular microbiota before and after various stages of treatment.

The effect of the "mechanical preparation" of the canal(s) on the microbiota has been tested using only water or saline as the irrigant. Taken collectively, the studies show that negative cultures were achieved on a weighted pooled average in 31% of the cases (range 0% to 79%). When sodium hypochlorite (concentration range 0.5% to 5%) irrigation supplemented the "mechanical preparation," the frequency of negative cultures immediately increased to a weighted pooled average of 52% (range 13% to 95%) (see Table 17.9).

Most studies report culture reversals during the interappointment period when active antibacterial dressing is not used in the root canal system between appointments. The reversals are due to regrowth of residual bacteria or recontamination by bacteria from coronal restoration leakage. When active interappointment antibacterial dressing is used, negative cultures in the subsequent visit were achieved on average in 71% of cases (range 25% to 100%) (see Table 17.9).

Effect of Persistent Bacteria on Root Canal Treatment Outcome. The bacteria present in preobturation cultures have included *Enterococcus*, *Streptococcus*, *Staphylococcus*, *Lactobacillus*, *Veillonella*, *Pseudomonas*, *Fusobacterium* species, and yeasts. Studies have been variable as to the relationship between individual species and treatment failure. Although the overall failure rate for cases with positive cultures was 31%, teeth testing positive for *Enterococcus* species had a failure rate of 55%, and teeth with positive cultures for *Streptococcus* species had 90% failures.[70] In another study, good-quality root canal treatment on 54 teeth with asymptomatic periapical disease gave an overall success rate of 74%, but teeth with positive cultures for *Enterococcus faecalis* achieved a success rate of only 66%.[231] The success rate for teeth with no bacteria was 80%, whereas that for teeth with bacteria in the canal before obturation was 33%. These associations cannot be regarded as direct cause-and-effect associations, but they further emphasize the need to determine a relationship between microbial diversity and treatment outcome.

A monkey-model study[63] used a four- or five-strain infection to test the effect of debridement and obturation procedures on outcome. When bacteria remained after chemomechanical debridement, 79% of the root canals

were associated with nonhealed periapical lesions, compared with 28% when no bacteria were found to remain. Combinations of several residual bacterial species were more frequently related to nonhealed lesions than were single strains. When no bacteria remained at the end of chemomechanical debridement, healing occurred independently of the quality of the root filling. In contrast, when bacteria remained in the canal system, there was a greater correlation with nonhealing associated with poor-quality root fillings than in technically well-performed fillings. In root canals where bacteria were found after removal of the root filling, 97% had not healed, compared with 18% for those root canal systems with no bacteria detected upon removal of root filling. The study emphasizes the importance of reducing bacteria below detection limits before permanent root filling in order to achieve optimal healing conditions for the periapical tissues. It also reinforces the view that obturation does indeed play a role when residual infection is present.

Regardless of the technique used for obtaining a sample for a canal culture, the presence of a negative culture seems to have a positive impact on treatment outcome. The association of specific species with treatment failure is not well established but the identity of the small group of species isolated from positive cultures is relatively constant and may hold answers to treatment resistance and failure. However, it is important to understand that there are many other factors that can influence root canal treatment outcome.

Root Filling Material and Technique. The interrelationship among the core root filling material, sealer (for filling the gaps between the core material and canal surface), and technique for their placement complicates the investigation of the effect of obturation and technique on treatment outcome. In previous studies on treatment outcome, the most commonly used core root filling material was gutta-percha with various types of sealer or gutta-percha softened in chloroform (chloropercha).[153] The sealers used may be classified into zinc oxide–eugenol–based, glass ionomer–based, and resin-based types.[153] Materials such as Resilon, SmartSeal, and MTA have been adopted but have not significantly penetrated the market, except for the use of MTA in surgical repairs or repairs for immature apices. In any case, there is no evidence to show that the nature of root filling material and the technique used for placement has any significant influence on treatment outcome.

Apical Extent of Root Filling. Of the many intraoperative factors associated with success and failure of root canal treatment, the apical extent of the root canal filling material has been the most frequently and thoroughly investigated. In these previous studies, the apical extent of root fillings has been classified into three categories for statistical analyses: more than 2 mm short of radiographic apex (short), 0 to 2 mm within the radiographic apex (flush), and extended beyond the radiographic apex (long).[153] The apical extent of root filling was found to have a significant influence on the success rates of treatment, regardless of the periapical status.[150,153] Flush root fillings were associated with the highest success rates, whereas long root fillings were associated with the lowest success rates.

Most previous retrospective studies could not distinguish between the effects of apical extent of instrumentation versus the apical extent of obturation; however, the London Eastman study[150] was able to separate the effect of these two factors and found them both to independently and significantly affect periapical healing. The factors correlated with each other, consistent with the fact that canals are normally filled to the same extent as canal preparation.

Extrusion of cleaning, medication, or filling materials beyond the apical terminus into the surrounding tissues may result in delayed healing or even treatment failure due to a foreign body reaction.[105,146,218,260] Magnesium and silicon from the talc-contaminated extruded gutta-percha were found to induce a foreign body reaction, resulting in treatment failure.[146] An animal study has shown that large pieces of subcutaneously implanted gutta-percha in guinea pigs were well encapsulated in collagenous capsules, but fine particles of gutta-percha induced an intense, localized tissue response.[218] The inference that perhaps extrusion of large pieces of gutta-percha may not impact periapical healing was not supported by data from previous studies.[150,153] The discrepancy may possibly be accounted for by bacterial contamination of the extruded gutta-percha in the clinical data.

The radiographic evidence of "sealer puffs" extruding through the main apical foramina and lateral/accessory canals has been pursued with vigor by some endodontists based on the undaunted belief of its value as "good practice." Their perception is that this represents a measure of root canal system cleanliness, and they ardently argue that healing would follow, albeit with some delay. The published evidence on the effects of sealer extrusion into the periapical tissues has been contradictory. Friedman and colleagues[68] found that extrusion of a glass ionomer-based sealer significantly reduced success rates. In contrast, Ng and associates[150] reported that extrusion of a zinc oxide–eugenol based–sealer had no significant effect on periapical healing. The discrepancy may be attributed to the difference in sealer type and the duration of treatment follow-up. The radiographic assessment of the presence or resorption of sealer may be complicated by the radiolucent property of its basic components and the insufficient sensitivity of the radiographic method used to detect small traces of it.[150] It is possible that, in some cases, the radiographic disappearance of extruded sealer may simply be due to resorption of the radio opaque additive, barium sulfate, or its uptake by macrophages, still resident in the vicinity.[146]

Extruded glass ionomer–based[68] zinc oxide–eugenol–based,[98] silicone-based[98] sealers or endomethasone[22] were not found to be resorbed/absorbed by periapical tissues after 1 year. Traces of calcium hydroxide–based sealer (Sealapex) could still be detected after 3 years.[194] In the latter study, treatments were carried out on primary molar teeth and the canals were obturated with Sealapex without gutta-percha. With longer duration of follow-up, complete resorption of extruded zinc oxide–eugenol–based sealers (Procosol, Roth Elite)[8] and a resin-based sealer (AH Plus, Dentsply/DeTrey, Konstanz, Germany)[194] was demonstrated in 69% and 45% of the cases after 4 and 5 years, respectively. Ng and associates[150] advanced two explanations for the difference between the effect of extruded core gutta-percha and the zinc oxide–eugenol sealer: the latter is

antibacterial and may kill residual microorganisms, whereas it is also more soluble and readily removed by host cells compared to gutta-percha.

Quality of Root Filling. Another much investigated parameter of obturation in retrospective studies has been the radiographic measure of the "quality of root filling." The rationale for complete obturation of the root canal system is to prevent recontamination by colonization from the residual infection or newly invading bacteria. Both are supposedly prevented by a "tight" seal with the canal wall and an absence of voids within the body of the material. Quality of root filling may therefore be regarded either as poor root-filling technique or as a surrogate measure of the quality of the entire root canal treatment, because good obturation relies on properly executed preliminary steps in canal preparation. A systematic review[153] reported that the criteria for judging the quality of root fillings have not been well defined in previous studies.[43,93,216] An unsatisfactory root filling has been defined as "inadequate seal," "poor apical seal," or "radiographic presence of voids." Nevertheless, satisfactory root fillings were found to be associated with significantly higher success rates than unsatisfactory root fillings.[153]

Acute Exacerbation During Treatment. The causes for interappointment "flare-up" or pain have not been precisely determined, and several hypothetical mechanisms involving chemical, mechanical, or microbial injury to the periradicular tissues, as well as psychological influences, have been suggested as contributory to postpreparation pain or swelling.[202,203] Although these factors have not been specifically studied in the context of periapical healing, acute "flare-ups" during treatment were not found to be significantly associated with periapical healing in two studies.[102,216] In contrast, the London Eastman study[150] found that pain or swelling occurred in 15% of cases after chemomechanical debridement and was found to significantly reduce success as measured by periapical healing. This interesting finding may be explained by the hypothesis that "flare-ups" were caused by extrusion of contaminated material during canal preparation. Such material may elicit a foreign body reaction or (transient) extraradicular infection, resulting in treatment failure in a proportion of such cases. Alternatively, acute symptoms may be the result of incomplete chemomechanical debridement at the first appointment, leading to a shift in canal microbial ecology favoring the growth of more virulent microorganisms, leading to further postpreparation pain and treatment failure. The exact biologic mechanisms of failure in these cases remain obscure and warrant further investigation.

Number of Treatment Visits. The number of treatment visits for completing root canal treatment and its effect on periapical healing remains an ongoing controversy. Generally, the argument for single-visit treatment centers around better patient acceptability and cost-effectiveness versus the preference of multiple-visit treatments based on biologic rationale.[224] The premise for multiple-visit treatments has been that primary debridement is not completely effective in eliminating all the adherent bacterial biofilm,[147] and the residual bacteria may multiply and recolonize the canal system.[25,29] Therefore the proponents consider it desirable to use the interappointment period to dress the canal with a long-lasting or slow-release antibacterial agent capable of destroying or incapacitating residual bacteria, as well as to take the opportunity to gauge the initial periapical response before root filling. Calcium hydroxide has served in this capacity for many years because of its ability to dissolve organic tissue, kill bacteria, detoxify antigenic material, and act as a slow-release agent because of its low solubility-product in an aqueous environment. However, its antibacterial ability has come under close scrutiny, with advocates suggesting that the material is not suitable for this purpose.[195] A final resolution to this debate is awaited based on robust clinical evidence. Most of the published randomized controlled trials found no significant influence of healing attributable to number of treatment visits, but they all lacked robust statistical power.

The debate about the merits of single- or multiple-visit treatments will continue unabated given the respective strengths of the motivational drivers among the opposing groups. The issue may only be resolved by properly documented, large randomized controlled trials (which are currently unavailable) because undocumented variables (i.e., operator skill, biologic or technical case complexity, and patient compliance) would continue to bias the outcome.

Post Root Canal Treatment Restorative Factors

Effect of Quality and Type of Restoration. The placement of a coronal restoration after root canal obturation is the final step in the management of teeth undergoing root canal treatment. It has been shown to have a major influence on endodontic outcomes. Teeth with satisfactory coronal restorations were found to have significantly better periapical healing compared with those with unsatisfactory restorations (OR = 3.31; 95% CI: 1.07, 10.30).[153] The term *satisfactory restorations* has been defined as a restoration with no evidence of marginal discrepancy, discoloration, or recurrent caries with absence of a history of decementation.[93,182]

Given that one of the roles of coronal restorations is to prevent postoperative root canal reinfection via coronal leakage, the criteria for unsatisfactory restoration given by Hoskinson and colleagues[93] could not infer coronal leakage when the inner core was still intact. Consequently, the London Eastman study[150] adopted a different classification and definition for unsatisfactory restorations in order to illustrate obvious and potential coronal leakage more accurately. The two groups of unsatisfactory restorations were defined as those with (1) obvious signs of exposed root filling and (2) potential leakage indicated by marginal defects and history of decementation. It is perhaps this strategy that contributed to the finding of a profound effect (OR = 10.7; 95% CI: 3.7, 31.5) of coronal leakage on the endodontic outcome.

A number of investigations have been performed based on comparisons between the types of post root canal treatment restorations, including permanent versus temporary restorations,[43,68,150] crown versus acrylic restorations,[43,68,150,216] presence versus absence of posts,[68,150] and non-abutment versus abutment.[150,216] Teeth that had been permanently restored were associated with significantly higher success rates than their temporarily restored counterparts

in some studies[43,68] but not in others.[38,150] The type of permanent restoration[43,68,150,216] was found to have no significant influence on the outcome of treatment.

It has often been recommended that it would be wise to provide a subseal over the root filling in case of loss of a permanent or temporary restoration; the subseal would be glass ionomer (GIC) or zinc oxide–eugenol cement.[196] The placement of a GIC or zinc oxide–eugenol (IRM) cement lining coronal to the gutta-percha filling and underneath the permanent core in order to provide an additional antibacterial coronal seal was found in a prospective study to have no beneficial effect on treatment success.[150]

In summary, the preceding findings overall support the ESE guidelines[61] that an adequate restoration should be placed after root canal treatment to prevent subsequent bacterial recontamination. Therefore the provision of a good-quality coronal restoration, regardless of type, should be considered the final part of the root canal treatment procedure following obturation.

Use of Root Treated Teeth as Abutments for Prostheses and Occlusal Contacts. Mechanical stress on restorations is a function of the role of individual teeth in the occlusal scheme. The pattern of occlusal loading both in static and dynamic occlusion is dictated by whether the teeth are involved as single units or abutments (bridge/denture) and whether they have holding or guiding contacts. It is reasonable to expect that bridge and denture abutments may be placed under unfavorable loads, as may last-standing teeth in the dental arch.[127] These teeth may therefore be expected to have lower success rates because of a potential increase in the development of cracks and fractures due to fatigue. This observation has been confirmed for teeth functioning as bridge abutments compared with those restored as individual units following root canal treatment.[216]

Summary of Factors Influencing Periapical Healing Following Nonsurgical Root Canal Treatment

The following factors are considered as having a major impact on periapical health subsequent to root canal treatment:

- Presence (Fig. 17.20) and size (Fig. 17.21) of periapical lesion
- Patency at the canal terminus (achieving patency significantly increased the chance of success twofold)[150]
- Apical extent of chemomechanical preparation in relation to the radiographic apex (Fig. 17.22)
- Outcome of intraoperative culture test (see Fig. 17.19)
- Iatrogenic perforation (if present, reduces the odds of success by 30%)[150]
- Quality of root canal treatment judged by the radiographic appearance of the root filling (Fig. 17.23)
- Quality of the final coronal restoration (Fig. 17.24)

The following factors are considered as having minimal impact on root canal treatment outcome:

- Age of patient
- Gender of patient
- Tooth morphologic type
- Specific root canal treatment protocol and technique (preparation, irrigation, and obturation material and technique)

Contemporary improvements in mechanical and chemical canal preparation have not resulted in an increase in the success rate for root canal treatment over the past century (Fig. 17.25). This observation may be explained by the currently available techniques not being effective in eliminating the infection in the apical canal anatomy.

It is notable that all factors that have a strong influence on periapical health after root canal treatment are associated in

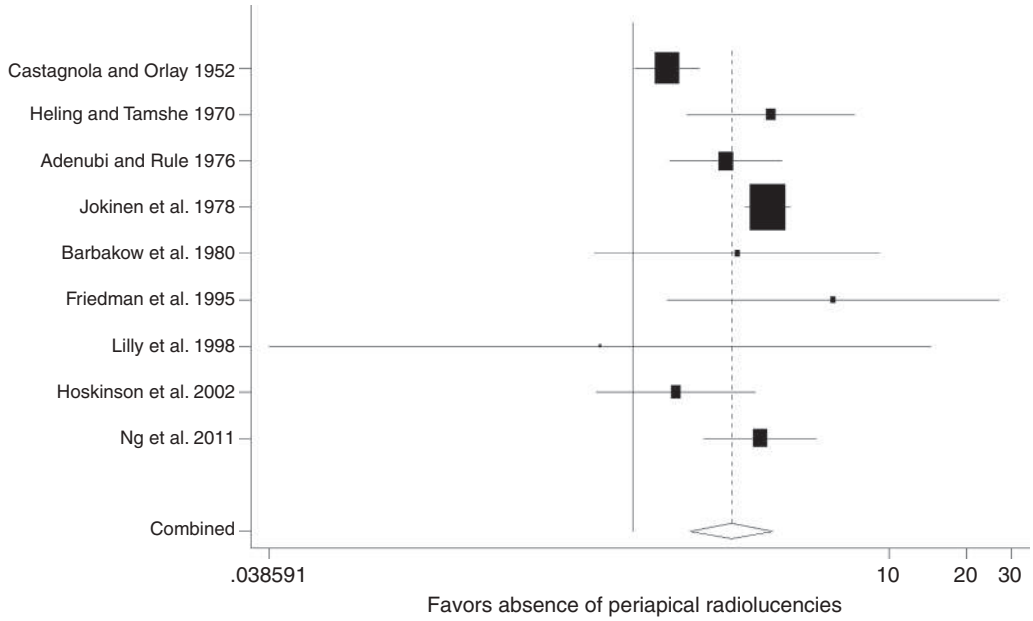


Fig. 17.20 Forest plot showing pooled and individual study's odds ratio *(OR)* for periapical health of teeth undergoing root canal treatment with preoperative nonvital pulps and absence of periapical radiolucencies versus teeth with nonvital pulps and presence of periapical radiolucencies (pooled OR = 2.4; 95% CI: 1.7, 3.5).

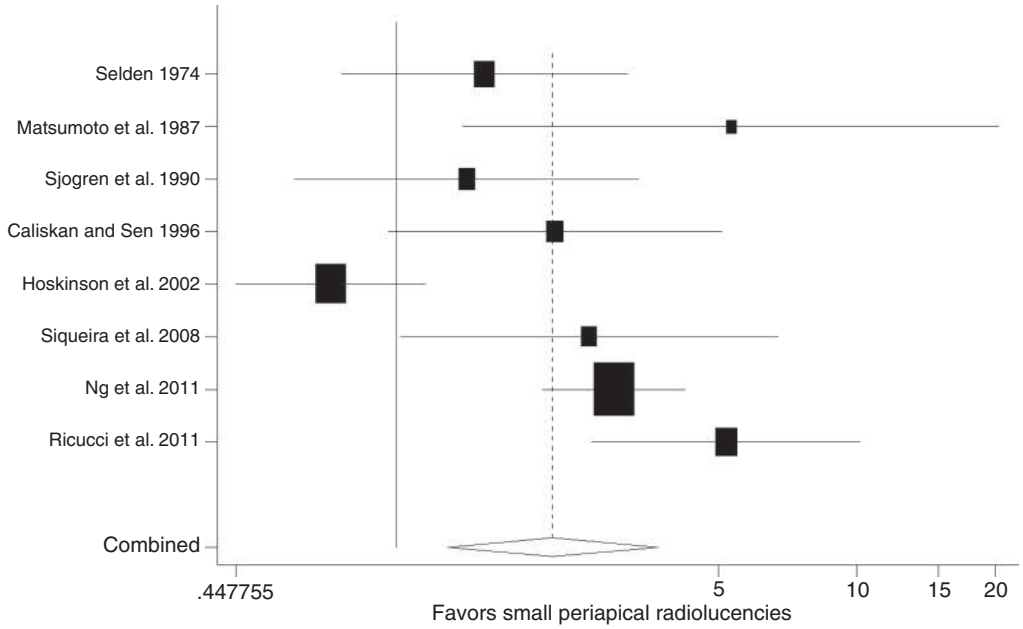
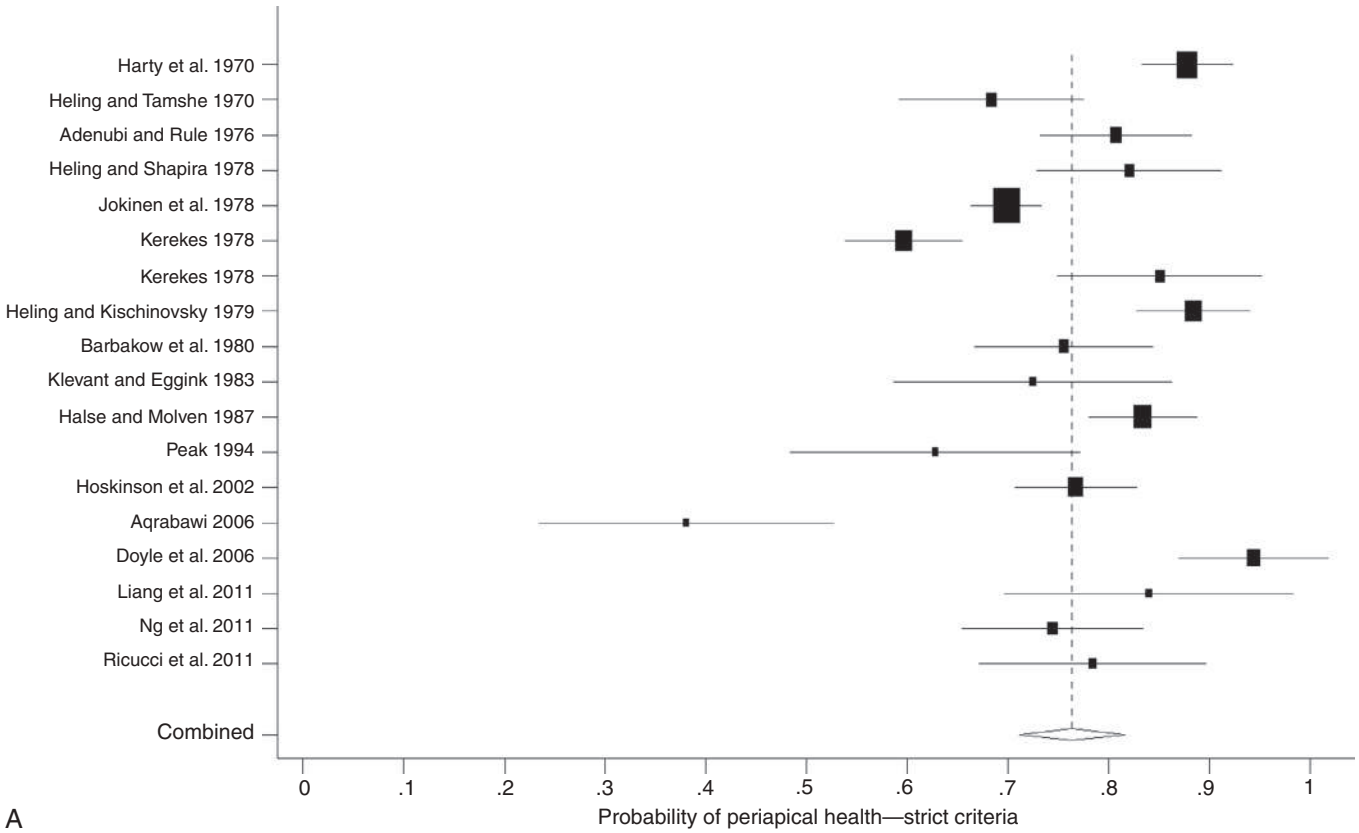
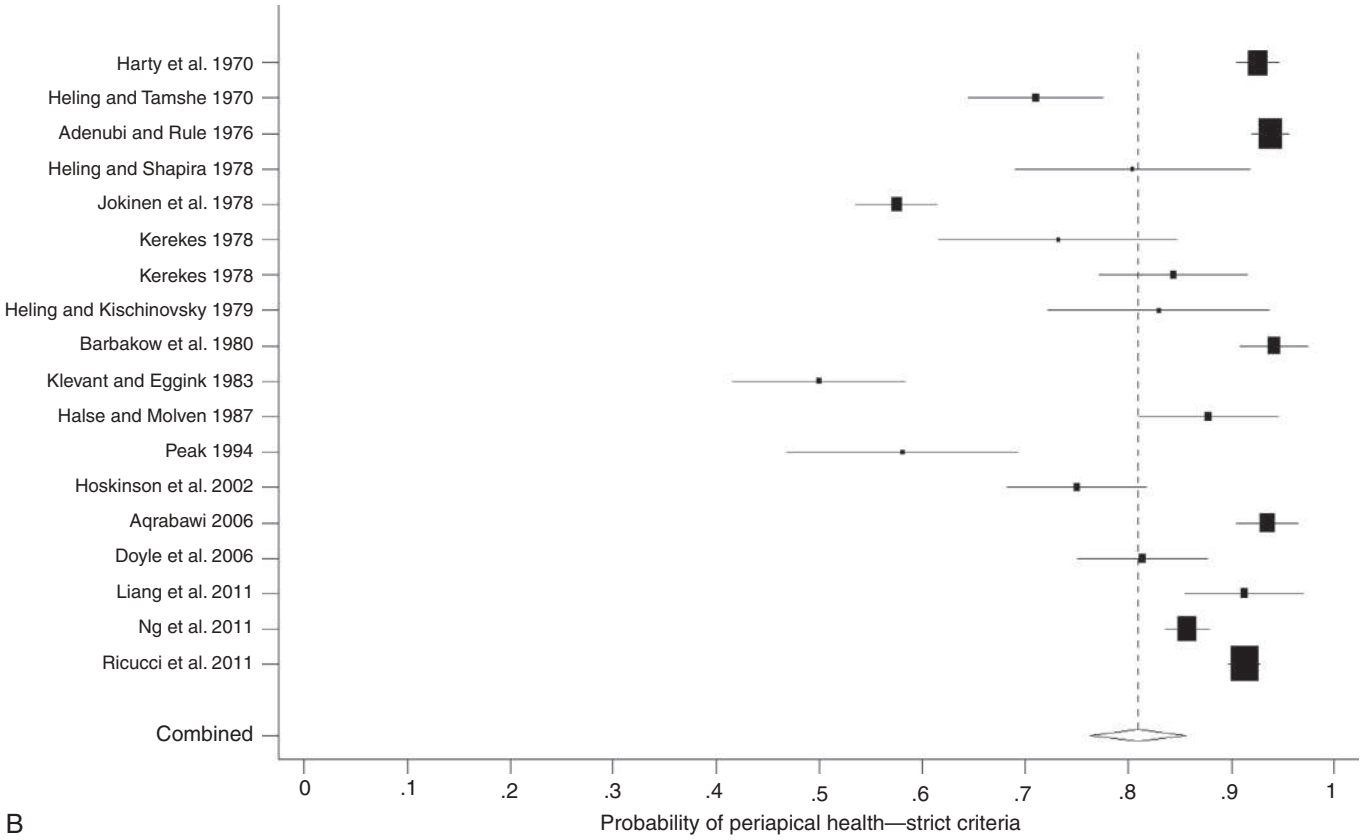


Fig. 17.21 Forest plot showing pooled and individual study's odds ratio *(OR)* for periapical health of teeth undergoing root canal treatment with preoperative large (>5 mm) versus small (<5 mm) periapical radiolucencies (pooled OR = 2.2; 95% CI: 1.3, 3.7).

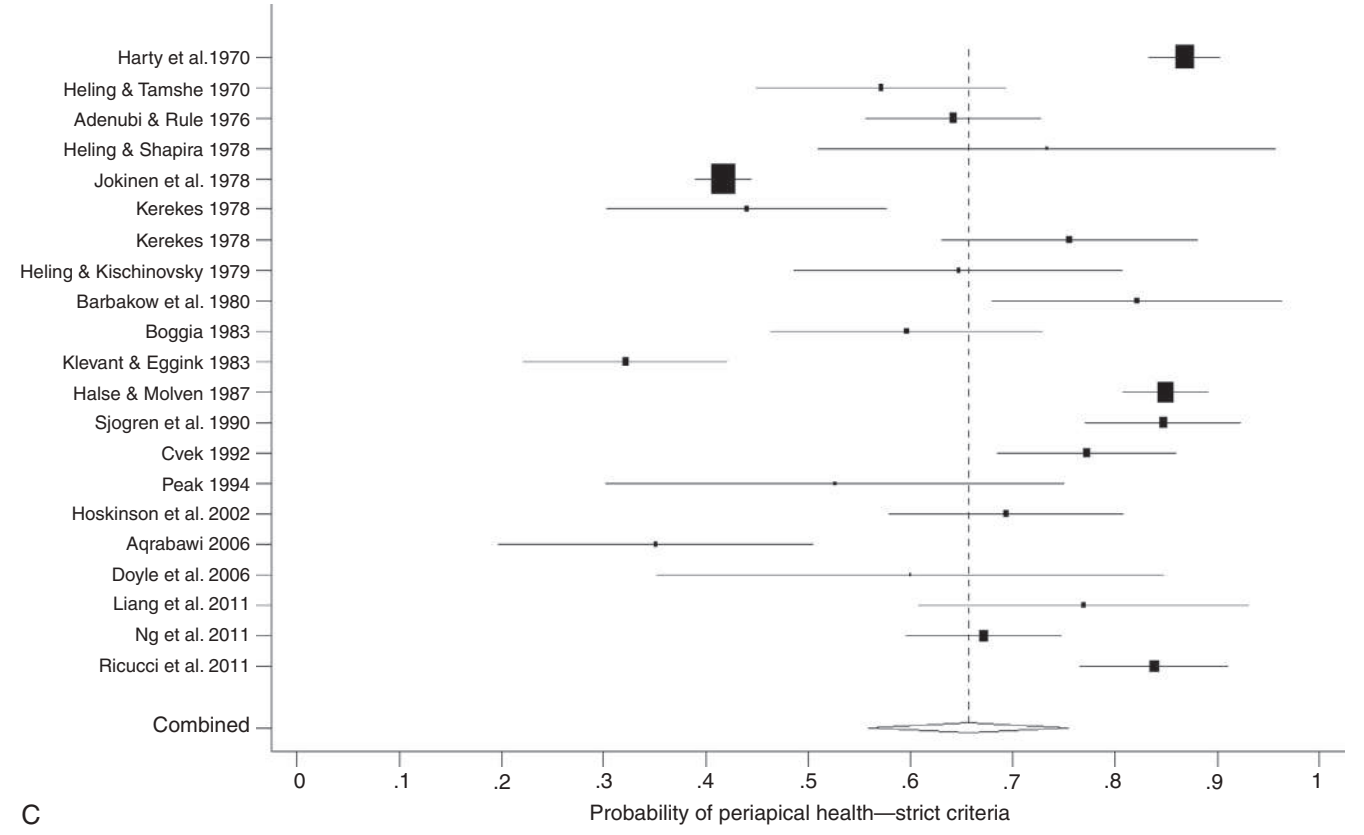


A

Fig. 17.22 Forest plot showing pooled and individual study's probability of periapical health for teeth undergoing root canal treatment with underextended (0.76 [0.71, 0.82]) **(A)**, flush (0.81 [0.76, 0.86]) **(B)**, or overextended (0.66 [0.56, 0.75]) (**C)** root fillings.



B



C

Fig. 17.22, cont'd

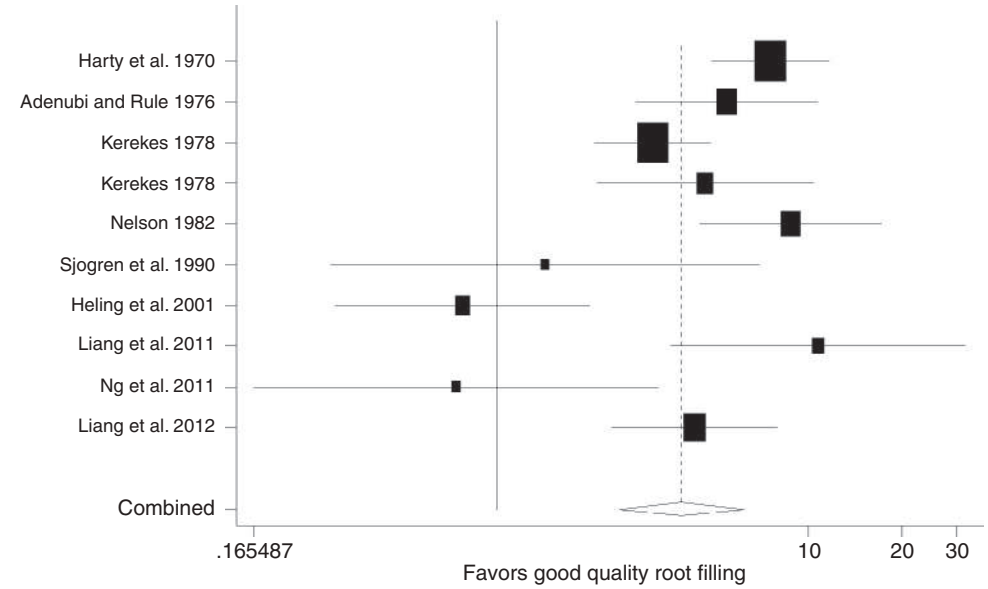


Fig. 17.23 Forest plot showing pooled and individual study's odds ratio (OR) of periapical health for teeth undergoing root canal treatment with good quality versus suboptimal quality root filling (pooled OR = 3.9; 95% CI: 2.5, 6.2).

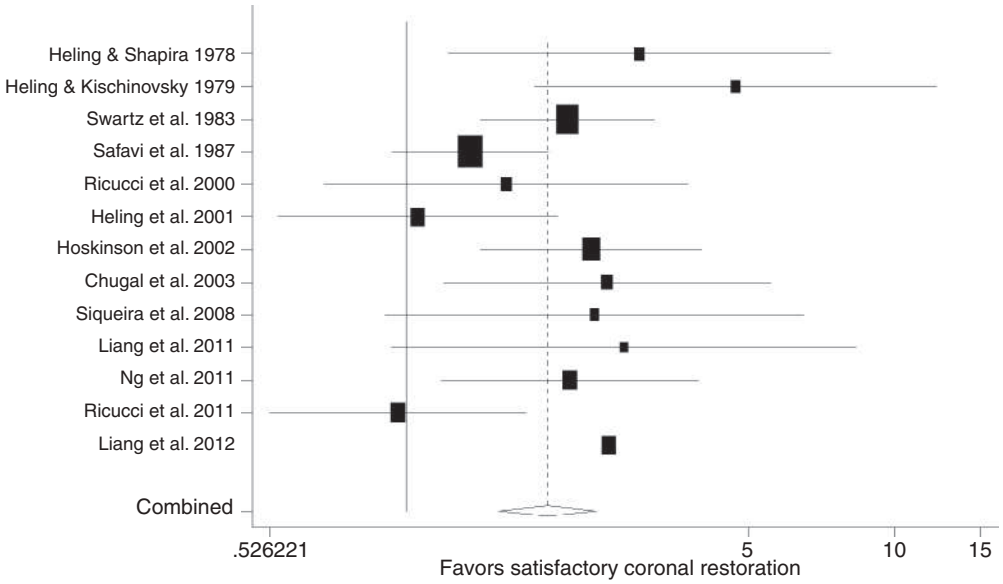


Fig. 17.24 Forest plot showing pooled and individual study's odds ratio (OR) of periapical health for teeth undergoing root canal treatment with satisfactory versus unsatisfactory coronal restoration at follow-up (pooled OR = 1.9; 95% CI: 1.5, 2.5).

some way with a persistent root canal infection. Further improvements in root canal treatment outcomes may therefore be achieved by understanding the nature of the root canal infection (especially apically) and the manner in which the microbiota is altered or eliminated during root canal treatment.

FACTORS AFFECTING TOOTH SURVIVAL FOLLOWING ROOT CANAL TREATMENT

A systematic review and meta-analysis has shown that 93% of endodontically treated teeth survive at 2 years postoperatively; but this survival reduced to 88% at 10 years following treatment (Fig. 17.26). The most common reasons for such tooth loss were due to problems of endodontic origin, tooth/root fracture, or restoration failure.[149,151] Consistent with the studies on periapical healing, the considerations affecting the survival of endodontically treated teeth may be divided into patient, intraoperative, and restorative factors.

Patient Factors

Ng and colleagues[151] found that teeth in patients suffering from diabetes or receiving systemic steroid therapy had a higher chance of being extracted after root canal treatment. The negative influence of diabetes on tooth survival is consistent with the report by Mindiola and associates,[135] whereas the influence of steroid therapy had never been

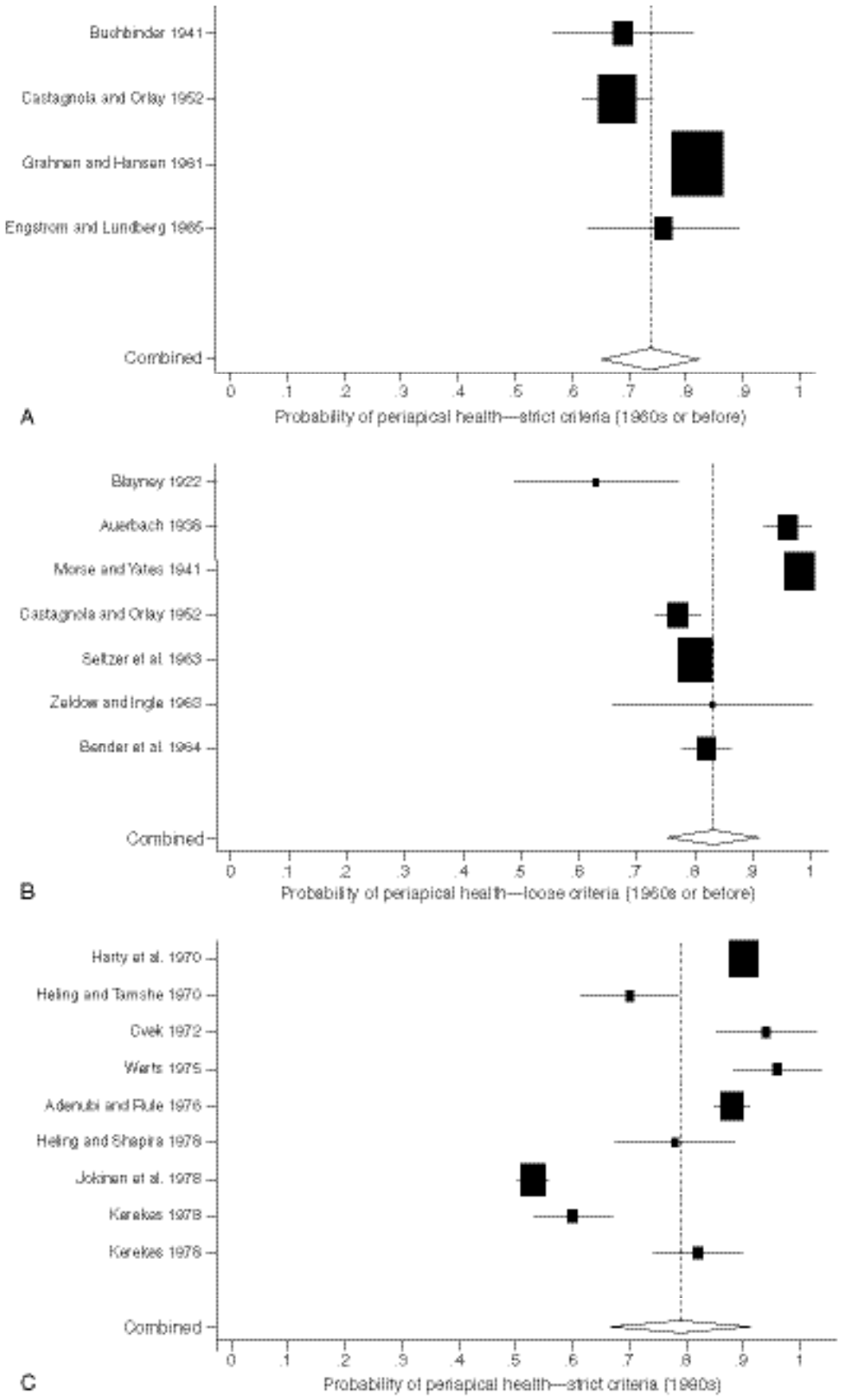
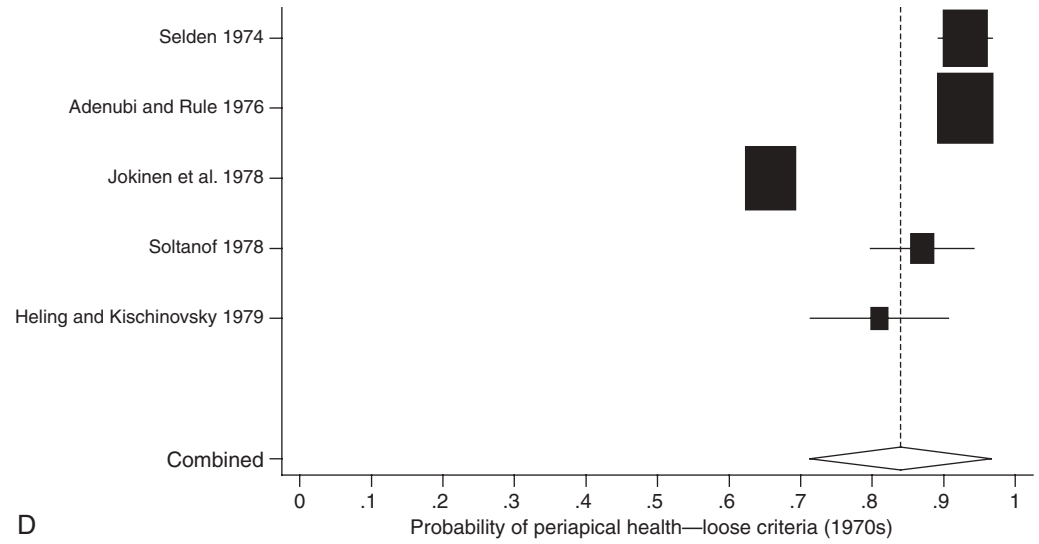
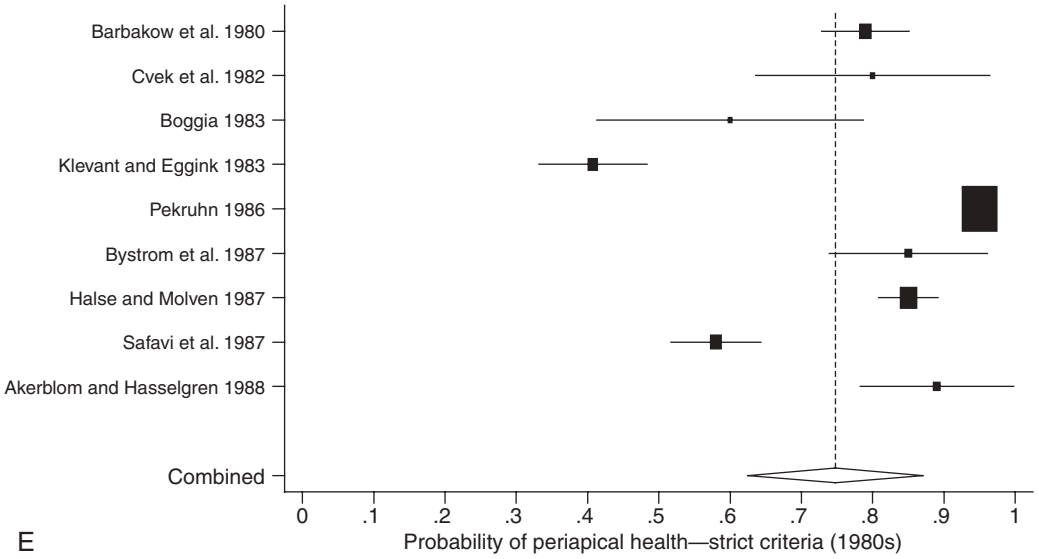


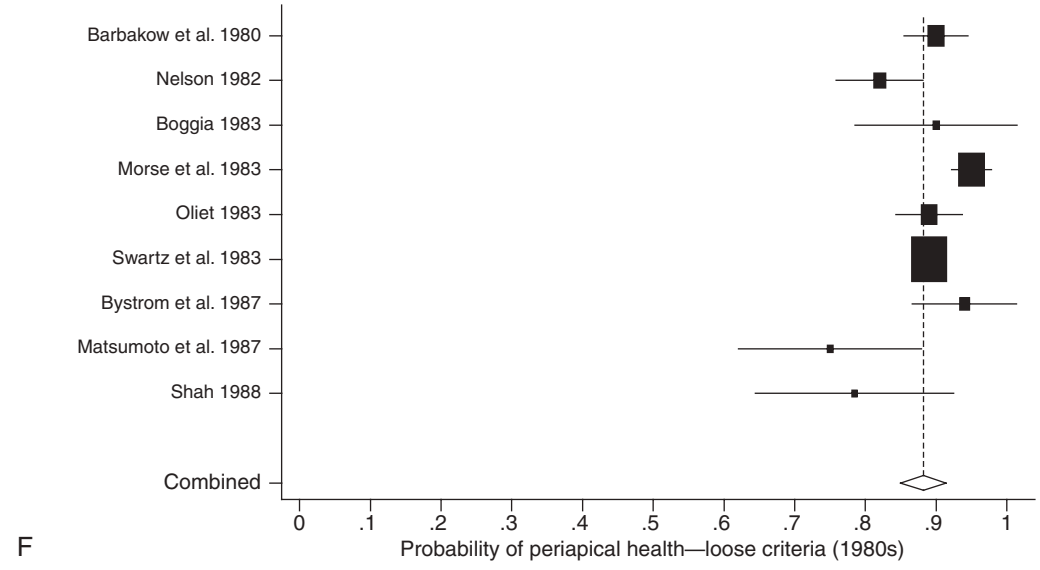
Fig. 17.25 Forest plot showing pooled and individual study's probability of periapical health for teeth undergoing root canal treatment by "decade of publication" and "criteria for success."



D

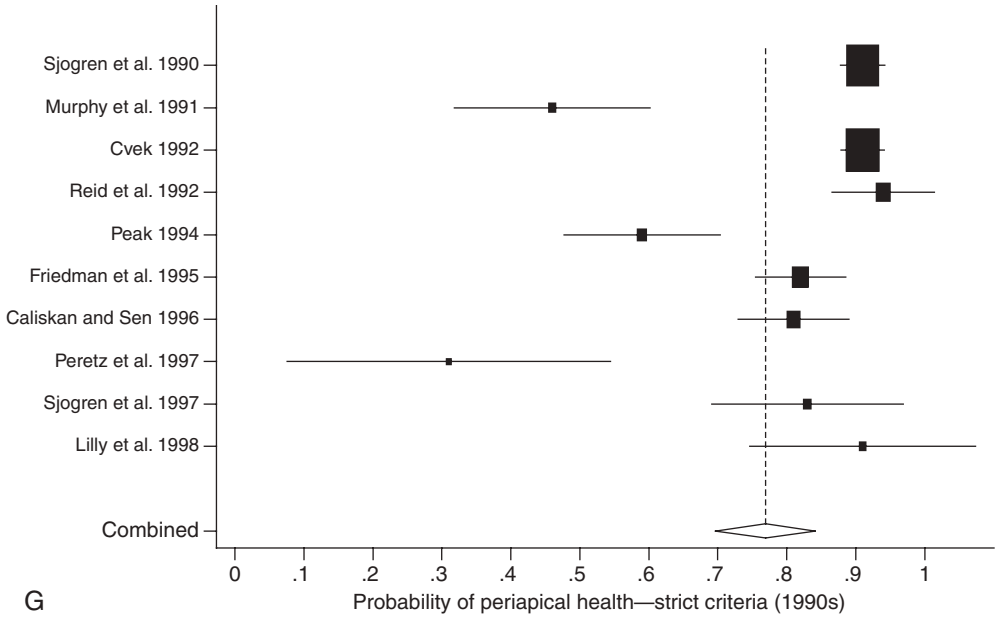


E

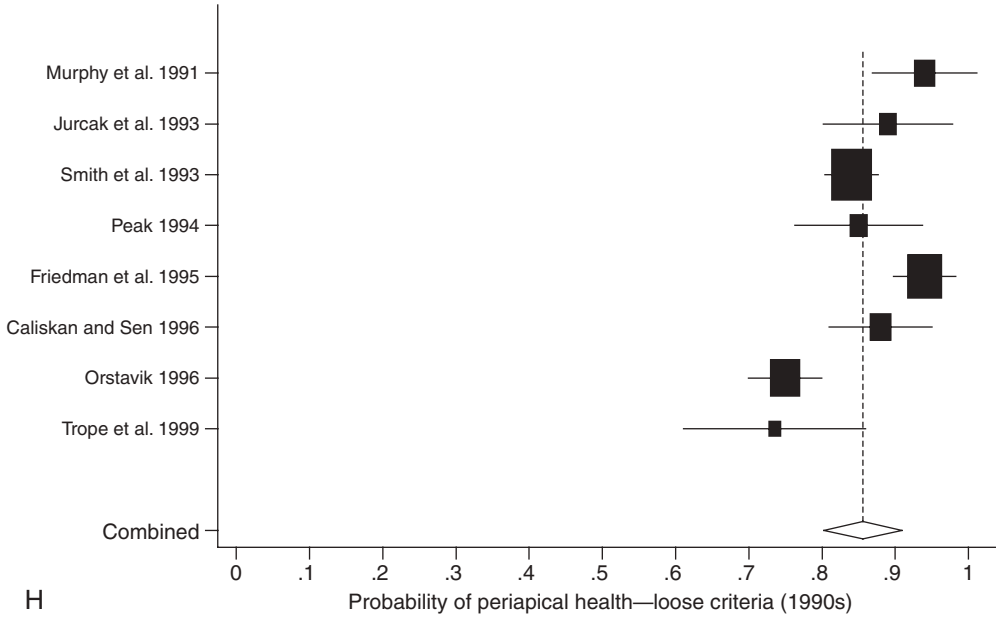


F

Fig. 17.25, cont'd



G



H

Fig. 17.25, cont'd

reported previously. It may be argued that patients with diabetes are more susceptible to periodontal disease[74] or have a lower success rate of root canal treatment[67] because of being immunologically compromised. However, the researchers reported that over 50% of such teeth were extracted due to persistent pain. Some of these observations may be explained by the presence of neuropathy, a debilitating painful complication of diabetes.[57] It is interesting to note that systemic steroid therapy is often prescribed to control such chronic pain.[39,51,101]

Tooth Morphologic Type and Location

Tooth types (i.e., location within the arch) may vary with respect to their susceptibility to tooth fracture, a common reason for tooth loss after endodontic treatment. Ng and coworkers[149] found that tooth type had a significant influence on survival. Maxillary premolars and mandibular molars were found to have the highest frequency of extraction, with tooth fracture being the most common reason. The observation is consistent with previous reports on fracture incidence of maxillary premolars and mandibular molars.[56,109] The factors "proximal contacts" and "terminal (last standing) teeth" were found to affect tooth survival considerably,[149,151] but they were significantly correlated with molar teeth. Most of the extractions of terminal teeth or teeth with one or fewer proximal contacts were due to tooth fracture.[151] The observation may be explained by the unfavorable distribution of occlusal forces and higher nonaxial stress on terminal teeth and those with fewer than two proximal contacts. Other possible reasons for their

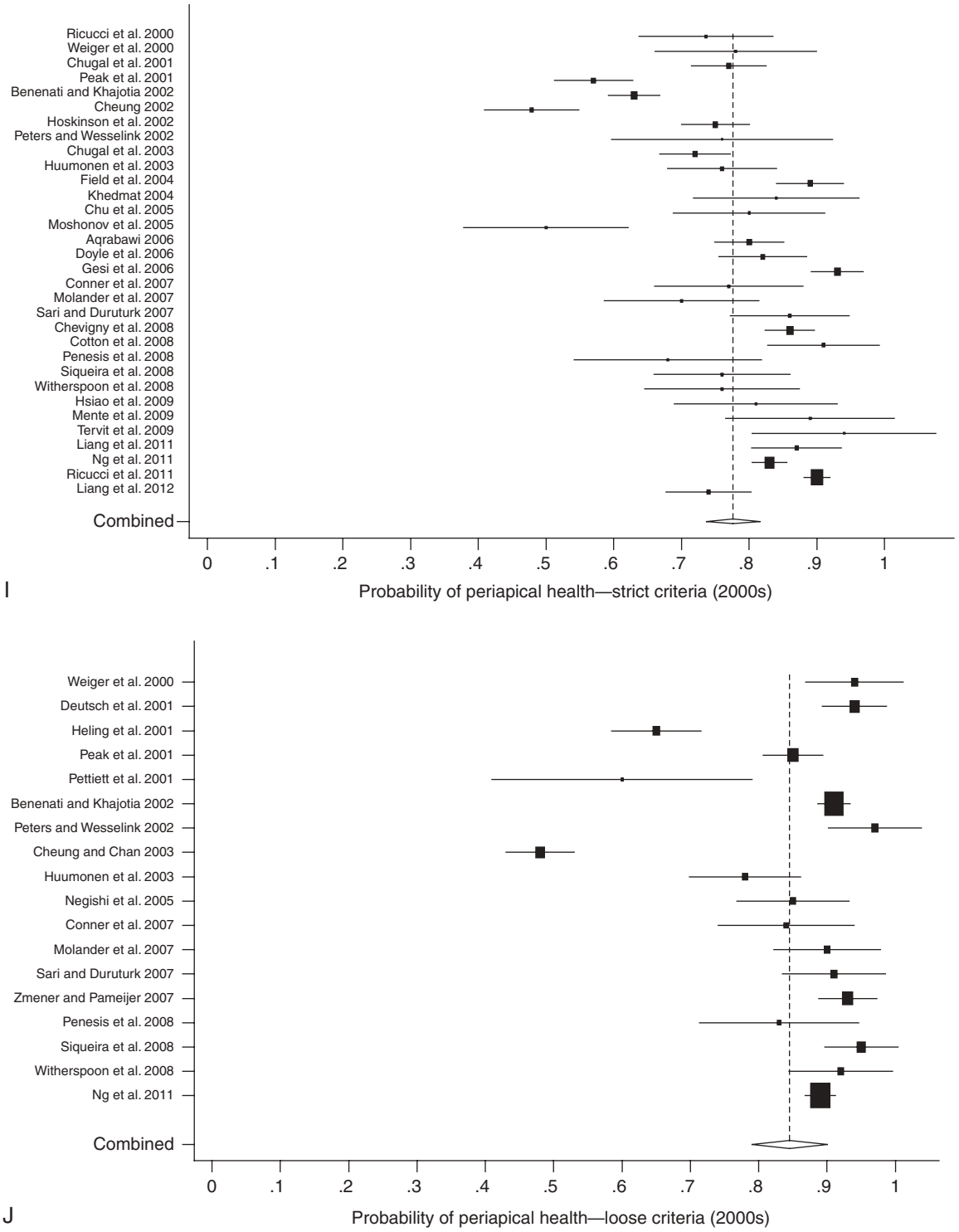
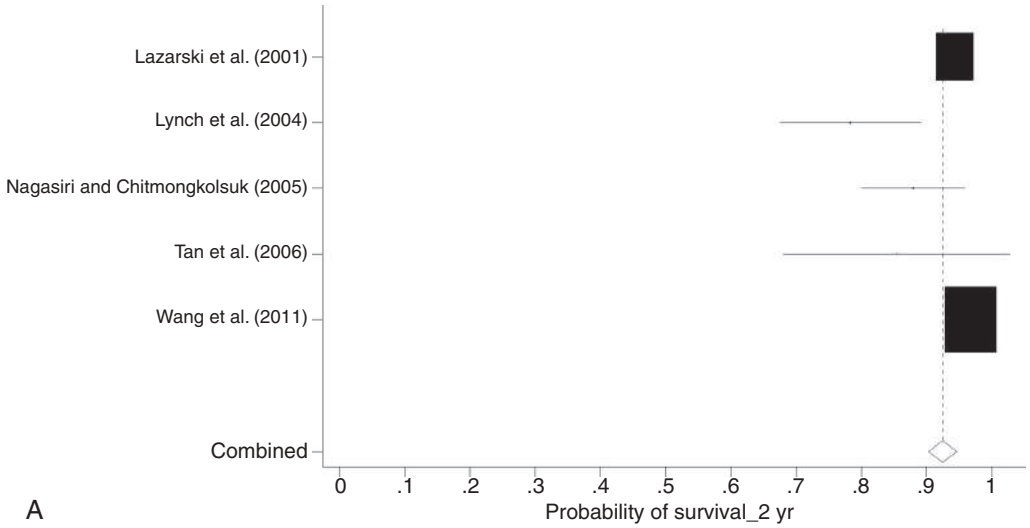


Fig. 17.25, cont'd

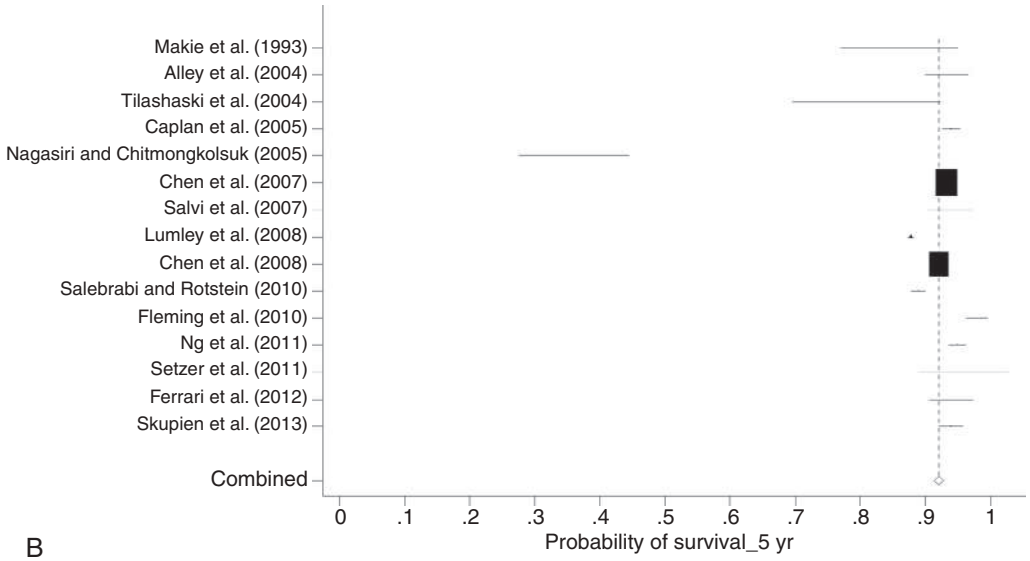
higher rate of loss are (1) failure of root canal treatment in a terminal tooth may be accepted more willingly because of little perceived aesthetic value and (2) clinicians may be less likely to offer further treatment on terminal molar teeth due to difficult access. Therefore when restoring molar teeth, the favorable distribution of occlusal forces must be considered, especially on teeth with one or fewer adjacent teeth or on terminal teeth.

Preoperative Conditions of Teeth

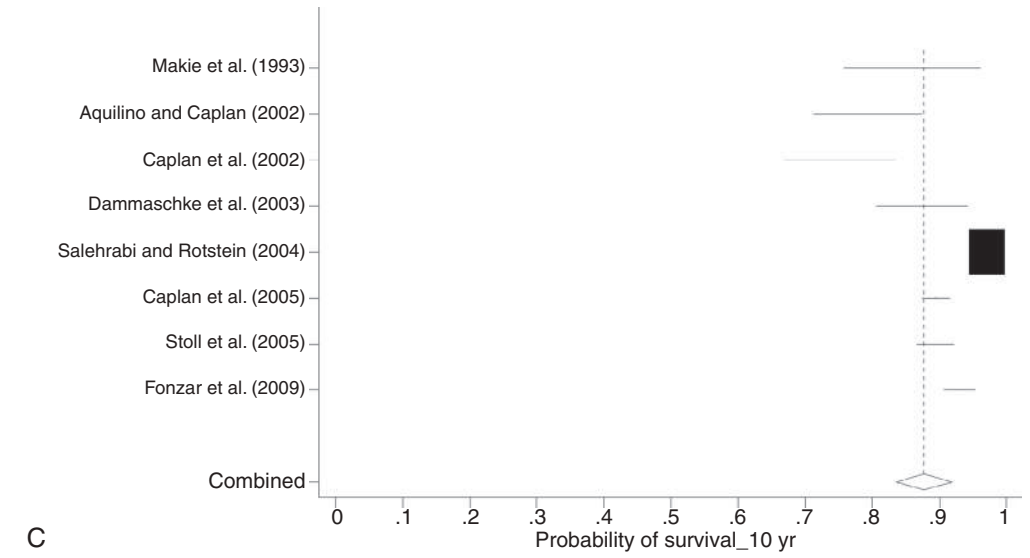
The presence of preoperative periapical lesions, which is the most significant prognostic factor for periapical healing, was found to have no significant influence on tooth survival.[151] On the other hand, the presence of preoperative periodontal probing defects of endodontic origin, preoperative pain, and preoperative sinus tracts were found to reduce tooth survival.[151] These observations are consistent



A



B



C

Fig. 17.26 Forest plot showing pooled and individual study's probability of 2-year (pooled probability = 0.93; 95% CI: 0.90, 0.95) **(A)**, 5-year (pooled probability = 0.92; 95% CI: 0.91, 0.93) **(B)**, and 10-year (pooled probability = 0.88; 95% CI: 0.84, 0.92) **(C)** survival of teeth undergoing root canal treatment.

with a previous report that the mere presence of a persistent periapical lesion was not a sufficient reason for dentists and patients to opt for further treatment, either retreatment or extraction.[180] The negative impact of preoperative pain on survival outcome highlights the importance of accurate pain diagnosis. In some instances, the pain may have been of nonendodontic origin and therefore would persist after treatment.[176] In other instances, preoperative pain of endodontic origin may persist following treatment, as a result of peripheral or central sensitization. Therefore effective pain diagnosis and management for patients presenting with preoperative pain are crucial.

The presence of preoperative cervical resorption and perforation was also found to significantly reduce tooth survival.[151] This was as expected because tooth fracture and reinfection due to leakage through the resorption and perforating defects are likely sequelae in such cases. In the presence of reinfection, clinicians are more inclined to suggest extraction due to the intuitive perception of poor long-term prognosis of such teeth.

Treatment Factors

Considering all of the intraoperative factors, the "lack of patency at apical foramen" and the "extrusion of gutta-percha root filling" were found to be the most significant intraoperative factors in reducing tooth survival.[151] In the presence of persistent problems and knowing that the treatment objective of cleaning to the canal terminus could not be achieved, patients and dentists may be more likely to opt for extraction sooner than later.

Restorative Factors

Protection of teeth with crowns or cast restorations has not been shown to influence periapical healing; however, the placement of good cores had a positive effect on endodontic outcome. In contrast, placement of crowns or cast restorations was found to improve tooth survival.[149,151,189] This suggests that crowns and cast restorations help prevent tooth fracture, whereas the mere placement of a satisfactory core is sufficient to prevent reinfection after endodontic treatment. Unfortunately, the study was not able to investigate the interrelationship between tooth morphologic type, the extent of tooth tissue loss after treatment, or the type of final restoration. Although the clinical inference from these findings is that cast restorations should preferably be placed on all teeth after root canal treatment, this is probably a gross exaggeration of the true need. On the basis of laboratory[178] and clinical findings,[144] posterior teeth with compromised marginal ridges (mesial or distal), together with heavy occlusal loading evidenced by faceting, may benefit from full coverage restorations. The restoration design should attempt to preserve as much remaining tooth tissue as possible; the implication is that the so-called nonaesthetic but technically demanding partial veneer onlays and partial coverage crowns would be the restorations of choice for root-treated teeth. In anterior teeth, the missing tooth tissue may often be adequately replaced with composite restorative materials. A crown is only indicated when tooth structure or aesthetics become compromised.

The use of cast post and cores for retention of restorations have also been found to reduce tooth survival.[149,151] It may be speculated that the presence of a post has different effects on anterior versus posterior teeth as they are subjected to different directions and amount of occlusal force. It has been reported that only 12% of the extracted teeth with cast post and cores were incisors or canines. Therefore the inference is that the use of such retention should be particularly avoided in premolar and molar teeth. Alternative treatment options should be considered for molar or premolar teeth lacking sufficient tooth structure.

Ng and colleagues[151] observed that teeth functioning as prosthetic abutments had poorer survival rates; however, the number of teeth ($n = 94$) in the study that functioned as abutments was too small a sample to be considered statistically significant. As previously described, the explanation may reside in the excessive and unfavorable distribution of occlusal stresses on abutment teeth. If possible, root-treated teeth should be avoided as abutments for prostheses or in provision of occlusal guidance in excursive movements.

Summary of Factors Influencing the Survival of Teeth Following Root Canal Treatment

The following conditions have been found to significantly improve tooth survival following root canal treatment:

- Nonmolar teeth (Fig. 17.27)
- Teeth with both mesial and distal adjacent teeth (Fig. 17.28)
- Teeth not located as the distal-most tooth in the arch[151]
- Teeth (molar) with cast restorations after treatment (Fig. 17.29)
- Teeth not requiring cast post and core for support and retention of restoration[111,151]
- Teeth not functioning as abutments for fixed prosthesis[3,111,191,151]
- Absence of preoperative deep periodontal probing defects, pain, sinus tract, or perforation[151]
- Achievement of patency at canal terminus and absence of root-filling extrusion during treatment[151]

In addition, it is important to ensure favorable distribution of occlusal forces when designing restorations for endodontically treated teeth.

IMPACT OF ROOT CANAL TREATMENT ON QUALITY OF LIFE

The impact of root canal treatment on the OHRQoL of patients has been evaluated using the short form (OHIP-14) or the modified version (OHIP-17) of the Oral Health Impact Profile (OHIP-14) (Table 17.10).[219] The distinctly positive impact of root canal treatment was apparent, regardless of cultural background of the patient group or the measure used.[55,73,87,115] As expected, physical pain, psychological discomfort (feeling tense), and disability (difficulty in relaxing) were the most improved domains following treatment.

Outcome of Nonsurgical Retreatment

When root canal treatment fails to resolve periapical disease, it is often considered appropriate to retreat the tooth using conventional approaches first, especially when the previous treatment is technically deficient (Fig. 17.30).

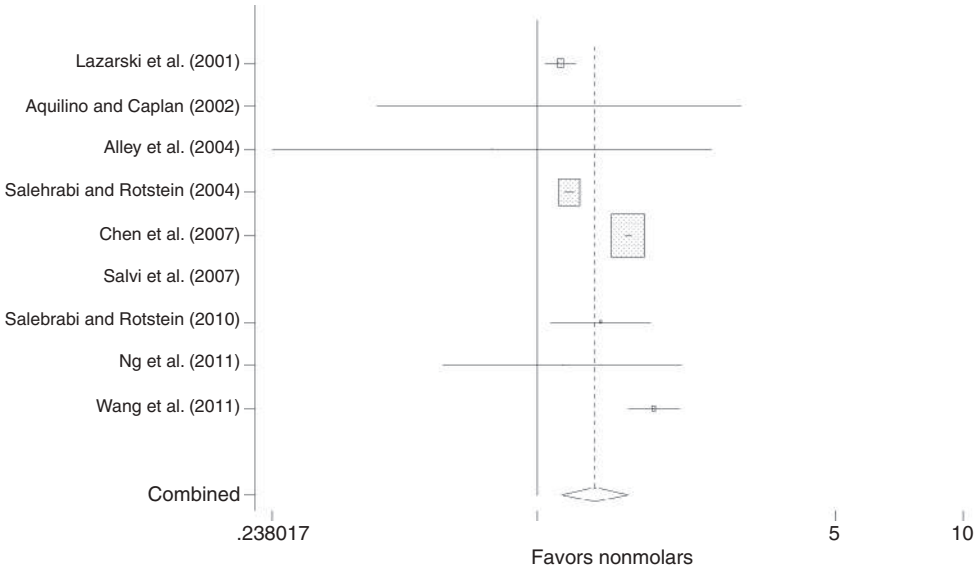


Fig. 17.27 Forest plot showing pooled and individual study's odds ratio (OR) of survival probability for nonmolar versus molar teeth (pooled OR = 1.4; 95% CI: 1.1, 1.6).

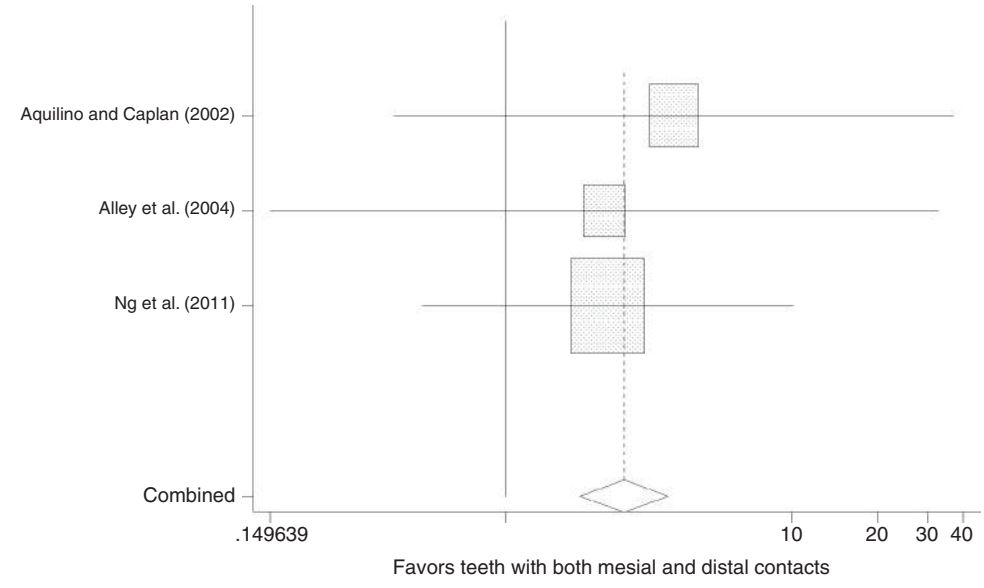


Fig. 17.28 Forest plot showing pooled and individual study's odds ratio (OR) of survival probability for teeth with both mesial and distal contacts present versus absent (pooled OR = 2.6; 95% CI: 1.8, 3.7).

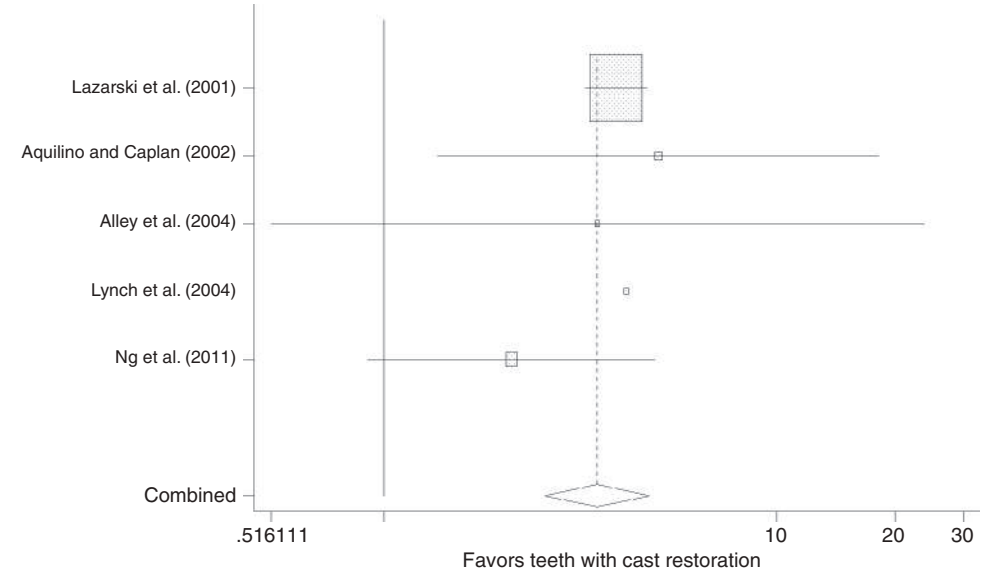


Fig. 17.29 Forest plot showing pooled and individual study's odds ratio (OR) of survival probability for teeth restored with cast versus plastic restoration (pooled OR = 3.5; 95% CI: 2.6, 4.7).

Table 17.10 The OHIP-14 and Modified Oral Health Impact Profile Instrument (OHIP-17)

OHIP-#	Item Question
OHIP-1	Have you had trouble pronouncing words because of your teeth and mouth?
OHIP-2	Have you felt that your sense of taste has worsened because of your teeth or mouth?
OHIP-3	Have you had painful aching in your mouth?
OHIP-4	Have you found it uncomfortable to eat any foods because of your teeth or mouth?
OHIP-5*	Have you had to alter the temperature of the foods that you eat because of your teeth or mouth?
OHIP-6	Have you been self-conscious because of your teeth or mouth?
OHIP-7	Have you felt tense because of your teeth or mouth?
OHIP-8	Has your diet been unsatisfactory because of your teeth or mouth?
OHIP-9	Have you had to interrupt meals because of your teeth or mouth?
OHIP-10	Have you found it difficult to relax because of your teeth or mouth?
OHIP-11*	Have you found it difficult to fall asleep because of your teeth or mouth?
OHIP-12*	Have you ever been awakened by problems with your teeth or mouth?
OHIP-13	Have you been embarrassed because of your teeth or mouth?
OHIP-14	Have you been irritable with other people because of your teeth or mouth?
OHIP-15	Have you had difficulty doing your usual jobs because of problems with your teeth or mouth?
OHIP-16	Have you felt that life in general was less satisfying because of your teeth or mouth?
OHIP-17	Have you been totally unable to function because of your teeth or mouth?

OHIP-14 contains all the items except those with an asterisk (*).
OHIP, Oral Health Impact Profile.
From Dugas NN, Lawrence HP, Teplitsky P, Friedman S: Quality of life and satisfaction outcomes of endodontic treatment, *J Endod* 28:819-827, 2002.

This requires removal of the previous root-filling material and any other material placed for restorative reasons. Correction of any procedural errors may also be required, if possible. All materials must be removed in their entirety to ensure delivery of antibacterial agents to all surfaces of the root canal dentin (see Fig. 17.30). The periapical healing rates of root canal retreatment are generally perceived to be lower compared to primary treatment for the following reasons:

- Obstructed access to the apical infection
- A potentially more resistant microbiota

The outcomes from a range of studies show that the mean weighted success rate of nonsurgical root canal retreatment is 66% (Figs. 17.31 and 17.32), which is about 6% lower than in the case of primary treatment on teeth with apical periodontitis.[148,150] However, it has also been shown that the survival rate of teeth having undergone nonsurgical root canal retreatment may be similar to that for primary root canal treatment.[151]

The factors affecting outcomes of periapical health and tooth survival following root canal retreatment are otherwise identical to those affecting primary root canal treatment. Of the potential prognostic factors unique to retreatment cases, the most significant factor influencing the outcome of treatment is the ability to remove or bypass preexisting root-filling material or separated instruments during retreatment. This is understandable because it would have a direct impact on the ability to achieve canal patency and canal disinfection at the apical terminus.[150]

Outcome of Surgical Retreatment

FACTORS AFFECTING PERIAPICAL HEALTH OR HEALING FOLLOWING PERIAPICAL SURGERY AND ROOT-END FILLING

There are a number of published systematic reviews on the prognostic factors for periapical surgery with apical filling.[45,47,48,155,204,205,239-241,250] The major limitations of this pooled data include the variable length of time for evaluating success following treatment as well as the radiographic criteria used to assess healing. An unpublished meta-analysis[130] of data from the studies listed in Table 17.11 reveals that the weighted pooled probability of success from periapical surgery with retrograde restorations, based on complete radiographic healing, is 67.5% (95% CI: 62.9%, 72.0%) (Fig. 17.33). The trend shows higher success rates reported in more recent studies. This observation is consistent with the much higher pooled success rate of 92% (95% CI: 86%, 95%) revealed in the meta-analysis of prospective outcome data of surgical endodontic treatments performed by a "modern" technique (using magnification, root-end resection with minimal or no bevel, retrograde cavity preparation with ultrasonic tips, and modern retrograde root canal filling materials).[240,241] In congruence with these findings, Setzer and colleagues reported that the pooled success rate of treatment using a microsurgical approach (94%; 95% CI: 89%, 98%) was more favorable than traditional root-end surgery (59%; 95% CI: 55%, 63%).[205] However, the studies included in the latter meta-analysis differed in design, case selection, duration after treatment, and the provision of preoperative nonsurgical treatment, which may bias the higher success rate of the contemporary treatment.

The unpublished meta-analysis[130] stratified by duration following treatment reveals that the pooled probability of success based on complete healing plateaued after 2 years postoperatively, being 51% (95% CI: 42%, 60%) after 6 months, 68% (95% CI: 63%, 73%) after 12 months, 76% (95% CI: 67%, 84%) after 24 months, 74% (95% CI: 52%, 95%) after 48 months, and 74% (95% CI: 66%, 82%) after more than 48 months. It is therefore advisable to follow up on cases that have undergone periapical surgery for a minimum period of 2 years and up to 4 years, as suggested in quality guidelines.[61]

The factors having a major impact on outcome of periapical surgery with retrograde cavities and fillings in teeth with preoperative periapical radiolucencies revealed in the

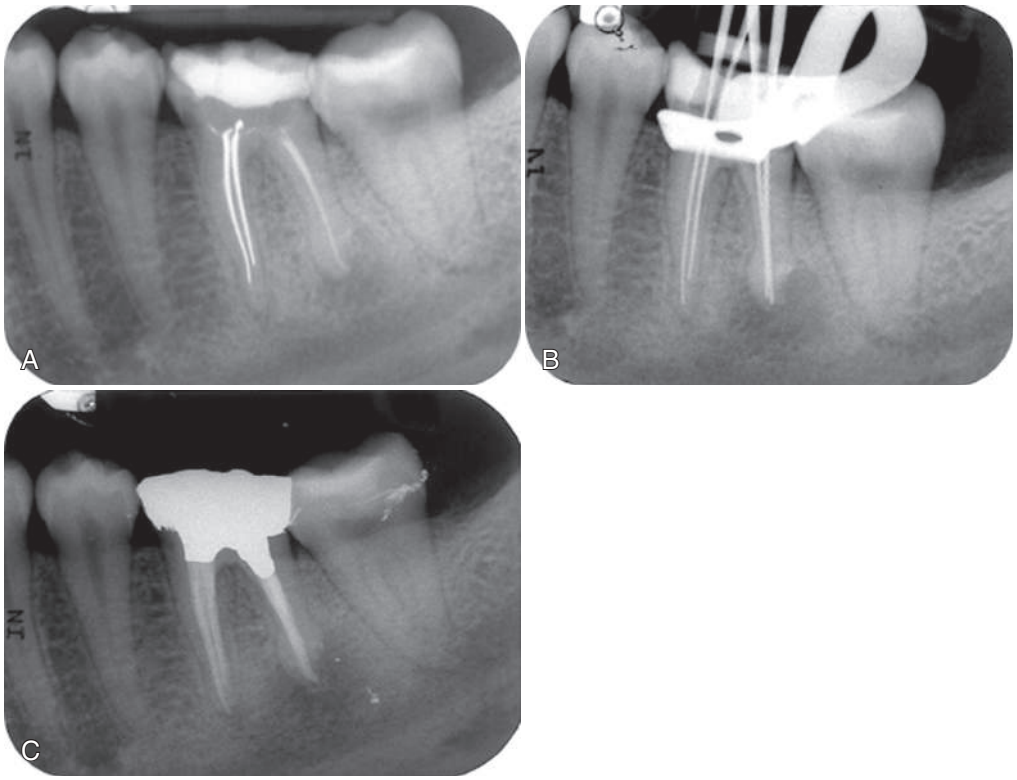


Fig. 17.30 Tooth with technically deficient root canal treatment **(A)**; having undergone (**B** and **C**) root canal retreatment.

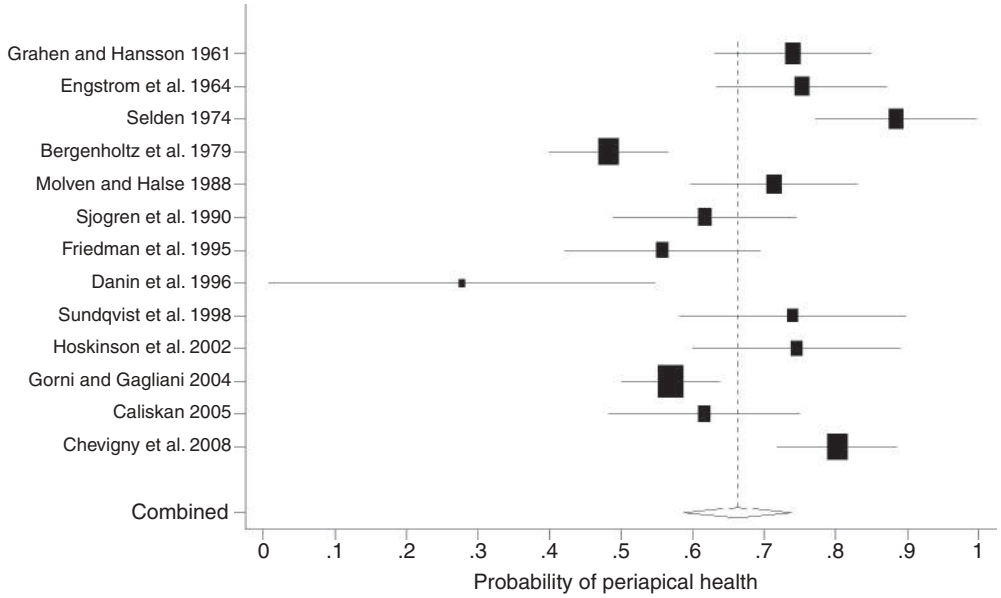


Fig. 17.31 Forest plot showing pooled and individual study's probability of complete periapical healing following root canal retreatment.

(as yet) unpublished meta-analyses[130] of data from studies listed in Table 17.11 are as follows:

- Small (≤5 mm) versus large (>5 mm) periapical lesion (risk ratio = 1.2; 95% CI: 1.1, 1.3)
- Periapical lesion involving one versus both cortical plates (risk ratio = 1.2; 95% CI: 1.0, 1.5)
- Absence versus presence of previous surgery (risk ratio = 1.2; 95% CI: 1.1, 1.3)
- Using magnification versus without the use of magnification during surgery (risk ratio = 1.5; 95% CI: 1.3, 1.8)
- Root-end resection with minimum versus obvious bevel (risk ratio = 1.3; 95% CI: 1.2, 1.4)
- Use of ultrasonic tip versus bur for retro-cavity preparation (risk ratio = 1.3; 95% CI: 1.2, 1.4)
- Use of retro-filling material with mineral trioxide aggregate (MTA) cement, super ethoxybenzoic acid (EBA)

Table 17.11 Studies Investigating Periapical Healing Following Apical Surgery

Author	Design	Examination	Type of Radiograph	Radiographic Criteria for Success	SAMPLE SIZE			Duration After Treatment (Months)
					Patients	Teeth	Roots	
Harty et al., 1970	Retrospective	C & R	Pa	Other		169		6–60
Nordendram, 1970	Prospective	C & R	Pa	Rud et al., 1972	66	66		6–24
Rud et al., 1972	Prospective	R	Pa	Rud et al., 1972		237		12–180
Finne et al., 1977	Prospective	C & R	Pa	Persson, 1973	156	218		36
Hirsch et al., 1979	Prospective	C & R	Pa	Rud et al., 1972	467	467		6–36
Ioannides and Borstlap, 1983	Retrospective	C & R	Pa	Other	50	50	45	6–60
Allen et al., 1989	Retrospective	C & R	Pa	Rud et al., 1972		175		12–60
Amagasa et al., 1989	Prospective	C & R	Pa	Other	42	64		12–90
Dorn and Gartner, 1990	Retrospective	R	Pa	Other		488		6–120
Grung et al., 1990	Prospective	C & R	Pa	Rud et al., 1972		161		12–96
Lustmann et al., 1991	Retrospective	C & R	Pa	Rud et al., 1972			123	6–96
Rapp et al., 1991	Retrospective	R	Pa	Rud et al., 1972	331	226		6–24
Rud et al., 1991	Retrospective	C & R	Pa	Rud et al., 1972	388	388		12
Waikakul et al., 1991	Prospective	C & R	Pa	Other	34	62		6–24
Pantschev et al., 1994	Prospective	C & R	Pa	Persson, 1973	79	103		36
Jesselen et al., 1995	Prospective	C & R	Pa	Other	67	82		12–60
Rud et al., 1996	Prospective	R	Pa	Rud et al., 1972			347	12–48
Sumi et al., 1996	Retrospective	C & R	Pa	Other	86	157		6–36
Jansson et al., 1997	Retrospective	C & R	Pa	Other	59	59		11–16
Testori et al., 1999	Retrospective	R	Pa	Rud et al., 1972	130	181	302	12–72
Von Arx and Kurt, 1999	Prospective	C & R	Pa	Von Arx and Kurt, 1999	38	43		12
Zuolo et al., 2000	Prospective	C & R	Pa	Molven et al., 1987	106	102		12–48
Pecora et al., 2001	RCT	R	Pa	Rud et al., 1972	20	20		6
Penarrocha et al., 2001	Retrospective	R	Pa	Von Arx and Kurt, 1999	30	31	71	12
Rahbaran et al., 2001	Retrospective	C & R	Pa	Other	154	154		48–108
Rud et al., 2001	Prospective	C & R	Pa	Rud et al., 1972		520	834	6–150
Von Arx et al., 2001	Prospective	C & R	Pa	Other	24	25	39	12
Jensen et al., 2002	RCT	C & R	Pa	Rud et al., 1972		122		
Rubinstein and Kim, 2002	Prospective	C & R	Pa	Rud et al., 1972	52	59	59	68
Tobon et al., 2002	RCT	R	Pa	Rud et al., 1972	25	26		12
Vallecillo et al., 2002	Prospective	R	Pa	Other	29	29		12
Chong et al., 2003	RCT	C & R	Pa	Molven et al., 1987	86	86		12–24
Maddalone and Gagliani, 2003	Prospective	C & R	Pa	Molven et al., 1987	79	120		3–36
Schwartz-Arad et al., 2003	Prospective	R	Pa	Other	101	122		6–45
Platt et al., 2004	RCT	C & R	Pa	Molven et al., 1987	28	34		12
Gagliani et al., 2005	Prospective	C & R	Pa	Rud et al., 1972	164	168	231	60
Lindeboom et al., 2005	RCT	C & R	Pa	Rud et al., 1972	100	100		12
Marti-Bowen et al., 2005	Retrospective	C & R	DPT	Von Arx and Kurt, 1999	52	71	95	6–12
Taschieri et al., 2005	Prospective	C & R	Pa	Rud et al., 1972	32	46		12
Taschieri et al., 2007	Prospective	C & R	Pa	Molven et al., 1987	28	28		12

Table 17.11 Studies Investigating Periapical Healing Following Apical Surgery—cont'd

Author	Design	Examination	Type of Radiograph	Radiographic Criteria for Success	SAMPLE SIZE			Duration After Treatment (Months)
					Patients	Teeth	Roots	
Tsesis et al., 2006	Retrospective	C & R	Pa	Rud et al., 1972	71	88		6–48
Marin-Botero et al., 2006	RCT	C & R	Pa	Rud et al., 1972	30	30		12
Taschieri et al., 2006	RCT	C & R	Pa	Molven et al., 1987	53	71		12
De Lange et al., 2007	RCT	C & R	Pa	Rud et al., 1972	290	290		12
Leco-Berrocal et al., 2007	Prospective	R	Pa and DPT	Other	45	45		6–24
Penarrocha et al., 2007	Prospective	C & R	Pa	Von Arx and Kurt, 1999	235	333	384	6–144
Taschieri et al., 2007a	Prospective	C & R	Pa	Molven et al., 1987	17	27		12
Taschieri et al., 2007b	Prospective	C & R	Pa	Molven et al., 1987	41	59		12
Von Arx et al., 2007	Prospective	C & R	Pa	Rud et al., 1972	200	177		12
Walivaara et al., 2007	Prospective	C & R	Pa	Rud et al., 1972	54	55		12
Garcia et al., 2008	Prospective	C & R	DPT	Von Arx and Kurt, 1999	92	106	129	6–12
Kim et al., 2008	Prospective	C & R	Pa	Molven et al., 1987		148		24
Penarrocha et al., 2008	Prospective	C & R	DPT	Von Arx and Kurt, 1999	278	278		12
Taschieri et al., 2008b	Prospective	C & R	Pa	Molven et al., 1987	27	31		12
Taschieri et al., 2008a	RCT	C & R	Pa	Molven et al., 1987	61	100		25
Christiansen et al., 2009	RCT	C & R	Pa	Molven et al., 1987		25		12
Dominiak et al., 2009	Retrospective	C & R	Pa	Other	106	106		12
Ortega-Sanchez et al., 2009	Retrospective	C & R	DPT	Von Arx and Kurt, 1999	30	30	37	
Pantschev et al., 2009	Retrospective	C & R	Pa	Other		147		12
Waalivaara et al., 2009	RCT	C & R	Pa	Molven et al., 1987	131	147		12
Barone et al., 2010	Prospective	C & R	Pa	PAI		129		48–120
Garcia-Mira et al., 2010	Retrospective	C & R	DPT	Von Arx and Kurt, 1999	75	87		12
Taschieri et al., 2010	Retrospective	C & R	Pa	Molven et al., 1987	76	112		48
Von Arx et al., 2010	Prospective	C & R	Pa	Rud et al., 1972		339		12
Goyal et al., 2011	RCT	C & R	Pa	Rud et al., 1972	25	25		12
Penarrocha et al., 2011	Retrospective	C & R	DPT	Von Arx and Kurt, 1999	150	178	178	12
Song et al., 2011a	Retrospective	C & R	Pa	Molven et al., 1987		441		3–12
Song et al., 2011b	Prospective	C & R	Pa	Molven et al., 1987	42	42		12
Taschieri et al., 2011	Retrospective	C & R	Pa	Molven et al., 1987	33	43		12–48
Waalivaara et al., 2011	RCT	C & R	Pa	Molven et al., 1987	153	194		12–21
Penarrocha et al., 2012	Prospective	C & R	DPT	Von Arx and Kurt, 1999	23	31		12–19
Song and Kim, 2012	RCT	C & R	Pa	Molven et al., 1987	192	192		12
Von Arx et al., 2012	Prospective	C & R	Pa	Rud et al., 1972		170		12–60
Kreisler et al., 2013	Prospective	C & R	Pa	Rud et al., 1972	255	281		6–12
Penarrocha et al., 2013	Retrospective	C & R	Pa and DPT	Von Arx and Kurt	96	139		6–12
Song et al., 2013a	Prospective	C & R	Pa	Molven et al., 1987		344		12–120
Song et al., 2013b	Prospective	C & R	Pa	Molven et al., 1987		135		12
Song et al., 2014	Retrospective	C & R	Pa	Rud et al., 1972		115		12–96
Taschieri et al., 2013	Retrospective	C & R	Pa	Molven et al., 1987		86		6–12
Villa-Machado et al., 2013	Retrospective	C & R	Pa	PAI	154	171		12–192
Li et al., 2014	Retrospective	C & R	Pa	Molven et al., 1987	82	101		48

C, Clinical; *DPT*, dental pantomogram; *Pa*, periapical; *PAI*, periapical index; *R*, radiographic; *RCT*, randomized controlled trial.

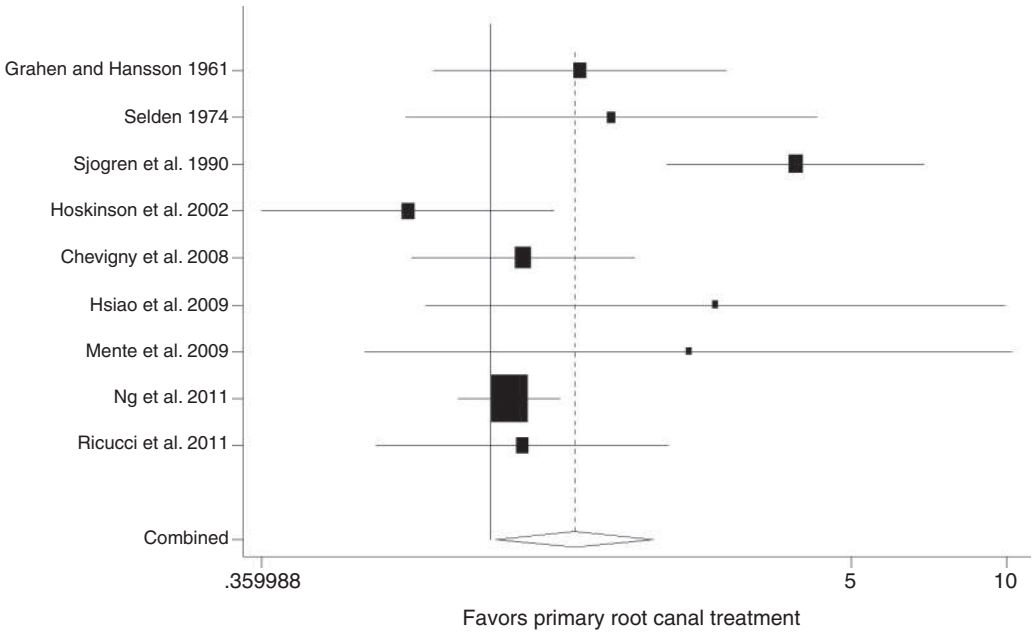


Fig. 17.32 Forest plot showing pooled and individual study's odds ratio of periapical health for teeth with preoperative periapical radiolucency undergoing primary root canal treatment versus root canal retreatment (pooled OR = 1.5; 95% CI: 1.0, 2.1).

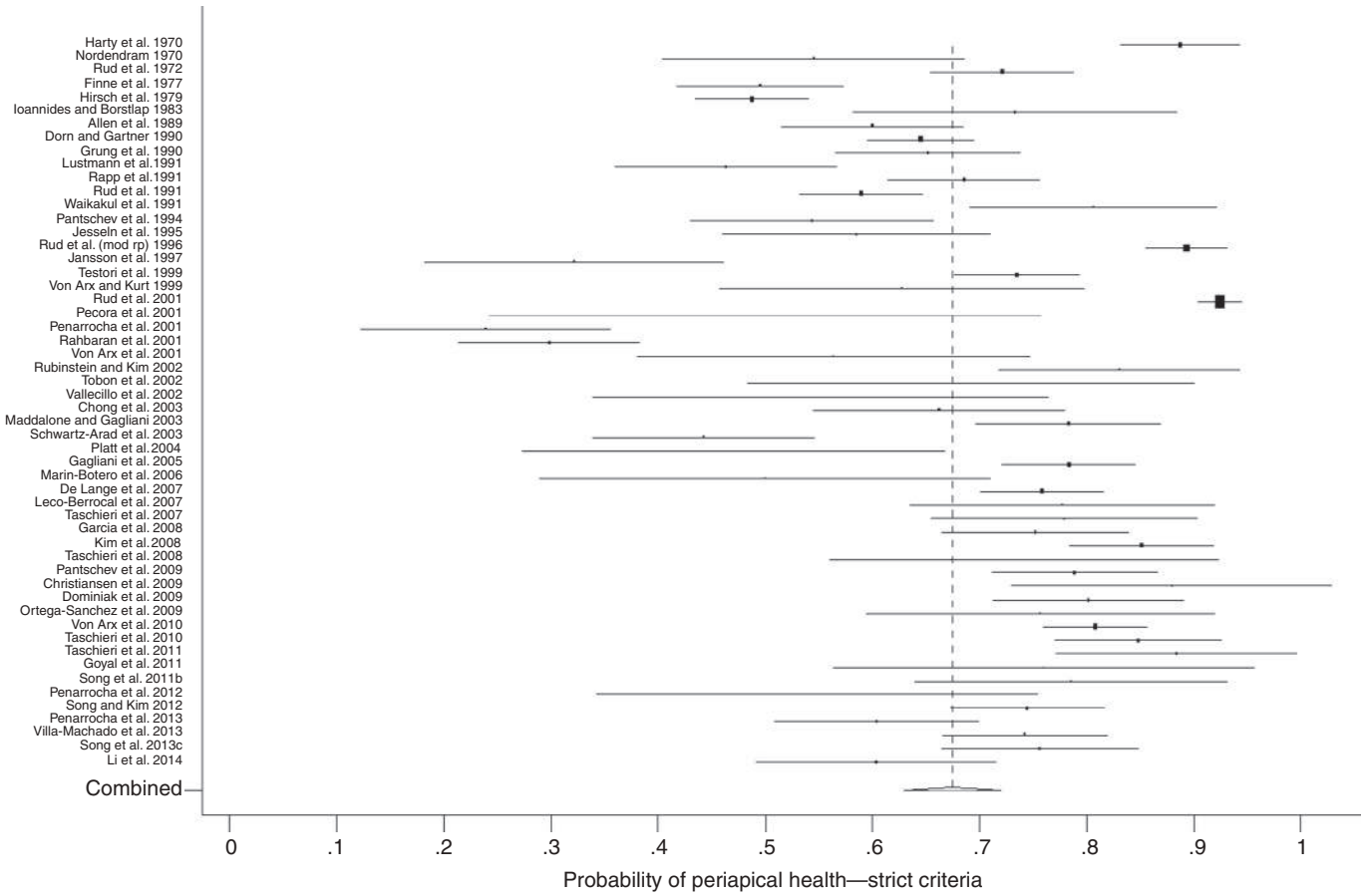


Fig. 17.33 Forest plot showing pooled and individual study's probability of complete periapical healing following apical surgery.

cement, or intermediate restorative material (IRM) versus amalgam; the use of MTA resulted in a similar outcome to SuperEBA (risk ratio = 1.0; 95% CI: 0.99, 1.1) or IRM (risk ratio = 1.1; 95% CI: 0.98, 1.1); SuperEBA and IRM were associated with significantly higher chances of success than amalgam (risk ratio = 1.2; 95% CI: 1.1, 1.3) as the retro-filling material

The preoperative presence of signs or symptoms (risk ratio = 1.2; 95% CI: 1.1, 1.3),[249] periodontal status (risk ratio = 2.1; 95% CI: 1.1, 3.8),[222] and quality of the coronal restoration (risk ratio = 1.6; 95% CI: 1.2, 2.1)[249] have also been revealed as significant prognostic factors by individual studies.

The following factors have minimal effect on surgical retreatment[43]:

- Age of patient
- Gender of patient
- General health of patient
- Tooth type
- Quality of the preexisting root canal filling, as judged radiographically
- Histologic diagnosis of the biopsied periapical lesion (cyst or granuloma)

The use of guided periodontal tissue regenerative membrane or grafting material has been advocated for cases with through-and-through defects (missing buccal and palatal cortical plates), but there are conflicting reports as to their benefits.[53,162,170,232-234,237,249] The unpublished meta-analyses[130] of data from these studies did not reveal a significant influence of such approaches on the periapical healing outcome.

FACTORS AFFECTING PERIODONTAL INCISIONAL WOUND HEALING

Periodontal attachment level after periapical surgery is an additional physical outcome measure compared to nonsurgical root canal treatment. Studies have compared the effect of different soft-tissue incision techniques (e.g., intrasulcular with or without involving interproximal papilla, submarginal, papilla-base).[106,245-247] As expected, all concluded that negligible marginal recession could be achieved by adopting a flap design that avoided reflection of the interproximal papilla.

The improvement in outcomes of periapical surgery has been attributed to modern surgical techniques plus greater biologic awareness of clinicians.[205,250] In addition, and perhaps even more importantly, case selection may be more critical in excluding potential failures, inferring that prognostication may have improved.

FACTORS AFFECTING TOOTH SURVIVAL FOLLOWING PERIAPICAL SURGERY AND ROOT-END FILLING

Unlike nonsurgical root canal treatment, the outcome measure of tooth survival had only been adopted by one study[253] on surgical cases at the time of this writing. The reported median survival time for first-time surgery was 92.1 months (95% CI: 40.9, 143.4) and that for resurgery was 39.1 months (95% CI: 6.1, 72.1).[253] The failure events, however, included tooth extraction and clinical and radiographic signs of periapical disease after treatment.

IMPACT OF PERIAPICAL SURGERY ON QUALITY OF LIFE

The impact of periapical surgery on the patient's quality of life has only been evaluated using a questionnaire including three domains: physical function (chewing, talking, sleeping, daily routine, and work), physical pain, and other physical symptoms (swelling, bleeding, nausea, bad taste/breath).[49] It was concluded that the papilla-base incision flap design resulted in a lower impact on physical pain and other symptoms within the first week postoperation. The impact on physical pain has also been explored by two other studies[35,36]; both reported the postoperative pain was of relatively short duration, with the intensity peaking at 3 to 5 hours postoperatively and progressively decreasing with time with no significant influencing factors identified.

Concluding Remarks

The procedures used to maintain pulp vitality and for prevention and treatment of periapical disease are able to achieve excellent outcomes. The outcome data and potential prognostic factors should be considered during treatment options appraisal and planning. Despite the fact that most important prognostic factors are beyond the control of clinicians, optimal outcomes for individual cases may still be achieved by performing the procedure to guideline standards. From a health and economic perspective, conventional root canal treatment is a highly cost-effective treatment as a first-line intervention to extend the life of the tooth with a preoperative periapical lesion.[103,171] If a root canal treatment subsequently fails, nonsurgical and surgical retreatments are also more cost-effective than replacement with a prosthesis.[103,171] Ultimately, all sources of evidence must be assessed for biasing influences based on the local cultures, expertise, treatment predilection, and funding sources.

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